

A UNIFORM HILBERT TRANSFORM ESTIMATE FOR LIPSCHITZ DIRECTIONS

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ABSTRACT. We prove a uniform L^2 bound for the Hilbert transform along a Lipschitz unit vector field in the plane, with integration restricted to a fixed absolute multiple of the reciprocal Lipschitz seminorm. The bound is uniform in the inner truncation and holds for fields depending on both coordinates. This gives an affirmative answer to Stein's weak- $(2, 2)$ conjecture.

CONTENTS

1. Short transforms and band reduction	3
1.1. The problem and its predecessors	3
1.2. The analytic and geometric reductions	4
1.3. The phase gain and its iteration	5
1.4. Organization, inputs and uniformity	6
1.5. Fourier conventions and the local reduction	6
1.6. A maximal theorem and elementary analytic tools	6
1.7. The local slope operator and its endpoints	7
1.8. The band theorem and summation of amplitude levels	8
1.9. A Lipschitz commutator	9
1.10. Square-function reassembly	12
1.11. Removing long and very short times	13
1.12. From a spatial truncation to a finite scale sum	14
1.13. Narrow vertical bins	15
1.14. The admissible packet forms	16
1.15. The precise remaining model estimates	19
2. Ordinary tiles and the popular forest	20
2.1. Density and the model estimates	21
2.2. Maximal functions and separated synthesis	21
2.3. Removing large size	23
2.4. Removing large density	25
2.5. The tree estimate	26
2.6. Proof of the model estimates	28
2.7. The improvement problem and popular tops	29
2.8. The assignment to popular tops	29
2.9. The adjoint sum and exceptional subsets	30
2.10. A weak L^2 estimate for the top count	30
2.11. Parameters and masks	33
3. Reference-line concentration and significant directions	34
3.1. Positive estimates and the main coefficients	35
3.2. Fourier sums and a dyadic martingale	37

3.3.	Every-line moment bounds	38
3.4.	Uniform top bounds and neighborhood derivatives	41
3.5.	The masked sum and the count threshold	42
3.6.	A simultaneous count of separated significant directions	43
3.7.	A counting lemma on a line	44
3.8.	Transferring witnesses to a reference line	46
3.9.	The common exceptional set and the choice of D	47
4.	A strict finite Fourier inequality with constant one	48
5.	Angular hierarchy and predicted factors	52
5.1.	Eligible masses and shifted blocks	52
5.2.	The analytic family and its block comparisons	54
5.3.	Full-root variation and amplitude bounds	56
5.4.	Spatial meshes for prediction	57
5.5.	Positive-mass motion and the locality of other shifts	58
5.6.	Variation of moving endpoint weights	59
5.7.	Amplitude motion from one good snapshot	60
5.8.	One deterministic residual sequence for each path	61
5.9.	Mass stopping and the exact path expansion	63
5.10.	Path drops and marked spatial cells	65
5.11.	Prediction and averaging on marked cells	66
5.12.	Snapshot data and denominator shrinkage	67
5.13.	Exact averaging of the prediction	68
5.14.	Comparison to the actual factor	69
5.15.	Absorbing the baseline error by blending	71
5.16.	Changing to uniform measure without a loss per level	71
5.17.	Step supremums and normalization	72
6.	Residual terms and integration along paths	73
6.1.	Marked residuals	74
6.2.	Unary segments and their energy telescope	75
6.3.	Common residual weights and variation	77
6.4.	The path product and its integral	79
7.	Interpolation, boundary averaging, and completion	80
7.1.	From the path estimate to analytic interpolation	80
7.2.	The central weight and the exponent budget	82
7.3.	Boundary averaging with the eligibility condition	83
7.4.	Recovering the original adjoint sum	84
7.5.	Choice of parameters and completion of the reductions	85
	References	87

1. SHORT TRANSFORMS AND BAND REDUCTION

Let $v : \mathbb{R}^2 \rightarrow S^1$ be a Lipschitz unit vector field. For $0 < \epsilon < a$, define the short directional Hilbert transform by

$$H_{v,a}^\epsilon f(x) = \int_{\epsilon < |t| < a} f(x - tv(x)) \frac{dt}{t}.$$

The direction is held fixed at the output point x : the integral samples the straight line through x in direction $v(x)$. For $f \in \mathcal{S}(\mathbb{R}^2)$, the symmetric principal value $H_{v,a} f = \lim_{\epsilon \downarrow 0} H_{v,a}^\epsilon f$ exists at every point, since $f(x - tv(x)) - f(x + tv(x)) = O_x(|t|)$ near zero.

Theorem 1.1. *There are absolute constants $a_* \in (0, 1/2)$ and $C_* < \infty$ such that every Lipschitz field $v : \mathbb{R}^2 \rightarrow S^1$ with $\text{Lip}(v) \leq 1$ satisfies*

$$\sup_{0 < \epsilon < a_*} \|H_{v,a_*}^\epsilon f\|_{L^2(\mathbb{R}^2)} \leq C_* \|f\|_{L^2(\mathbb{R}^2)} \quad (f \in \mathcal{S}(\mathbb{R}^2)).$$

Its symmetric principal value therefore satisfies

$$\|H_{v,a_*} f\|_2 \leq C_* \|f\|_2, \quad |\{x : |H_{v,a_*} f(x)| > \lambda\}| \leq C_*^2 \lambda^{-2} \|f\|_2^2 \quad (\lambda > 0).$$

Both coordinates may enter the field arbitrarily subject to the Lipschitz bound. The supremum in the theorem is outside the norm, so its first assertion is a uniform operator bound for each inner truncation. The principal-value assertion is stated on Schwartz functions and gives an L^2 -bounded extension by density.

The outer length has a natural scaling. If $L_v = \text{Lip}(v) > 0$, put $\tilde{v}(X) = v(X/L_v)$ and $\tilde{f}(X) = f(X/L_v)$. Then $\text{Lip}(\tilde{v}) = 1$, and changing variables $s = L_v t$ gives

$$H_{v,a_*/L_v}^\epsilon f(x) = H_{\tilde{v},a_*}^{L_v \epsilon} \tilde{f}(L_v x), \quad \|\tilde{f}\|_2 = L_v \|f\|_2.$$

The theorem thus has the same constant at outer length a_*/L_v . If $L_v = 0$, the field is constant. Rotation and Fubini reduce its transform, for any finite outer length, to the one-dimensional Hilbert transform. These conventions distinguish the short-scale assertion from an estimate at arbitrary outer lengths for varying fields.

1.1. The problem and its predecessors. Stein's question places Lipschitz regularity at the center of the planar directional singular-integral problem. In the formulation [21, Conjecture 1.5], it asks for a uniform weak-(2,2) bound with the outer length controlled by the reciprocal Lipschitz seminorm. Theorem 1.1 gives an affirmative answer with a strong L^2 bound. The scale restriction limits the turning of the field, but it does not make the operator a Fourier multiplier: the line used by the integral still varies with the output point.

Lacey and Li trace Stein's question to Zygmund's conjecture on almost-everywhere differentiation by averages over shrinking straight segments in direction $v(x)$. They also recall a Besicovitch obstruction showing that Hölder regularity of any exponent below one does not by itself give the corresponding uniform weak-(2,2) estimate for the short positive averaging maximal operator [21, Preface, Conjecture 1.2 and Chapter 2]. Analytic fields admit additional structure: Bourgain proved an L^2 theorem for their averaging maximal operator [4], and Stein and Street proved local L^p , $1 < p < \infty$, estimates for single-parameter singular Radon transforms associated with real-analytic maps [26]. Their singular-integral theorem includes analytic frozen-direction Hilbert transforms, with localization depending on the analytic map. The uniform Lipschitz estimate considered here asks for constants controlled by the Lipschitz seminorm alone.

Several distinct advances explain the remaining difficulty. Lacey and Li proved an annular weak-(2,2) estimate, together with strong L^p estimates for $p > 2$, for directional Hilbert transforms with measurable directions [23, Theorem 1.1]. An annular restriction isolates a

frequency scale; it does not give a uniform strong L^2 bound for the complete transform. For Lipschitz unit fields, Guo obtained annular weak- $(2, 2)$ and strong L^p , $p > 2$, bounds, uniformly in the annulus scale, for the r -variation, $r > 2$, of smooth dyadic Hilbert-transform partial sums over the short scale range determined by the reciprocal Lipschitz length; the varying dyadic cutoff is distinct from the hard inner time cutoff in [Theorem 1.1](#) [[15](#), [Theorem 1.2](#) and (1.6)–(1.8) of the author version]. Lacey and Li’s Lipschitz Keakeya theorem controls averages over short rectangles on which the field has a prescribed popular direction [[20](#), [Theorem 1.4](#)]. For popularity δ , its weak- L^2 norm costs $C\delta^{-1/2}$. We use precisely this estimate.

The Lacey–Li memoir develops the relation between those maximal operators and the Hilbert-transform problem [[21](#)]. Its [Theorem 1.18](#) assumes a stronger maximal estimate below L^2 for the strong annular conclusion; its full-transform conclusion also requires $C^{1+\eta}$ regularity. These are conditional results with hypotheses different from those of [Theorem 1.1](#). For bounded measurable slopes depending on one coordinate, Bateman proved single-annulus L^p estimates for $1 < p < \infty$ [[2](#)]. Bateman and Thiele proved strong L^p bounds for the full-line principal-value transform when $3/2 < p < \infty$, for every nonvanishing measurable field depending on one coordinate [[3](#), [Theorem 1](#)]. The one-coordinate L^2 case is already connected to Carleson’s theorem by a partial Fourier transform [[6](#), [3](#)]; Bateman and Thiele obtain the stated wider p -range. The one-coordinate hypothesis remains a substantive restriction on the direction field. Guo extended the one-coordinate setting to fields constant along suitable Lipschitz foliations, with a uniform transversality assumption [[13](#), [14](#)].

A different family of results combines Lipschitz control with lacunary directions. Guo and Thiele proved strong L^p bounds without a frequency restriction for short principal-value transforms associated with dyadic roundings of a positive Lipschitz slope [[16](#)]; Di Plinio and Parissis extended this framework to finite lacunary order for fields generated by separated dyadic roundings of Lipschitz angular functions [[10](#)]. The rounded direction field can itself be discontinuous. These results exploit directional structure beyond that assumed in the present theorem.

A further reduction separates frequency reassembly from the main single-band estimate. Di Plinio, Guo, Thiele and Zorin-Kranich established that uniform strong L^2 bounds on vertical frequency bands imply the full short principal-value bound under a small uniform Lipschitz bound in the vertical variable [[9](#), [Corollaries 1.3–1.4](#) of the author version]. We give a commutator proof adapted to the two truncation endpoints and to the narrow frequency bins used later. The new estimate to be proved is the logarithmically improved restricted bound on a band; the principle of band-to-full reassembly is already present in that work.

1.2. The analytic and geometric reductions. We describe the proof before fixing its technical parameters. Localization and rotation write a short piece of the operator in slope form

$$H_{u,a}^\epsilon h(x, y) = \int_{\epsilon < |t| < a} h(x - t, y - tu(x, y)) \frac{dt}{t}, \quad |u| \leq 1, \quad \text{Lip}(u) \leq C,$$

with both input and test supported in a short horizontal strip. Writing η for the frequency dual to y , a vertical frequency band at scale w means that $|\eta|$ is comparable to w^{-1} . For its smooth projection P_w , the central restricted estimate is

$$|\langle H_{u,a}^\epsilon P_w f, g \rangle| \lesssim (1 + |\log(|F|/|E|)|)^{-3} \sqrt{|F||E|}, \quad |f| \leq \mathbf{1}_F, \quad |g| \leq \mathbf{1}_E.$$

Here F, E are sets of positive finite measure in the horizontal strip. Its constants are uniform in w and ϵ . The logarithmic factor makes amplitude-level summation converge. A Lipschitz commutator then permits Littlewood–Paley reassembly across vertical bands. These implications,

including all endpoint errors, are proved in this section before the restricted estimate itself is used.

The difficult regime is $|E| \gg |F|$. Write $L = \log \sqrt{|E|/|F|}$. Positive operator bounds discard all but very short integration times. The vertical band is then divided into narrow intervals centered at frequencies η_j , with common width comparable to W^{-1} , where $W = mw$ and m grows subexponentially in L . The components form one finite-dimensional Hilbert space. A wave packet has horizontal length ℓ , transverse width W , and a direction label with precision $d = w/\ell$. Smooth packet forms retain the original field only through scalar selectors of its direction.

Ordinary size and density estimates remove the small-density packets. Each remaining packet is assigned to a top box whose direction is popular on a measured subset of the test set. The popularity witnesses are disjoint. Lacey–Li’s theorem controls a decaying weighted count of the tops, after every thin parallelogram has been enclosed by an admissible orthogonal rectangle. This step is geometric: it supplies a polynomial cost in L , and does not provide the final oscillatory saving.

1.3. The phase gain and its iteration. Two statements make the oscillation usable. First, on every reference line the packet sums have polynomial moment bounds, square-function bounds and 3-variation bounds. The variation argument combines Lépingle’s martingale inequality with the comparison between smooth Fourier truncations and conditional expectations developed by Jones, Seeger and Wright [24, 19]. All three packet estimates hold for every fixed deterministic subcollection. Popularity and Lipschitz control then produce one exceptional set outside which each separated list of significant top directions in the required angular cluster and precision range has at most $D = L^C$ members, simultaneously over angular resolutions.

Second, a finite Fourier sum with at most D separated real frequencies has a strict improvement over the usual coefficient exponent. For suitable band-limited probability measures ν , we prove

$$\left\| \sum_{i=1}^d a_i e^{i\lambda_i t} \right\|_{L^p(\nu)} \leq \left(\sum_{i=1}^d |a_i|^{r_*} \right)^{1/r_*}, \quad \frac{1}{r_*} = 1 - \frac{1 + c/\log(2D)}{p}, \quad p \geq 4.$$

The operator constant is exactly one, for complex coefficients and for Hilbert-valued coefficients acted on by diagonal phases. The proof separates the case of one dominant coefficient from its complement. In the latter case, separated quantiles give a strict fourth-moment deficit, which survives a small improvement of the coefficient exponent. An unspecified constant in this inequality would accumulate at every angular level and destroy the argument.

The iteration organizes directions into nested angular intervals. Each interval carries the positive mass of the tops whose direction precision is sufficiently fine for that interval; these are its eligible tops. It also carries an oscillatory amplitude assembled from its packets. The significant-direction count enters phase averaging through a weighted selection of separated block children, whose analytic weights cancel the probability cost of retaining them. We compare mass-normalized p -energies after adding the mass itself as a baseline. On an appropriate spatial interval, a finer amplitude can be predicted from one good point by diagonal phase motion. The finite Fourier estimate therefore controls the average of the predicted energy factor. The comparison errors are proportional to the relative mass lost on the path, so their sum is controlled by a logarithmic mass telescope.

Packets that stop between two angular endpoints create residual terms. Square summation handles blocks with a definite mass drop. Across nearly unary portions of a path, the energies telescope and the remaining residual sums are estimated by 3-variation of one fixed sequence for that path. The sequence is chosen before the spatial point or mass stop is selected. Normalized

factors on nested spatial cells can then be integrated without any cost proportional to the angular depth.

Finally we interpolate among shifted groupings of angular levels. At the central parameter the packet coefficients contain boundary weights. Averaging one common angular-grid translation removes these weights, using eligibility as well as angular proximity in the boundary telescope. With $p = L^{1/2}$, the strict phase gain leaves a decrease of order $(\sqrt{L} \log L)^{-1}$ in the exponent of $\sqrt{|E|/|F|}$. This dominates all fixed polynomial losses and supplies the logarithmic restricted estimate above.

1.4. Organization, inputs and uniformity. Section 1 gives the analytic reduction and the precise packet estimates needed for the band theorem. Section 2 proves the ordinary tile estimates and constructs the popular forest. Section 3 establishes every-line concentration and the simultaneous count of significant directions. Section 4 proves the constant-one finite Fourier inequality. Section 5 constructs the angular hierarchy and the predicted factors. Section 6 controls residuals and integrates path products. Section 7 performs interpolation, angular-grid averaging and the final strong-type assembly.

There are two specialized external inputs: Lacey–Li’s short Lipschitz-popularity maximal theorem, stated in Theorem 1.2, and the unweighted scalar martingale 3-variation inequality of Lépingle, used in the form [29, Theorem 1.1, $p = 2$, $r = 3$, weight one]. The latter enters the locally proved Fourier/martingale comparison in Theorem 3.3; we credit its origin to [24]. Standard one-dimensional Hilbert-transform, maximal-function and interpolation facts are recalled where used. The commutator, tile, concentration, finite-phase and hierarchy arguments are included in full.

All packet collections, scale ranges and angular depths are finite until estimates independent of those truncations have been proved. The bin Hilbert spaces are finite-dimensional and complex, and every constant is independent of their dimension. Pairings are linear in the first variable. A factor L^C always has a fixed exponent; it never conceals the number of packets, scales or hierarchy levels.

1.5. Fourier conventions and the local reduction. We use the Fourier convention

$$\widehat{f}(\xi, \eta) = \int_{\mathbb{R}^2} f(x, y) e^{-i(x\xi + y\eta)} dx dy, \quad f(x, y) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \widehat{f}(\xi, \eta) e^{i(x\xi + y\eta)} d\xi d\eta.$$

The same convention is used in one dimension. Pairings are linear in the first entry. A constant depending on a fixed finite list of smooth cutoffs or on their prescribed seminorms is an absolute constant. No such list will depend on a frequency scale, an inner truncation, the number of packets, or the dimension of the bin Hilbert space.

1.6. A maximal theorem and elementary analytic tools. We first record precisely the specialized maximal theorem used later. If an orthogonal rectangle R has long side $L(R)$, short side $W(R)$, and long-side direction $e_R \in S^1$, let $\text{EX}(R)$ be the arc centered at e_R of total length $W(R)/L(R)$.

Theorem 1.2 (Lacey–Li). *Let $v : \mathbb{R}^2 \rightarrow S^1$ be Lipschitz and $0 < \delta < 1$. Restrict to rectangles satisfying*

$$L(R) \leq \frac{1}{100 \text{Lip}(v)}, \quad \left| R \cap v^{-1}(\text{EX}(R)) \right| \geq \delta |R|,$$

where the first restriction is void when $\text{Lip}(v) = 0$. Then

$$\mathcal{M}_{v,\delta} h(z) = \sup_R \frac{\mathbf{1}_R(z)}{|R|} \int_R |h| \quad \text{satisfies} \quad \|\mathcal{M}_{v,\delta} h\|_{L^{2,\infty}} \leq C \delta^{-1/2} \|h\|_2.$$

Here the supremum is tested at every point of R , and the constant is independent of v and δ .

This is [20, Theorem 1.4]. Requiring the field to lie in a smaller fixed subarc only restricts the collection of admissible rectangles. In the bounded slope range, the map $t \mapsto (1, t)/\sqrt{1+t^2}$ changes slope distances into arc distances by comparable absolute factors. These observations explain the slope version of Theorem 1.2 used below. When an intermediate box is a parallelogram, the application of the theorem will explicitly enclose it by a comparable orthogonal rectangle of admissible length.

We also use the following familiar one-dimensional estimates, with their indicated uniformities. The Hardy–Littlewood maximal operator is bounded on L^p , $1 < p < \infty$, and

$$\sup_{0 < r < R < \infty} \left\| \int_{r < |t| < R} h(\cdot - t) \frac{dt}{t} \right\|_p \leq C_p \|h\|_p. \quad (1.1)$$

The supremum in (1.1) is outside the norm; the usual maximal truncation theorem gives this assertion as well. See [17, Theorems 1.1 and 7.6 and the proof of (7.7)]. Fubini gives the corresponding horizontal estimates on $\mathbb{R}^2$. The L^2 square-function identities used here are consequences of Plancherel and finite overlap of smooth frequency annuli. The standard smooth Littlewood–Paley and random-sign multiplier formulations are also recorded in [27, Proposition 5.3, Corollary 5.4, and Remark 4.5]. We give the particular Carleson embedding and commutator arguments needed for our reduction below. The John–Nirenberg and martingale variation estimates are stated in the sections where their local forms are used.

1.7. The local slope operator and its endpoints. Fix an absolute bound C_{Lip} large enough for the slope fields obtained in the following localization; for example, a bound of ten suffices after taking the squares small enough. Choose an absolute horizontal length $b_0 > 0$ with $b_0 C_{\text{Lip}} < 1/10$, and subsequently choose $0 < a_* < b_0/100$. These constants may be decreased to accommodate the fixed geometric enlargements in later applications of Theorem 1.2. All such enlargements are chosen absolutely. For the remainder of the local argument write $a = a_*$, and put

$$H_{u,a}^\epsilon h(x, y) = \int_{\epsilon < |t| < a} h(x - t, y - tu(x, y)) \frac{dt}{t}, \quad 0 < \epsilon < a, \quad (1.2)$$

where $|u| \leq 1$ and $\text{Lip}(u) \leq C_{\text{Lip}}$. The local inputs and tests will be supported in $I \times \mathbb{R}$, where I is any interval of length at most b_0 .

Lemma 1.3. *For $|t| C_{\text{Lip}} < 1$, define*

$$A_t h(x, y) = h(x - t, y - tu(x, y)).$$

For $1 \leq p < \infty$,

$$\|A_t h\|_p \leq (1 - |t| C_{\text{Lip}})^{-1/p} \|h\|_p, \quad (1.3)$$

and A_t is a contraction on L^∞ . Consequently, replacing either symmetric endpoint at scale r by a measurable endpoint between cr and Cr , with $c, C > 0$ fixed and $\max(1, C)rC_{\text{Lip}} < 1/2$, costs a bounded L^p operator, independently of r .

Proof. For each fixed x , the map $\varphi_{x,t}(y) = y - tu(x, y)$ is increasing, onto, and satisfies

$$(1 - |t| C_{\text{Lip}}) |y - y'| \leq |\varphi_{x,t}(y) - \varphi_{x,t}(y')| \leq (1 + |t| C_{\text{Lip}}) |y - y'|.$$

The change-of-variables formula for a one-dimensional bi-Lipschitz map, followed by translation in x , proves (1.3). A difference of endpoint truncations is pointwise dominated by

$$\int_{c'r < |t| < Cr} |A_t h| \frac{dt}{|t|}$$

with $c' = \min(1, c)$ and $C' = \max(1, C)$. Minkowski and (1.3) bound this by $C_p \log(C'/c') \|h\|_p$. The same proof applies to inner and outer endpoints, and permits endpoints depending on (x, y) . $\square$

We explain the localization of the original unit-field operator. Partition the plane into squares Q of a sufficiently small fixed side length. For output in Q , only a fixed enlargement Q^+ of that square contributes to a transform of length a_* . Rotate coordinates so that the vector at the center of Q points along the positive first coordinate axis. Since $\text{Lip}(v) \leq 1$, on the rotated output square we have

$$v = (v_1, v_2), \quad v_1 \geq c > 0, \quad u = v_2/v_1, \quad |u| \leq 1, \quad \text{Lip}(u) \leq C_{\text{Lip}}.$$

The last inequality follows by subtracting the two ratios and using the lower bound for v_1 . Extend u to the plane with the same Lipschitz bound, using $\inf_{z \in Q} (u(z) + C_{\text{Lip}} |\cdot - z|)$, and then clip the extension to $[-1, 1]$. The infimum extension agrees with u on Q , and clipping preserves both agreement and the Lipschitz bound.

At an output point, the substitution $s = tv_1(x, y)$ changes dt/t to ds/s and changes the two endpoints to $\epsilon v_1(x, y)$ and $a_* v_1(x, y)$. By Theorem 1.3, each endpoint can be replaced by ϵ or a_* , respectively, at a uniform L^2 cost. The rotated sets Q and Q^+ fit in a common horizontal interval of length at most b_0 , after choosing the square side and a_* in that order. Thus a uniform bound for $\mathbf{1}_{I \times \mathbb{R}} H_{u,a}^\epsilon \mathbf{1}_{I \times \mathbb{R}}$ implies the required bound on this output square. The output squares are disjoint, the sets Q^+ have bounded overlap, and rotations preserve area. Squaring the local estimates and summing over Q therefore proves the global estimate. Notice that both endpoint errors are uniform in the inner radius ϵ .

1.8. The band theorem and summation of amplitude levels. A vertical band cutoff at length $w > 0$ is a multiplier $P_w = p(wD_y)$, where p is smooth and supported in a fixed compact annulus $\{r : c_\eta \leq |r| \leq C_\eta\}$, with $0 < c_\eta < C_\eta < \infty$ absolute. We allow a fixed bounded family of such cutoffs, including slightly enlarged cutoffs equal to one on the supports of others. All constants below are uniform when the required finitely many cutoff seminorms are bounded. The dyadic inverse frequency scale w belongs to $2^\mathbb{Z}$.

Theorem 1.4 (Restricted estimate on a vertical band). *For the fixed absolute constants in (1.2), there is an absolute C such that, for every field u with $|u| \leq 1$ and $\text{Lip}(u) \leq C_{\text{Lip}}$, every $0 < \epsilon < a$, every dyadic w , and every vertical band cutoff P_w ,*

$$\left| \langle H_{u,a}^\epsilon P_w f, g \rangle \right| \leq C(1 + |\log(|F|/|E|)|)^{-3} \sqrt{|F||E|} \quad (1.4)$$

whenever $F, E \subset I \times \mathbb{R}$ have positive finite measure, $|f| \leq \mathbf{1}_F$, $|g| \leq \mathbf{1}_E$, and $|I| \leq b_0$.

The proof of Theorem 1.4 will be completed after the packet estimates and their iteration have been established. The next results show, without using that theorem as a premise elsewhere in its proof, that it implies Theorem 1.1. The last part of this section then gives explicit sufficient packet estimates for Theorem 1.4.

Lemma 1.5. *Suppose a bilinear form $\mathcal{B}$ satisfies*

$$|\mathcal{B}(f, g)| \leq C_0(1 + |\log(|F|/|E|)|)^{-3} \sqrt{|F||E|} \quad (|f| \leq \mathbf{1}_F, |g| \leq \mathbf{1}_E)$$

on finite-measure sets in a fixed measurable domain. Then

$$|\mathcal{B}(f, g)| \leq CC_0 \|f\|_2 \|g\|_2$$

for finite amplitude decompositions, and hence for all L^2 inputs whenever the form is obtained by an L^2 -continuous truncation or by completion from those inputs.

Proof. Use disjoint amplitude sets $F_i = \{2^{i-1} < |f| \leq 2^i\}$ and write $f = \sum_i 2^i f_i$, where $|f_i| \leq \mathbf{1}_{F_i}$. For the moment all sums are finite. Put $a_i = 2^i \sqrt{|F_i|}$, so $\sum_i a_i^2 \leq 4 \|f\|_2^2$. Group the indices into measure classes

$$\mathcal{I}_k = \{i : 2^k \leq |F_i| < 2^{k+1}\}, \quad A_k = \left(\sum_{i \in \mathcal{I}_k} a_i^2 \right)^{1/2}.$$

If i_0 is the largest index in a nonempty class, then

$$\sum_{i \in \mathcal{I}_k} a_i \leq 2^{(k+1)/2} \sum_{i \leq i_0} 2^i \leq 2\sqrt{2} a_{i_0} \leq 2\sqrt{2} A_k.$$

Make the analogous decomposition $g = \sum_j 2^j g_j$, with measure classes $\mathcal{J}_l$ and square sums B_l . For $i \in \mathcal{I}_k$, $j \in \mathcal{J}_l$,

$$(1 + |\log(|F_i|/|E_j|)|)^{-3} \leq C(1 + |k - l|)^{-3}.$$

The assumed bound, followed by the within-class estimates, gives

$$|\mathcal{B}(f, g)| \leq CC_0 \sum_{k, l} (1 + |k - l|)^{-3} A_k B_l \leq CC_0 \|A\|_{\ell^2} \|B\|_{\ell^2}.$$

The final step is Young's inequality for the summable sequence $(1 + |k|)^{-3}$. This proves the assertion. Truncating the amplitude decompositions and using the asserted continuity or completion gives the last statement, with no dependence on the number of classes. $\square$

1.9. A Lipschitz commutator. The estimate below belongs to the Lipschitz commutator theory initiated by Calderón and developed by Coifman and Meyer [5, 8]. We give a dyadic proof with the precise multiplier seminorms needed for Littlewood–Paley reassembly and for the later narrow-bin reduction. The argument separates low output frequencies from the annular interactions, because the constant must remain uniform as the number of frequency scales increases. Let χ be a real even smooth cutoff equal to one on $[-1, 1]$ and supported in $[-2, 2]$, and define

$$S_k = \chi(2^{-k}D), \quad \Delta_k = S_k - S_{k-1}.$$

Thus Δ_k is supported in $\{2^{k-1} \leq |\zeta| \leq 2^{k+1}\}$, and Plancherel gives

$$\|h\|_2^2 \asymp \sum_k \|\Delta_k h\|_2^2. \quad (1.5)$$

Lemma 1.6 (The required Carleson embedding). *For every fixed integer c , every $q \in L^\infty(\mathbb{R})$, and every $h \in L^2(\mathbb{R})$,*

$$\sum_k \|(\Delta_k q) S_{k+c} h\|_2^2 \leq C_c \|q\|_\infty^2 \|h\|_2^2. \quad (1.6)$$

Proof. For a dyadic interval J of length 2^{-j} , split $q = q\mathbf{1}_{3J} + q\mathbf{1}_{(3J)^c}$. By (1.5), the first part satisfies

$$\sum_{k \geq j} \int_J |\Delta_k(q\mathbf{1}_{3J})|^2 \leq C \|q\|_\infty^2 |J|.$$

The kernels of Δ_k have Schwartz decay at scale 2^{-k} . For $x \in J$ and $k \geq j$, this gives

$$|\Delta_k(q\mathbf{1}_{(3J)^c})(x)| \leq C \|q\|_\infty 2^{j-k}.$$

After summing the squares, we obtain

$$\sum_{k \geq j} \int_J |\Delta_k q|^2 \leq C \|q\|_\infty^2 |J|. \quad (1.7)$$

For each dyadic interval I of length 2^{-k} , set $\alpha_I = \int_I |\Delta_k q|^2$. Then (1.7) says $\sum_{I \subset J} \alpha_I \leq C \|q\|_\infty^2 |J|$.

Write M for the Hardy–Littlewood maximal operator. Kernel decay, summed on dyadic annuli about a point of I , gives

$$\sup_{x \in I} |S_{k+c} h(x)| \leq C_c \operatorname{ess\,inf}_{y \in I} Mh(y).$$

The shift c is fixed, so moving the kernel center across I changes the bound by only a fixed factor. If $F \geq 0$ belongs to L^2 , the dyadic Carleson condition implies

$$\sum_I \alpha_I (\operatorname{ess\,inf}_I F)^2 \leq C \|q\|_\infty^2 \|F\|_2^2.$$

Indeed, for each $t > 0$, sum over the maximal dyadic intervals on which $\operatorname{ess\,inf}_I F > t$, apply the Carleson condition on each such interval, and integrate against $2t \, dt$. Their union lies, up to null sets, in $\{F > t\}$. Apply this assertion with $F = Mh$, and use the L^2 maximal theorem. This proves (1.6). This is also the usual continuous Littlewood–Paley Carleson embedding; compare the proof of [28, Proposition 4.1]. $\square$

Lemma 1.7. *Let $b \in W^{1,\infty}(\mathbb{R})$, and let $T = \mathfrak{m}(D)$, where $\mathfrak{m}$ is smooth away from zero and*

$$A(\mathfrak{m}) = \max_{0 \leq r \leq 4} \sup_{\zeta \neq 0} |\zeta|^r |\mathfrak{m}^{(r)}(\zeta)| < \infty.$$

Then

$$\|[b, T] \partial h\|_2 \leq CA(\mathfrak{m}) \|b'\|_\infty \|h\|_2. \quad (1.8)$$

The commutator is initially defined on the Sobolev space $W^{1,2}(\mathbb{R})$; the estimate gives its bounded extension to L^2 . Its constant is independent of $\|b\|_\infty$ and of all frequency truncations.

Proof. Put $L_b = \|b'\|_\infty$ and $b_k = \Delta_k b$. Cancellation and the first absolute moment of the smooth kernels give

$$\|b_k\|_\infty \leq C 2^{-k} L_b, \quad \|\partial S_k b\|_\infty \leq C L_b, \quad b'_k = \Delta_k b'. \quad (1.9)$$

Also $\|S_N b - b\|_\infty \leq C 2^{-N} L_b$. Consequently, for each fixed j , the exact identity

$$b = S_{j-4} b + \sum_{k \geq j-3} b_k \quad (1.10)$$

has a uniformly convergent high-frequency sum. This identity retains the constant and low-frequency part of b ; no homogeneous reconstruction of a bounded function at frequency zero is needed.

For any fixed smooth annular cutoff ρ , let K_j be the kernel of $\mathfrak{m}(D) \rho(2^{-j} D)$. Rescaling and three integrations by parts on its fixed compact frequency support show

$$\int_{\mathbb{R}} |z| |K_j(z)| \, dz \leq CA(\mathfrak{m}) 2^{-j}. \quad (1.11)$$

In fact the rescaled kernel is bounded by $CA(\mathfrak{m})(1 + |z|)^{-3}$, which proves the assertion.

First suppose that $\hat{h}$ is smooth and compactly supported away from zero. Put $h_j = \Delta_j h$; only finitely many h_j are nonzero. Apply (1.10) to each of them. There are three interactions.

For the low frequencies of b , put $B_j = S_{j-4} b$. The supports of both h_j and $B_j h_j$ lie in a fixed annulus of radius 2^j . A slightly enlarged annular multiplier T_j therefore satisfies $[B_j, T] \partial h_j = [B_j, T_j] \partial h_j$. The kernel formula and (1.11) give

$$\|[B_j, T_j] \partial h_j\|_2 \leq \operatorname{Lip}(B_j) \int |z| |K_j(z)| \, dz \|\partial h_j\|_2 \leq CA(\mathfrak{m}) L_b \|h_j\|_2.$$

The outputs remain in fixed annuli of radius 2^j . Finite frequency overlap and (1.5) bound their sum by $CA(\mathbf{m})L_b \|h\|_2$.

For $k \geq j + 4$, the output of $[b_k, T]\partial h_j$ lies in a fixed annulus of radius 2^k . Plancherel, Bernstein's inequality, and (1.9) yield

$$\|[b_k, T]\partial h_j\|_2 \leq CA(\mathbf{m})L_b 2^{j-k} \|h_j\|_2.$$

Group by k , use frequency overlap, and apply Young's inequality on $\ell^2(\mathbb{Z})$ to the summable sequence $2^{-n}\mathbf{1}_{n \geq 4}$. This bounds the high-low sum by the same quantity and proves its convergence in L^2 .

It remains to consider $|k - j| \leq 3$. Fix one such difference and write $H_k = h_{k+r}$, $|r| \leq 3$. The exact identity

$$[b_k, T]\partial H_k = \partial([b_k, T]H_k) - b'_k TH_k + T(b'_k H_k) \quad (1.12)$$

treats the possible low output frequencies. Indeed, $\partial([b_k, T]H_k)$ has spectrum in a ball of radius $C2^k$. Its Δ_i projection vanishes for $i > k + c_0$, with c_0 fixed, and otherwise

$$\|\Delta_i \partial([b_k, T]H_k)\|_2 \leq CA(\mathbf{m})L_b 2^{i-k} \|H_k\|_2.$$

Another application of (1.5) and Young's inequality to $2^{-n}\mathbf{1}_{n \geq -c_0}$ bounds the sum of these derivative terms. Thus arbitrarily low output frequencies cause no logarithmic loss.

Choose a fixed c_1 so that S_{k+c_1} equals one on the spectrum of every product $b'_k H_k$. For $g \in L^2$,

$$\begin{aligned} \left| \left\langle \sum_k b'_k H_k, g \right\rangle \right| &\leq \left(\sum_k \|H_k\|_2^2 \right)^{1/2} \left(\sum_k \|(\Delta_k b') S_{k+c_1} g\|_2^2 \right)^{1/2} \\ &\leq CL_b \left(\sum_k \|H_k\|_2^2 \right)^{1/2} \|g\|_2, \end{aligned}$$

by Theorem 1.6. Since $\|T\|_{2 \rightarrow 2} \leq A(\mathbf{m})$, this bounds $T \sum_k b'_k H_k$. Replacing H_k by TH_k in the same calculation bounds $\sum_k b'_k TH_k$: the frequency supports are unchanged and $\sum_k \|TH_k\|_2^2 \leq A(\mathbf{m})^2 \sum_k \|H_k\|_2^2$. Equation (1.12), summed over its seven possible differences, completes the proof on the chosen dense class.

For $h \in W^{1,2}$, approximate by $(S_N - S_{-N})h$, followed if necessary by smooth compact Fourier approximations away from zero. The approximation converges in $W^{1,2}$. Since b is bounded and T is L^2 -bounded, the defining commutators converge in L^2 . The estimate passes to the limit, and then defines a bounded operator on all of L^2 . $\square$

For a vertical multiplier T , set

$$C_t h(x, y) = h(x, y - tu(x, y)).$$

Corollary 1.8. For $|t| \leq b_0$ and T as in Theorem 1.7,

$$\|TC_t h - C_t Th\|_2 \leq CA(\mathbf{m}) |t| \|h\|_2. \quad (1.13)$$

The constant is uniform over $|u| \leq 1$, $\text{Lip}(u) \leq C_{\text{Lip}}$.

Proof. Work first with a smooth field and on one fixed x -fiber. Let $\varphi_s(y) = y - su(x, y)$. Direct differentiation gives

$$C_s^{-1} C'_s = b_s \partial_y, \quad b_s = -u(x, \cdot) \circ \varphi_s^{-1},$$

and hence, since T commutes with ∂_y ,

$$\frac{d}{ds} (C_s T C_s^{-1}) = C_s [b_s, T] \partial_y C_s^{-1}. \quad (1.14)$$

The bi-Lipschitz estimates in [Theorem 1.3](#) imply

$$\|C_s\|_{2 \rightarrow 2} + \|C_s^{-1}\|_{2 \rightarrow 2} \leq C, \quad \text{Lip}(b_s) \leq \frac{C_{\text{Lip}}}{1 - |s|C_{\text{Lip}}} \leq C.$$

Apply [Theorem 1.7](#) inside (1.14), integrate from 0 to t , and multiply by C_t . The estimate is uniform in the fiber, so integration in x proves (1.13).

A bounded Lipschitz field is approximated by convolutions with smooth approximate identities, preserving the bounds on its size and Lipschitz seminorm. For each fixed small s , the corresponding composition operators converge strongly on L^2 : this follows first for continuous compactly supported inputs by uniform convergence of the maps, and then by density and their uniform operator bounds. Passing to the limit proves the assertion for the original field. $\square$

1.10. Square-function reassembly. The preceding composition estimate pays a factor of $|t|$, which cancels the singular measure $dt/|t|$ when commutators are integrated. This is the mechanism that permits the band estimates to be assembled. Di Plinio, Guo, Thiele and Zorin-Kranich established this band-to-full reduction under Lipschitz regularity [[9](#), Corollaries 1.3–1.4 of the author version]. We retain the argument here to track the fixed outer endpoint, arbitrary inner endpoint and horizontal localization used throughout this paper.

Choose a real smooth annular function p so that $\sum_{w \in 2^{\mathbb{Z}}} p(w\eta)^2 = 1$ for $\eta \neq 0$. Such a function is obtained by dividing an ordinary smooth dyadic partition by the square root of its sum of squares. Set $P_w = p(wD_y)$. For every finite set $\mathcal{W}$ of scales and every choice of signs $\varepsilon_w \in \{-1, 1\}$, the multiplier $T = \sum_{w \in \mathcal{W}} \varepsilon_w P_w$ satisfies $A(\mathbf{m}_T) \leq C$, by finite overlap of the annuli. Writing $S_t^x h(x, y) = h(x - t, y)$, we have

$$H_{u,a}^\epsilon = \int_{\epsilon < |t| < a} C_t S_t^x \frac{dt}{t}.$$

The vertical multiplier T commutes with S_t^x . Thus [Theorem 1.8](#) implies

$$\|[T, H_{u,a}^\epsilon]\|_{2 \rightarrow 2} \leq C \int_{\epsilon < |t| < a} |t| \frac{dt}{|t|} \leq Ca. \quad (1.15)$$

All these operators are already bounded on L^2 for fixed $\epsilon > 0$, with the elementary bound $C \log(a/\epsilon)$; no conclusion of [Theorem 1.4](#) is needed to form their square functions.

Randomize the signs independently and use their orthogonality. For $f = \mathbf{1}_{I \times \mathbb{R}} f$, Equation (1.15) and the triangle inequality in the random-sign L^2 space give

$$\left(\sum_{w \in \mathcal{W}} \|\mathbf{1}_{I \times \mathbb{R}} P_w H_{u,a}^\epsilon f\|_2^2 \right)^{1/2} \leq C \|f\|_2 + \left(\sum_{w \in \mathcal{W}} \|\mathbf{1}_{I \times \mathbb{R}} H_{u,a}^\epsilon P_w f\|_2^2 \right)^{1/2}. \quad (1.16)$$

The horizontal restriction commutes with every vertical multiplier.

Proposition 1.9. *The uniform restricted estimate in [Theorem 1.4](#) implies [Theorem 1.1](#).*

Proof. Apply [Theorem 1.5](#) to the localized band form. It gives a uniform local L^2 bound for $H_{u,a}^\epsilon \tilde{P}_w$, where $\tilde{P}_w$ is an enlarged cutoff equal to one on the spectrum of P_w . Since $P_w f$ retains the same horizontal support,

$$\|\mathbf{1}_{I \times \mathbb{R}} H_{u,a}^\epsilon P_w f\|_2 = \|\mathbf{1}_{I \times \mathbb{R}} H_{u,a}^\epsilon \tilde{P}_w (P_w f)\|_2 \leq C \|P_w f\|_2.$$

Insert this in (1.16) and let $\mathcal{W}$ increase to all dyadic scales. Plancherel and the square partition give $\|\mathbf{1}_{I \times \mathbb{R}} H_{u,a}^\epsilon f\|_2 \leq C \|f\|_2$, uniformly in ϵ . The localization already proved above gives the asserted global bound for the unit field. Finally, for a Schwartz input the symmetric principal value exists pointwise; Fatou gives its L^2 bound, and Chebyshev gives the weak estimate in [Theorem 1.1](#). $\square$

1.11. Removing long and very short times. We now reduce [Theorem 1.4](#) to packet estimates. In this part only, abbreviate $H_{u,a}^\epsilon$ to H , and put

$$L = \log \sqrt{|E|/|F|}.$$

Fix a sufficiently large absolute threshold L_0 , which can be increased after the fixed constants in the argument are chosen. When $L \geq L_0$, choose dyadic numbers satisfying

$$\frac{1}{2}e^{-L^{0.97}} < \tau \leq e^{-L^{0.97}}, \quad e^{L^{0.9}} \leq m < 2e^{L^{0.9}}, \quad W = mw. \quad (1.17)$$

Here τ is the short-time cutoff. The parameter m sets the number of vertical frequency bins: their width will be comparable to W^{-1} , and $W = mw$ is the corresponding spatial width. For every fixed C , the product $m^C \tau$ tends to zero as $L \rightarrow \infty$; this pays for the binning errors. We take L_0 large enough that $\tau < a$. For all other values of L , let $m = 1$, $W = w$, and let $\tau \in (0, a)$ be a fixed small dyadic constant.

Lemma 1.10. *For any fixed $1 < p < \infty$, the contribution of $\max(\epsilon, \tau) < |t| < a$ to HP_w has L^p norm at most $C_p(1 + L)$ if $L \geq L_0$, and at most a fixed C_p otherwise. On the remaining times below w , replacing u by zero costs a uniformly bounded L^p operator. The resulting horizontal symmetric truncation is uniformly bounded on L^p .*

Proof. The first assertion follows from [Theorem 1.3](#), the uniform L^1 norm of the convolution kernel of P_w , and

$$\int_{\tau < |t| < a} \frac{dt}{|t|} = 2 \log(a/\tau) \leq C(1 + L^{0.97}).$$

For the second, the fundamental theorem of calculus gives

$$\begin{aligned} & (P_w h)(x - t, y - tu(x, y)) - (P_w h)(x - t, y) \\ &= -tu(x, y) \int_0^1 (\partial_y P_w h)(x - t, y - \theta tu(x, y)) d\theta. \end{aligned}$$

The intermediate maps satisfy [Theorem 1.3](#), and the kernel of $\partial_y P_w$ has L^1 norm $O(w^{-1})$. Consequently,

$$\|(A_t - S_t^x)P_w h\|_p \leq C_p |t| w^{-1} \|h\|_p. \quad (1.18)$$

Integrating over any subinterval of $0 < |t| < w$ against $dt/|t|$ gives a constant. Equation [\(1.1\)](#) handles the horizontal part, uniformly in its inner and outer endpoints. $\square$

Every error with L^p norm B_p contributes at most

$$B_p |F|^{1/p} |E|^{1-1/p} = B_p \sqrt{|F||E|} \left(\frac{|F|}{|E|} \right)^{1/p-1/2} \quad (1.19)$$

to the restricted form. In the regime $L \geq L_0$, choose one fixed $p < 2$, for example $p = 3/2$; the normalized error is then at most $C(1 + L)e^{-L/3}$. In the opposite regime $|F| \geq |E|$, the fixed choice $p = 4$ gives the power $(|E|/|F|)^{1/4}$ for each uniformly bounded error. On a bounded range of L , fixed operator bounds suffice. These choices are made once and do not approach the endpoint $p = 2$.

1.12. From a spatial truncation to a finite scale sum. Choose an even smooth cutoff χ_0 , equal to one near zero and compactly supported, and set

$$\vartheta(r) = \chi_0(r) - \chi_0(2r), \quad \psi(r) = -i\pi \operatorname{sgn}(r)\vartheta(r).$$

The function ψ is smooth and supported in a fixed compact annulus. Its inverse Fourier transform k is Schwartz, and $k_\ell(t) = \ell^{-1}k(t/\ell)$ has multiplier $\psi(\ell\zeta)$. Since $\sum_{\ell \in 2^\mathbb{Z}} \vartheta(\ell\zeta) = 1$ for $\zeta \neq 0$,

$$\sum_{\ell \in 2^\mathbb{Z}} k_\ell(t) = \frac{1}{t} \quad (t \neq 0).$$

The identity follows by Fourier inversion in tempered distributions; away from zero the kernel sum is locally uniformly convergent by Schwartz decay for small ℓ and the bound C/ℓ for large ℓ .

Lemma 1.11. *For dyadic $0 < r \leq R$,*

$$\int_{\mathbb{R}} \left| \frac{\mathbf{1}_{\{r < |t| < R\}}}{t} - \sum_{r \leq \ell \leq R} k_\ell(t) \right| dt \leq C. \quad (1.20)$$

Also,

$$\int_{\mathbb{R}} |t| \left| \sum_{r \leq \ell \leq R} k_\ell(t) \right| dt \leq CR. \quad (1.21)$$

Both constants are independent of the number of scales. Changing either spatial endpoint by a fixed factor preserves the first bound.

Proof. For $r < |t| < R$, the omitted scales below r have sum bounded by $Cr/|t|^2$, using $|k_\ell(t)| \leq C\ell/|t|^2$ and a geometric sum. The omitted scales above R have sum at most C/R . The integrals of these bounds over the middle interval are bounded. For $|t| < r$, the retained sum is at most $C \sum_{\ell \geq r} \ell^{-1} \leq C/r$; for $|t| > R$, it is at most $CR/|t|^2$. These are integrable on their respective regions and prove (1.20). For the moment bound, use

$$\int |t| |k_\ell(t)| dt = \ell \int |s| |k(s)| ds$$

and $\sum_{\ell \leq R} \ell \leq 2R$. Finally, changing an endpoint by a fixed factor changes the spatial kernel only on an annulus having bounded $dt/|t|$ measure. $\square$

After Theorem 1.10, the only possible remaining times are between $r = \max(w, \epsilon)$ and τ . Round the lower endpoint to a comparable dyadic number; the resulting annular error has a uniform L^p bound. If this lower endpoint exceeds τ , there is no remaining model. Otherwise Theorem 1.11 reduces the multiplier, with u frozen at the output point, to

$$\mathfrak{a}_{u, \mathcal{L}}(x, y; \xi, \eta) = \sum_{\ell \in \mathcal{L}} \psi(\ell(\xi + \eta u(x, y))), \quad \mathcal{L} \subset \{\ell \in 2^\mathbb{Z} : w \leq \ell \leq \tau\}. \quad (1.22)$$

Here $\mathcal{L}$ is the consecutive set of retained scales; when $\epsilon \leq w$ it is the whole displayed range. Denote the associated operator by $\mathcal{A}_{u, \mathcal{L}}$, or simply $\mathcal{A}$. Its spatial representation is

$$\mathcal{A}h(x, y) = \int_{\mathbb{R}} K_{\mathcal{L}}(t) h(x - t, y - tu(x, y)) dt, \quad K_{\mathcal{L}} = \sum_{\ell \in \mathcal{L}} k_\ell.$$

The kernels in this representation are not compactly supported in t . This causes no large-time change-of-variables problem: if both the input and the test are supported in $I \times \mathbb{R}$, a nonzero contribution has $t \in I - I$, hence $|t| \leq b_0$. Every vertical cutoff preserves that horizontal support. Thus Theorem 1.3 applies to the localized kernel error in (1.20), and its L^p cost is uniform. Only the original input and test carry horizontal restrictions; no spatial support is subsequently imposed on their convolutions.

1.13. Narrow vertical bins. Split the vertical cutoff into its positive and negative frequency parts. Reflection $y \mapsto -y$ interchanges the two cases and replaces u by $-u$, preserving all assumptions. We work with positive frequencies, so $c_\eta \leq w\eta \leq C_\eta$.

Suppose first that $L \geq L_0$ and m is as in (1.17). Choose real smooth cutoffs Q_j , $1 \leq j \leq J \leq Cm$, supported in intervals of length C/W centered at $\eta_j \asymp w^{-1}$, with bounded overlap and

$$\sum_{j=1}^J Q_j(\eta)^2 = 1 \quad \text{on } \text{supp } p(w\eta).$$

One obtains them from a fixed smooth partition at spacing W^{-1} by dividing by the square root of its sum of squares on a slightly enlarged band. Their normalized profiles and all fixed derivatives are uniform. In particular, their convolution kernels have bounded L^1 norms, and

$$A(Q_j) \leq Cm^4. \quad (1.23)$$

Let $\tilde{Q}_j$ be a slightly enlarged cutoff equal to one on the spectrum of $Q_j P_w$. These cutoffs again have the stated bin width and normalized smoothness bounds.

Set $F_j^0 = Q_j P_w f$ and $G_j^0 = Q_j g$. The identity on the band and self-adjointness of the Q_j give

$$\langle \mathcal{A} P_w f, g \rangle = \sum_j \langle \mathcal{A} \tilde{Q}_j F_j^0, G_j^0 \rangle + \sum_j \langle [\mathcal{A}, Q_j] F_j^0, g \rangle. \quad (1.24)$$

Indeed, expand $P_w = \sum_j Q_j^2 P_w$, commute one Q_j past $\mathcal{A}$, and use $\tilde{Q}_j F_j^0 = F_j^0$. This identity does not require $\mathcal{A}$ to preserve individual bins.

On localized inputs and tests, Equations (1.13), (1.21), and (1.23) imply

$$|\langle [\mathcal{A}, Q_j] F_j^0, g \rangle| \leq Cm^4 \tau \|F_j^0\|_2 \|g\|_2.$$

In applying the composition estimate, only $|t| \leq b_0$ are tested, as explained above. Bounded frequency overlap gives $\sum_j \|F_j^0\|_2^2 \leq C \|f\|_2^2$, so the total error in (1.24) is at most

$$Cm^5 \tau \|f\|_2 \|g\|_2. \quad (1.25)$$

The harmless integer exponent five includes the factor $\sqrt{J}$. More generally, any fixed multiplier seminorm choice would give $Cm^C \tau$, and for every fixed C, N ,

$$m^C \tau \leq C \exp(C' L^{0.9} - L^{0.97}) = O(L^{-N}). \quad (1.26)$$

The large- L threshold in this statement is absolute after the fixed exponents are chosen.

Let $\mathcal{H} = \mathbb{C}^J$ with its Euclidean norm, and regard $F^0 = (F_j^0)_j$, $G^0 = (G_j^0)_j$ as $\mathcal{H}$ -valued functions. For some fixed power $Q = Cm^C$, their norms obey

$$\begin{aligned} \|F^0\|_{L^\infty(\mathcal{H})} &\leq Q, & \|F^0\|_{L^1(\mathcal{H})} &\leq Q|F|, & \|F^0\|_{L^2(\mathcal{H})}^2 &\leq C|F|, \\ \|G^0\|_{L^\infty(\mathcal{H})} &\leq Q, & \|G^0\|_{L^1(\mathcal{H})} &\leq Q|E|, & \|G^0\|_{L^2(\mathcal{H})}^2 &\leq C|E|. \end{aligned} \quad (1.27)$$

For the first two columns, apply the uniform scalar kernel bounds and sum over at most Cm components. For the last column, use Plancherel and bounded overlap. No compact support in y is asserted after the projections. Horizontal support in I is preserved.

When $m = 1$, we do not perform this commutation or project the test. Instead retain the original positive-frequency band cutoff on the input and use one scalar component. It fits in a vertical frequency interval centered at a fixed $\eta_1 \asymp w^{-1}$ of length $C/w = C/W$. Its positive annular support is retained.

Lemma 1.12 (Amplitude levels after bin projection). *Let $h : \mathbb{R}^2 \rightarrow \mathcal{H}$ satisfy*

$$\|h\|_\infty \leq Q, \quad \|h\|_1 \leq QM_0, \quad \|h\|_2^2 \leq C_0M_0,$$

where $M_0 > 0$ and $Q \leq C \exp(CL^{0.9})$. Write

$$A_i = \{2^{i-1} < \|h(\cdot)\|_{\mathcal{H}} \leq 2^i\}, \quad a_i = 2^i \sqrt{|A_i|/M_0}.$$

For sufficiently large absolute L ,

$$\sum_i a_i \leq C\sqrt{1+L}, \quad \sum_{\{i: |\log(|A_i|/M_0)| > L/4\}} a_i \leq Ce^{-cL}. \quad (1.28)$$

Empty level sets are omitted. The constants depend on the fixed constants in the hypotheses, not on the dimension of $\mathcal{H}$.

Proof. The amplitude upper bound implies $2^i \leq 2Q$ on every nonempty level. If $|A_i| < e^{-L/4}M_0$, then $a_i \leq 2^i e^{-L/8}$, and summing the geometric series in i gives $CQe^{-L/8}$. If $|A_i| > e^{L/4}M_0$, the L^1 bound gives

$$2^i \leq 2Qe^{-L/4}, \quad a_i^2 = 4^i |A_i|/M_0 \leq 2Q2^i.$$

Summing the square-root geometric series over the allowed amplitudes again gives $CQe^{-L/8}$. Since $\log Q = O(L^{0.9})$, both tails are $O(e^{-cL})$.

The remaining level sets occupy at most $C(1+L)$ dyadic measure classes. Within a class, the geometric estimate in the proof of [Theorem 1.5](#) bounds the sum of a_i by a constant times its square sum. Cauchy–Schwarz over the classes and $\sum_i a_i^2 \leq 4C_0$ give the first bound in (1.28). The already bounded tails can be included in that bound. $\square$

Apply [Theorem 1.12](#) to the two vectors in (1.27). Decomposing them into amplitude levels produces vector inputs and tests bounded by indicators of new sets. The extra cutoffs $\tilde{Q}_j$ in (1.24) remain part of the operator after this decomposition; thus no band support is being assumed for an individual amplitude slice. If both new sets have retained measures, then their logarithmic disparity is in

$$\left[\frac{3}{4}L, \frac{5}{4}L\right] \subset \left[\frac{1}{2}L, 2L\right]. \quad (1.29)$$

Indeed, the two changes of logarithmic measure are each at most $L/4$, and the logarithm of the square-root ratio uses one half of their difference. The product of the two sums of normalized amplitude weights is $O(1+L)$.

1.14. The admissible packet forms. We next describe the packet model with all geometric and analytic uniformities that will be used. Fix $w \in 2^{\mathbb{Z}}$, $m \geq 1$, $W = mw$, a finite dyadic scale set $\mathcal{L} \subset [w, \tau]$, and at most Cm bin centers $\eta_j \asymp w^{-1}$. The scalar case $m = 1$ just described is included. Constants in this definition are fixed throughout.

Definition 1.13. A tile s has a length $\ell_s \in \mathcal{L}$, a bounded slope θ_s , a center $z_s = (x_s, y_s)$, and precision $d_s = w/\ell_s$. Its box is

$$R_s = \left\{ (x, y) : |x - x_s| \leq \frac{\ell_s}{2}, \quad |(y - y_s) - \theta_s(x - x_s)| \leq \frac{W}{2} \right\}, \quad |R_s| = \ell_s W.$$

A t -box is defined by the same formula with slope t . The nominal horizontal length need not be the Euclidean long side when $W > \ell_s$.

At a fixed scale, slope labels have bounded multiplicity on every interval of length d_s . For each scale and label, the centers lie in one translated box lattice

$$z_0 + \{(n\ell_s, \theta_s n\ell_s + kW) : n, k \in \mathbb{Z}\}.$$

The lattice and center of a tile are common to every bin component. An admissible collection is any finite subcollection of such lattices.

Its analyzing and synthesizing waves, denoted $\Phi_{s,j}$ and $\Psi_{s,j}$, are bounded-normalized modulated Schwartz functions adapted to R_s . More precisely, each can be written as

$$e^{in_j((y-y_s)-\theta_s(x-x_s))}\phi_{s,j}\left(\frac{x-x_s}{\ell_s}, \frac{(y-y_s)-\theta_s(x-x_s)}{W}\right),$$

where, for all fixed nonnegative integers a, b, N ,

$$|\partial_X^a \partial_Y^b \phi_{s,j}(X, Y)| \leq C_{a,b,N}(1 + |X| + |Y|)^{-N}. \quad (1.30)$$

The constants are uniform in s, j, w, m . Both waves also have Fourier support in

$$|\eta - \eta_j| \leq CW^{-1}, \quad c'_\eta \leq w\eta \leq C'_\eta, \quad |\xi + \theta_s\eta| \leq c_*\ell_s^{-1}, \quad (1.31)$$

with fixed positive c'_η, C'_η . The constant c_* is chosen sufficiently small compared with the selector separation constants below and the fixed annular support of ψ .

The scalar selector $b_{s,j}$ is smooth, with

$$\text{supp } b_{s,j} \subset \{t : c_b d_s \leq |t - \theta_s| \leq C_b d_s\}, \quad d_s^r \left\| b_{s,j}^{(r)} \right\|_\infty \leq C_r \quad (1.32)$$

for every fixed $r \geq 0$, where $0 < c_b < C_b$ are absolute. Finally, the collection uses one angular dyadic grid. Each tile has an interval Ω_s in that grid, of length $C_\Omega d_s$, where C_Ω is a fixed large dyadic number, containing both the wave slope support $\{-\xi/\eta : (\xi, \eta) \in \text{supp } \widehat{\Phi}_{s,j} \cup \text{supp } \widehat{\Psi}_{s,j}\}$ and every selector support, with a margin at least $c_\Omega d_s$, where $c_\Omega > 0$ is fixed. The interval is common to all components. Its level depends only on ℓ_s .

The associated form is

$$\Lambda_S(f, g) = \sum_{s \in S} |R_s|^{-1} \sum_j \langle f_j, \Phi_{s,j} \rangle \langle b_{s,j}(u) \Psi_{s,j}, g_j \rangle. \quad (1.33)$$

It is linear in f and conjugate-linear in g . The definition makes sense for an arbitrary measurable bounded field u . In particular, the ordinary scalar and vector estimates in the next section do not require spatial Lipschitz regularity of u . The stronger estimates later in the proof use the Lipschitz hypothesis from (1.2).

Proposition 1.14 (Packet representation). *The diagonal form*

$$\sum_j \langle \mathcal{A}_{u, \mathcal{L}} \tilde{Q}_j f_j, g_j \rangle$$

is a sum, with absolutely summable scalar coefficients, of averages of forms (1.33). Each form satisfies [Theorem 1.13](#), uniformly in $w, m, \mathcal{L}$, and has one common angular grid and common spatial lattices across components. The same assertion holds for $m = 1$ with the original positive-frequency band cutoff in place of $\tilde{Q}_1$. The sums can be obtained as limits of finite packet collections.

Proof. We give the construction, including the normalization and the separation of the different series forms. At each length ℓ , partition the frequency slope $-\xi/\eta$ smoothly into intervals of width cd , $d = w/\ell$, with centers θ on a lattice of spacing a fixed small multiple of d . Take c so small that on every such frequency piece $|\ell(\xi + \theta\eta)| \leq c_*/4$. Introduce normalized coordinates

$$X' = \ell(\xi + \theta\eta), \quad Y' = W(\eta - \eta_j), \quad U' = (u - \theta)/d. \quad (1.34)$$

Then

$$\ell(\xi + u\eta) = X' + (w\eta)U', \quad w\eta = w\eta_j + Y'/m. \quad (1.35)$$

The vertical cutoff restricts Y' to a fixed compact interval and $w\eta$ to a fixed positive annulus. On the support of ψ , its argument is bounded away from zero. In view of $|X'| \leq c_*/4$, this forces $c'_b \leq |U'| \leq C'_b$, with fixed positive bounds. Enlarge these bounds slightly, retaining separation from zero. Since $d \leq 1$ and $|u| \leq 1$, only bounded values of θ can contribute.

In the coordinates (1.34), the product of the annular multiplier, the slope partition, and the vertical cutoff is smooth on a fixed compact coordinate region, with all fixed derivatives bounded uniformly. For example the slope partition is a smooth function of $-X'/(w\eta)$, and (1.35) shows that its derivatives are uniform because $w\eta$ stays away from zero and $m \geq 1$. Choose slightly enlarged smooth plateaus in X', Y', U' , with the X' -plateau supported in $|X'| < c_*$, the U' -plateau supported away from zero, and an additional vertical plateau keeping $w\eta$ in a positive annulus. Fourier series on a larger fixed coordinate box now give

$$\sum_{\nu \in \mathbb{Z}^3} a_{\nu, \ell, \theta, j} A_{\nu, \ell, \theta, j}(\xi, \eta) B_{\nu, \ell, \theta, j}((u - \theta)/d),$$

where A_ν is a localized exponential in (X', Y') , B_ν is a localized exponential in U' , and for every fixed N ,

$$\sup_{\ell, \theta, j, w, m} |a_{\nu, \ell, \theta, j}| \leq C_N (1 + |\nu|)^{-N}. \quad (1.36)$$

This follows by integration by parts on the fixed coordinate box. Multiplication by the plateaus restores the original symbol exactly; their supports give the strict spectral and selector restrictions in Theorem 1.13.

Factor the frequency function A_ν into two smooth frequency functions by inserting a plateau equal to one on its support. Both factors retain $|X'| < c_*$, a vertical interval of length C/W , and a positive annular vertical support. Their inverse Fourier transforms are modulated Schwartz kernels at the scales ℓ, W . Their fixed seminorms, and those of the selector factor, grow at most polynomially in $|\nu|$. The coefficients in (1.36) absorb these losses uniformly. One can do this simultaneously for every fixed seminorm as follows. Let $A_\nu^* = \sup_{\ell, \theta, j, w, m} |a_{\nu, \ell, \theta, j}|$ and omit zero values. Extract the common coefficient $c_\nu = (A_\nu^*)^{1/4}$, multiply each of the two frequency factors by $(A_\nu^*)^{1/4}$, and multiply the selector factor by $a_{\nu, \ell, \theta, j}/(A_\nu^*)^{3/4}$. The product is unchanged. Each of the three factors now has uniform fixed seminorms, by (1.36), and $\sum_\nu c_\nu < \infty$. In particular, no derivative order in this construction grows with L or a truncation parameter.

For clarity, consider one such factored term $b(u)A(D)B(D)$. Write K_A, K_B for its two convolution kernels and $|R| = \ell W$. At an intermediate convolution center z , put

$$\Psi_z(x) = |R|K_A(x - z), \quad \Phi_z(y) = |R|\overline{K_B(z - y)}.$$

Then the contribution to the bilinear form is exactly

$$\frac{1}{|R|^2} \int_{\mathbb{R}^2} \langle f, \Phi_z \rangle \langle b(u)\Psi_z, g \rangle dz. \quad (1.37)$$

The conjugation in Φ_z keeps its Fourier support in the same positive-frequency region as that of Ψ_z . Both waves have the carrier $\eta_j(-\theta, 1)$ and satisfy (1.30)–(1.31).

Let $\mathcal{Q}$ be a fundamental cell of the box lattice, of area $|R|$. For any integrable function F ,

$$\int_{\mathbb{R}^2} F(z) dz = \int_{\mathcal{Q}} \sum_{n, k \in \mathbb{Z}} F(z + (n\ell, \theta n\ell + kW)) dz.$$

Applying this identity to (1.37) and normalizing the average over $\mathcal{Q}$ leaves precisely the factor $|R|^{-1}$ in (1.33). Use the same translation for every j at this scale and slope. There are only finitely many lengths and relevant slope labels, so a finite product of these normalized translation measures supplies a common average for their complete sum.

We also need one angular grid per form. The union of the wave slope supports and all selector supports for a label θ is contained in $[\theta - Cd, \theta + Cd]$, with C fixed and independent of j . Choose a sufficiently large fixed dyadic multiple Hd . Among three adjacent dyadic grids, this interval is contained in a cell of length Hd with a margin comparable to d . One may see this directly from the three grids

$$\{2^{-k}([0, 1) + n + (-1)^k \alpha) : n, k \in \mathbb{Z}\}, \quad \alpha \in \{0, 1/3, 2/3\} :$$

they are nested, and their boundary positions at any fixed scale are spaced by one third of that scale. Taking H large gives the claimed containment and margin. This is the usual adjacent-grid principle; see [25, Theorem 3.1 and Remark 3.2]. Assign each label to one suitable grid and split the sum into the three resulting forms. Since $d = w/\ell$ is dyadic, the chosen cell level depends only on ℓ .

At a fixed length and label, the lattice sums converge absolutely in a bilinear pairing of L^2 functions. Indeed, envelope decay and Cauchy–Schwarz give

$$\sum_z |R|^{-1} |\langle f, \Phi_z \rangle|^2 \leq C \|f\|_2^2, \quad \sum_z |R|^{-1} |\langle b_z(u) \Psi_z, g \rangle|^2 \leq C \|g\|_2^2.$$

For example, bound each squared pairing by $C|R|$ times the integral of $|f|^2$ against a decaying box envelope, and then sum those envelopes over the lattice. Their sum is uniformly bounded. The same argument works with the bounded selector and with vector components, by a further Cauchy–Schwarz inequality in j . Since the length and label sets are finite, the continuous representation is justified by these bounds and the summability in ν , first on regular inputs and then on L^2 . Exhausting each lattice by finite subsets gives the claimed finite packet approximations.

For each fixed series index and angular grid, all scales and labels are kept in one form before its estimate is applied. Different series indices remain separate forms and are summed only using $\sum_\nu c_\nu < \infty$. In particular, their copies of the same spatial lattice are never combined into a collection with unbounded multiplicity. This completes the representation. $\square$

1.15. The precise remaining model estimates. The following proposition states the forward dependency used to finish the band theorem. It also records exactly how much saving survives the bin and amplitude reductions.

Proposition 1.15. *Suppose that the following bounds hold, uniformly for every admissible finite form Λ_S in Theorem 1.13, with the fixed packet constants there. The field $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ is bounded and measurable, the sets $F, E \subset \mathbb{R}^2$ are measurable with positive finite measure, and the input and test satisfy $\|f(z)\|_{\mathcal{H}} \leq \mathbf{1}_F(z)$, $\|g(z)\|_{\mathcal{H}} \leq \mathbf{1}_E(z)$.*

(i) *For $m \geq 1$, there is a fixed exponent C_1 such that*

$$|\Lambda_S(f, g)| \leq C(1 + \log(2m))^{C_1} \sqrt{|F||E|}. \quad (1.38)$$

(ii) *For $L \geq L_0$, with τ, m, W as in (1.17), assume in addition that $|u| \leq 1$, $\text{Lip}(u) \leq C_{\text{Lip}}$, and $F, E \subset I \times \mathbb{R}$ for an interval I with $|I| \leq b_0$. Whenever $L/2 \leq \log \sqrt{|E|/|F|} \leq 2L$,*

$$|\Lambda_S(f, g)| \leq CL^{-20} \sqrt{|F||E|}. \quad (1.39)$$

(iii) *For $m = 1$, one scalar component, and $|E| \leq |F|$, there is a fixed $c > 0$ such that*

$$|\Lambda_S(f, g)| \leq C \sqrt{|F||E|} (|E|/|F|)^c. \quad (1.40)$$

Then Theorem 1.4 holds. The constants remain uniform in w, ϵ , and the number of retained scales.

Proof. We apply the assumed bounds to the local field and sets of [Theorem 1.4](#), for which $|u| \leq 1$, $\text{Lip}(u) \leq C_{\text{Lip}}$, and $F, E \subset I \times \mathbb{R}$ with $|I| \leq b_0$. Suppose first that $L = \log \sqrt{|E|/|F|} \geq L_0$. The time and scale reductions give exponentially small normalized errors by (1.19). The bin error is smaller than any fixed power of L^{-1} , by (1.26). It remains to estimate the diagonal form in (1.24).

Apply the amplitude decompositions from [Theorem 1.12](#) to F^0, G^0 . The projections defining these vectors preserve horizontal support. After omitting levels of measure zero, their level sets have positive finite measure by (1.27) and lie in $I \times \mathbb{R}$; the normalized slices satisfy the stated indicator bounds. First use finite decompositions. Each pair of slices has a packet representation by [Theorem 1.14](#); its cutoff is $\tilde{Q}_j$, so the slice itself need not have narrow spectrum. Summing the common series coefficients and averaging the translations preserve each assumed model bound. Pairs for which both measures are retained satisfy (1.29); apply (1.39) to them. Their total contribution, divided by $\sqrt{|F||E|}$, is at most

$$CL^{-20}(1+L) \leq CL^{-19}.$$

If either measure is discarded, use (1.38). The sum of those normalized weights is at most $C\sqrt{1+L}e^{-cL} + Ce^{-2cL}$; the factor $(1 + \log(2m))^{C_1}$ is a fixed power of L . These terms are therefore exponentially small. The estimates are uniform for the finite amplitude sums. Their tails converge absolutely by the same bounds, so they apply to the full diagonal form. All errors and the main term are bounded by $C(1+L)^{-3}\sqrt{|F||E|}$, as required.

If L ranges over a fixed bounded interval, use $m = 1$. No bin projections or their amplitude losses are needed. The time and scale errors have bounded operator norms, and (1.38), with the scalar packet representation, bounds the remaining form. The factor $(1 + |\log(|F|/|E|)|)^{-3}$ is bounded below by a positive absolute constant on this interval.

Finally suppose that L is large and negative. Again use $m = 1$, retain the original scalar input and test, and apply (1.40). It gives a positive power of $|E|/|F|$. The errors give the power $(|E|/|F|)^{1/4}$ by the fixed choice $p = 4$ in (1.19). Every fixed positive power of this ratio is bounded by a constant times $(1 + |\log(|F|/|E|)|)^{-3}$. This proves the remaining case.

All arguments include an arbitrary positive inner cutoff. If $\epsilon > w$, the lower retained scale is raised to a dyadic scale comparable to ϵ , which simply selects a finite subcollection of the allowed scales. If it exceeds τ , the middle form is empty and the error estimates suffice. If $w \geq \tau$, the short-time and horizontal bounds already cover the whole remaining integral. None of the bounds used here depends on how many scales lie between the endpoints. $\square$

The basic and reverse-disparity assertions of [Theorem 1.15](#) will follow from the ordinary tile estimates, in particular [Theorem 2.1](#). The remaining large-positive-disparity assertion is the improvement proved by the popular-forest argument and assembled in [Theorem 7.5](#). Thus the conditional use of [Theorem 1.4](#) in [Theorem 1.9](#) is separate from, and subsequent to, the proof of its model hypotheses.

2. ORDINARY TILES AND THE POPULAR FOREST

The purpose of this section is to separate the part of the packet form controlled by ordinary size and density estimates from the part requiring phase averaging. The size–density–tree decomposition follows the time–frequency framework of Lacey and Thiele for the Carleson operator [22] and its planar directional adaptation by Lacey and Li [23, 21]. We prove the estimates in the present normalization, including arbitrary subcollections and the Hilbert-valued summation needed for narrow vertical bins.

For the ordinary estimates, write the packet form (1.33) as $\Lambda_{\mathcal{S}}(f, g)$, where $\mathcal{S}$ is a finite collection from one of the forms constructed in the preceding section. The constants in these

estimates depend on fixed packet adaptation, separation, and multiplicity constants only. In particular, they do not depend on the number of tiles or lengths, on w , or on $m = W/w \geq 1$. These ordinary estimates hold for every measurable real-valued slope function u .

2.1. Density and the model estimates. Fix one of the angular dyadic grids. Denote by $\Omega(t, l)$ the interval in this grid at the level corresponding to the dyadic length l that contains t . The intervals are half open. Their length is a fixed constant times w/l ; in particular, $\Omega(t, \ell_s) = \Omega_s$ whenever $t \in \Omega_s$.

For a box R of length l , width W , slope t , and center $z_R = (x_R, y_R)$, put

$$B_R(x, y) = \left(1 + \frac{|x - x_R|}{l} + \frac{|(y - y_R) - t(x - x_R)|}{W}\right)^{-D_0}. \quad (2.1)$$

Here D_0 is a sufficiently large fixed integer, chosen once below. For a measurable set E of finite measure define

$$D_E(s) = \sup_{\substack{\ell_s \leq l \leq r, \, l \text{ dyadic} \\ t \in \Omega_s, \, z_s \in R}} \frac{1}{|R|} \int_{E \cap \{u \in \Omega(t, l)\}} B_R(z) \, dz. \quad (2.2)$$

The boxes in this supremum have the dimensions and slope used in (2.1). Since $\int B_R \leq C|R|$, all densities are bounded by a fixed constant. Density is monotone in E ; passing to a subcollection of tiles does not change the density of any remaining tile.

Proposition 2.1 (Ordinary model estimates). *Let the bin Hilbert space be $\mathcal{H} = \mathbb{C}^J$, where $J \leq C_{\text{bin}} m$. If $\|g(z)\|_{\mathcal{H}} \leq \mathbf{1}_E(z)$ and $D_E(s) \leq \gamma$ for every $s \in \mathcal{S}$, with $0 < \gamma \leq C$, then*

$$|\Lambda_{\mathcal{S}}(f, g)| \leq C(1 + \log(2m))^C \gamma^{1/4} |E|^{1/2} \|f\|_{L^2(\mathcal{H})}. \quad (2.3)$$

In the scalar case, if $|f| \leq \mathbf{1}_F$, $|g| \leq \mathbf{1}_E$, and $0 < |E| < |F| < \infty$, then

$$|\Lambda_{\mathcal{S}}(f, g)| \leq C \sqrt{|F||E|} \left(\frac{|E|}{|F|}\right)^{9/20}. \quad (2.4)$$

Both conclusions hold uniformly for every subcollection of $\mathcal{S}$.

We first prove the scalar estimates, suppressing the bin index j ; size and density selection and the tree bound provide the scalar model bounds. The final component summation gives the stated logarithmic vector loss in m .

2.2. Maximal functions and separated synthesis. For a slope t , let $\mathcal{M}_t$ be the uncentered strong maximal operator in the coordinates $(x, y - tx)$. We shall use

$$\|\mathcal{M}_t h\|_2 \leq C \|h\|_2, \quad (2.5)$$

with a constant independent of t . Here and below maximal operators act on the absolute value of their argument. To recall a proof, the one-dimensional uncentered maximal operator M is of weak type $(1, 1)$: a finite interval covering of a compact subset of $\{Mh > \lambda\}$ has a disjoint subfamily whose threefold enlargements cover that subset, giving measure at most $3\lambda^{-1} \|h\|_1$. Approximation gives the full inequality. Apply it to $h \mathbf{1}_{\{|h| > \lambda/2\}}$, since the complementary part has maximal function at most $\lambda/2$, and integrate the distribution estimate against $2\lambda \, d\lambda$. Fubini's theorem gives $\|Mh\|_2^2 \leq C \|h\|_2^2$. The strong maximal operator is bounded pointwise by the composition of the two one-dimensional maximal operators. Fubini and the determinant-one shear prove (2.5).

A consequence we use repeatedly is smooth reproduction on a box. Suppose a Schwartz function, after removal of a fixed carrier, has Fourier support

$$|\xi + t\eta| \leq A/r, \quad |\eta| \leq A/W.$$

A product of smooth low-pass cutoffs that equals one on this set reproduces the function. Its kernel is bounded, for any fixed N , by

$$\frac{C_{A,N}}{rW} \left(1 + \frac{|x|}{r}\right)^{-N} \left(1 + \frac{|y - tx|}{W}\right)^{-N}.$$

Decomposing this majorant into dilated rectangles proves the following statement. If Q is an $r \times W$ box of slope t , and F_α is any finite family with these common spectral bounds, then

$$\sup_{z \in Q} \sup_{\alpha} |F_\alpha(z)| \leq C_A \inf_{z \in Q} \mathcal{M}_t(\sup_{\alpha} |F_\alpha|)(z). \quad (2.6)$$

Indeed, replacing the center of any kernel rectangle by any other point of Q changes its required enlargement by only an absolute factor. A common carrier has modulus one and does not affect this argument.

A test tree has parameters $T = (z_T, t, l)$, where $l \leq \tau$ is dyadic, and a box R_T of length l , width W , and slope t . Its tiles satisfy

$$\ell_s \leq l, \quad z_s \in R_T, \quad t \in \Omega_s. \quad (2.7)$$

An arbitrary subset satisfying these conditions is also a tree. Choose a fixed $c_1 > 0$ so that $c_* \ll c_1 \ll \min(c_b, c_\Omega)$, where $c_b d_s$ is the lower distance of the selector support from θ_s . A tile is separated in this tree if $|t - \theta_s| \geq c_1 d_s$.

For a positive integer J_0 define the top polynomial

$$\mathfrak{p}_{T,J_0}(z) = \left(1 + \frac{(x - x_T)^2}{l^2} + \frac{((y - y_T) - t(x - x_T))^2}{W^2}\right)^{J_0}. \quad (2.8)$$

Lemma 2.2 (Separated synthesis). *Let $\mathcal{T}$ be separated tiles satisfying (2.7), and let Υ_s be either of their packet families Φ_s or Ψ_s . For every fixed J_0 ,*

$$\left\| \mathfrak{p}_{T,J_0} \sum_{s \in \mathcal{T}} c_s \Upsilon_s \right\|_2 \leq C_{J_0} \left(\sum_{s \in \mathcal{T}} |c_s|^2 |R_s| \right)^{1/2}, \quad (2.9)$$

$$\left\| \mathfrak{p}_{T,J_0} \sup_{a < b} \left| \sum_{\substack{s \in \mathcal{T} \\ a < \ell_s \leq b}} c_s \Upsilon_s \right| \right\|_2 \leq C_{J_0} \left(\sum_{s \in \mathcal{T}} |c_s|^2 |R_s| \right)^{1/2}. \quad (2.10)$$

The same statements hold for Hilbert-valued sums $(\sum_s c_{s,j} \Upsilon_{s,j})_j$, with $\sum_s \|c_s\|_{\mathcal{H}}^2 |R_s|$ on the right and constants independent of J . Bounded scalar factors depending on s, j , such as $b_{s,j}(t)$, can be incorporated into the coefficients.

Proof. From the packet support and $t \in \Omega_s$,

$$\xi + t\eta = (\xi + \theta_s \eta) + (t - \theta_s)\eta, \quad \frac{a_0}{\ell_s} \leq |\xi + t\eta| \leq \frac{a_1}{\ell_s},$$

where $0 < a_0 < a_1 < \infty$ are fixed. Positivity of a_0 uses $c_* \ll c_1$ and the two fixed bounds on $w\eta$. For envelopes, replacing slope θ_s by t changes the transverse coordinate over one tile length by at most $C d_s \ell_s = Cw \leq CW$. Thus every fixed Schwartz envelope remains a uniform envelope in the t -coordinates.

At one length there are boundedly many labels, since all labels of the tree lie in an interval of length $O(d_s)$. Within each label the centers lie on a translated box lattice. The inner product of two normalized packets at this length is bounded by a fixed rapidly decreasing function of their normalized center separation. Its row and column sums are bounded. The elementary Schur

inequality, obtained by applying $2|a_s a_q| \leq |a_s|^2 + |a_q|^2$ to these row and column sums, proves the fixed-length synthesis bound.

This argument also applies after multiplying packets by $\mathfrak{p}_{T,J_0}$. Their centers belong to R_T , their lengths are at most l , and their widths are W . Consequently the polynomial can be absorbed into a fixed larger number of packet decay powers. Multiplication by a polynomial differentiates the Fourier transform and does not enlarge its support.

Choose a fixed integer q_0 such that $2^{q_0} a_0 > 4a_1$, and split dyadic lengths according to their exponents modulo q_0 . Within each resulting list, the displayed annuli are disjoint and have gaps. Orthogonality in the first t -coordinate therefore proves (2.9), also for the polynomially weighted packets.

On one such list, the sum over all lengths above a given endpoint is exactly a smooth low-pass convolution in that coordinate applied to the complete sum: choose the cutoff to equal one on the lower-frequency annuli retained at that endpoint and zero on all higher-frequency annuli. The gap permits dilates of one fixed smooth cutoff. Their absolute kernels are dominated by the one-dimensional maximal operator. An interval of lengths is a difference of two endpoint sums. Apply this observation to the full weighted sum, use its just-proved L^2 bound, and sum over the fixed number of lists. This gives (2.10).

For Hilbert-valued sums, the fixed-length and orthogonality bounds follow by summing the scalar squared bounds over components. The same low-pass convolution acts on every component, since a_0, a_1 are common. Its norm is pointwise bounded by the positive kernel acting on the Hilbert norm of the full sum. The maximal-function proof is therefore dimension free. Alternatively one can sum the squared scalar maximal estimates. $\square$

For scalar coefficients $v_s = \langle f, \Phi_s \rangle / |R_s|$, define their size on a collection by

$$\text{size}(v) = \sup_T \left(\frac{1}{|R_T|} \sum_{\substack{s \in T \\ |t - \theta_s| \geq c_1 d_s}} |v_s|^2 |R_s| \right)^{1/2}. \quad (2.11)$$

A one-tile test gives $|v_s| \leq C \text{size}(v)$. If $\Phi_s = 0$, then $v_s = 0$. Otherwise choose $(\xi, \eta) \in \text{supp } \widehat{\Phi}_s$ and put $q = -\xi/\eta$. The angular margin and $c_1 \ll c_\Omega$ place both $q - 2c_1 d_s$ and $q + 2c_1 d_s$ in Ω_s . At least one is at distance at least $2c_1 d_s$ from θ_s ; use it as the direction of a top centered at z_s with length ℓ_s . This is a separated one-tile test of area $|R_s|$, which proves the bound. Moreover

$$\text{size}(v) \leq C \|f\|_\infty. \quad (2.12)$$

To prove this, for the separated portion of a fixed tree put $B_T = \sum_s |v_s|^2 |R_s|$ and $A_T = \sum_s v_s \Phi_s$. The pairing is linear in its first argument, so $B_T = \langle f, A_T \rangle$. By Theorem 2.2 and Cauchy–Schwarz with $\mathfrak{p}_{T,1}^{-1}$,

$$B_T \leq \|f\|_\infty \|A_T\|_1 \leq C \|f\|_\infty |R_T|^{1/2} B_T^{1/2}.$$

Division, with the zero case understood, proves (2.12).

2.3. Removing large size.

Lemma 2.3 (Size selection). *Suppose $\text{size}(v) \leq C_* \sigma$, with C_* fixed. One can remove a collection of trees so that the remaining size is at most σ and*

$$\sum_T |R_T| \leq C \sigma^{-2} \|f\|_2^2. \quad (2.13)$$

The removed tiles partition into ordinary trees satisfying (2.7); their total top area has the same bound.

Proof. First consider positive separated tests, with $t - \theta_s \geq c_1 d_s$. While a test has energy larger than $\sigma^2 |R_T|/2$, choose one whose direction is within ε of the infimum of the directions of all such tests. Here $\varepsilon > 0$ is a sufficiently small fixed multiple of $\min_{s \in \mathcal{S}} d_s$. Delete all remaining tiles with $\ell_s \leq l_T$, $t_T \in \Omega_s$, and $z_s \in A_0 R_T$, including nonseparated tiles. The fixed absolute dilation A_0 is chosen below. Record the positive separated test tiles present at selection as $\mathcal{Q}_T$. The procedure terminates because every selection deletes at least one tile. Thereafter perform the negative separated selection, using directions within ε of the supremum and reversing the order. At the end, each signed energy is at most $\sigma^2 |R_T|/2$; hence the size is at most σ .

We prove the area bound for the positive selection. The $\mathcal{Q}_T$ are disjoint and

$$B := \sum_T \sum_{s \in \mathcal{Q}_T} |v_s|^2 |R_s| \asymp \sigma^2 \sum_T |R_T|. \quad (2.14)$$

The lower comparison is the selection threshold and the upper one follows from the initial size bound, which persists on every remaining subcollection. We claim

$$\left\| \sum_T \sum_{s \in \mathcal{Q}_T} v_s \Phi_s \right\|_2^2 \leq CB. \quad (2.15)$$

Expand the squared norm. Pairs with lengths in a fixed ratio have total contribution $O(B)$. Indeed, spectral overlap forces their labels to lie within $O(d_s)$, so there are boundedly many neighboring labels. Their box shapes are comparable because the transverse discrepancy over either length is $O(w) \leq O(W)$. Packet decay and the spatial lattices then give the bounded row and column sums for the normalized Gram matrix, exactly as in the fixed-length proof above.

For the other pairs fix the top T of the longer member $q \in \mathcal{Q}_T$, and write $t = t_T$. Let $\mathcal{A}_T$ consist of the selected tiles s for which $\ell_q > C_0 \ell_s$ and $\langle \Phi_s, \Phi_q \rangle \neq 0$ for at least one such q . Choose the fixed C_0 sufficiently large. Spectral overlap and the common vertical frequency range give

$$|\theta_s - \theta_q| \leq C c_* (d_s + d_q), \quad |\theta_q - t| \leq C_\Omega d_q.$$

Thus, by the choices of c_* and C_0 ,

$$|\theta_s - t| \leq \frac{1}{4} c_1 d_s, \quad t \in \Omega_s. \quad (2.16)$$

The second assertion uses the fixed margin about the wave slope support inside Ω_s .

We next verify a bounded overlap fact for the boxes of $\mathcal{A}_T$ in the coordinates of slope t . These boxes can all be replaced by fixed comparable $\ell_s \times W$ boxes in those coordinates. Suppose two of the latter overlap, with ℓ_s much smaller than $\ell_{s'}$. Let $t_s, t_{s'}$ be the directions of their respective selecting tests. Because s was in a positive separated test,

$$t_s \geq \theta_s + c_1 d_s > t + \frac{1}{2} c_1 d_s, \quad |t_{s'} - t| \leq C d_{s'}.$$

After separating the lengths by another fixed factor, $t_{s'} < t + \frac{1}{4} c_1 d_s$. If the test selecting s' had not yet been selected when s was selected, all its eventual test tiles would still have been present at that time. It would therefore already have been failing, since deleting tiles only decreases each fixed test energy. The near-infimum rule, with $\varepsilon \ll c_1 \min d_s$, therefore forces the test of s' to have been selected first.

Its length is at least $\ell_{s'}$. Overlap of the comparable t -boxes bounds the horizontal center separation by $C \ell_{s'}$. In the coordinates of $t_{s'}$, the transverse center separation is $O(W + w) = O(W)$: use $|t_{s'} - t| \ell_{s'} = O(w)$, or equivalently compare each original tile over its own length. Also $t_{s'} \in \Omega_s$, by the margin in (2.16). For an absolute A_0 this places s in the enlarged deletion tree of s' , a contradiction. At comparable lengths, bounded label and lattice multiplicity give bounded overlap directly. Splitting lengths into a fixed number of lists proves the asserted

overlap bound for all of $\mathcal{A}_T$. The same reasoning applies to any fixed enlargement of these boxes by increasing A_0 once.

Let B_s^* be sufficiently rapidly decreasing uniform tile envelopes in the t -coordinates, and put $D_T = \mathfrak{p}_{T,J_0}^{-1}$, with J_0 a fixed large integer. Then

$$\left\| D_T \sum_{s \in \mathcal{A}_T} B_s^* \right\|_2 \leq C |R_T|^{1/2}. \quad (2.17)$$

Here centers of $\mathcal{A}_T$ need not belong to R_T . To prove the estimate, test against $\|h\|_2 \leq 1$. Since $\ell_s \leq l_T$, the top weight can be passed across a tile envelope by spending fixed decay powers. Decomposition into dilated tile boxes, followed by the overlap just proved, gives

$$\begin{aligned} \sum_s \int |h| D_T B_s^* &\leq C \sum_s |R_s| \inf_{R_s^t} D_T \inf_{R_s^t} \mathcal{M}_t h \\ &\leq C \int D_T \mathcal{M}_t h \leq C \|D_T\|_2 \|h\|_2 \leq C |R_T|^{1/2}. \end{aligned}$$

The notation R_s^t denotes the fixed comparable t -box; changing it by an absolute dilation is harmless. This proves (2.17).

For each $s \in \mathcal{A}_T$, the sum of its cross terms with $\mathcal{Q}_T$ can include all $q \in \mathcal{Q}_T$ with $\ell_q > C_0 \ell_s$, since the added inner products are zero. Therefore its absolute value is bounded using

$$U_T(z) = \sup_{r>0} \left| \sum_{\substack{q \in \mathcal{Q}_T \\ \ell_q > r}} v_q \Phi_q(z) \right|.$$

The coefficient bound $|v_s| \leq C\sigma$, (2.17), and Theorem 2.2 show that all such cross terms for this top have sum at most

$$C\sigma \int \left(\sum_{s \in \mathcal{A}_T} B_s^* \right) U_T \leq C\sigma |R_T|^{1/2} \|D_T^{-1} U_T\|_2 \leq C\sigma^2 |R_T|.$$

Summing and using (2.14) proves (2.15). Pairing its synthesis with f gives $B \leq \|f\|_2 (CB)^{1/2}$, hence $B \leq C \|f\|_2^2$. The negative selection has the identical proof with order reversed.

Finally, partition each enlarged deletion box $A_0 R_T$ into a fixed number of translated ordinary top boxes of length l_T and width W , assigning tiles by their centers. Their direction and length conditions are unchanged, and their total area is $O(|R_T|)$. This proves the last assertion and (2.13). $\square$

2.4. Removing large density.

Lemma 2.4 (Density selection). *For $0 < \eta \leq C$, all tiles with $D_E(s) > \eta$ can be assigned to keeper boxes R_T , of dyadic length $l_T \leq \tau$, width W , and slope t_T , with the following properties. There are pairwise disjoint measurable sets E_T such that*

$$E_T \subset E \cap K_\eta R_T \cap \{u \in \Omega(t_T, l_T)\}, \quad |E_T| \geq c\eta |R_T|, \quad (2.18)$$

$$\sum_T |R_T| \leq C\eta^{-1} |E|, \quad K_\eta \leq C(1 + \eta^{-1/100}), \quad (2.19)$$

$$\ell_s \leq l_T, \quad t_T \in \Omega_s, \quad z_s \in CK_\eta R_T \quad \text{for each tile assigned to } T. \quad (2.20)$$

After partitioning by centers, the assigned tiles form ordinary test trees of total top area at most

$$C\eta^{-1} (1 + \eta^{-1/20}) |E|. \quad (2.21)$$

Proof. For each tile to be removed select one box in (2.2) witnessing an average greater than η . Since

$$\int_{\mathbb{R}^2 \setminus KR} B_R \leq CK^{2-D_0}|R|,$$

choosing the fixed D_0 sufficiently large and then $K = K_\eta \leq C(1 + \eta^{-1/100})$ retains at least $\eta|R|/2$ of the weighted test inside KR . Its unweighted tested set has measure at least $\eta|R|/2$, because $B_R \leq 1$.

Process these finitely many witnesses in decreasing length order. Keep a witness unless both its K -dilated spatial box and its dyadic direction interval intersect the corresponding objects of an earlier keeper. Assign it to any such keeper when it is rejected. The tested sets of the keepers are disjoint: a common tested point would force both intersections and preclude keeping the later one. This proves (2.18) and (2.19).

Let a rejected witness have length l' , direction t' , and original associated tile s , and let its keeper have length $l_T \geq l'$ and direction t_T . Dyadic nesting gives

$$\Omega(t_T, l_T) \subset \Omega(t', l') \subset \Omega_s, \quad |t_T - t'| \leq Cw/l'.$$

Choose a point common to the two dilated spatial boxes. Moving from it to the rejected box center, or to z_s , costs horizontally $O(Kl')$ and transversely in the t_T -coordinates $O(KW + Kl'|t_T - t'|) = O(KW)$. This proves (2.20). A keeper's own associated tiles satisfy the same conclusion.

Partition CKR_T into $O(K^2)$ translated boxes of length l_T , width W , and slope t_T , assigning tiles by their centers. The length and angular requirements for a test tree persist. Their total area is at most $C(1 + \eta^{-1/50})\eta^{-1}|E|$, which is bounded by (2.21) after changing the absolute constant. $\square$

2.5. The tree estimate.

Lemma 2.5 (Tree bound). *Suppose a scalar tree $\mathcal{T}$ satisfies $D_E(s) \leq \eta$ on its tiles and has size at most σ . If $|g| \leq \mathbf{1}_E$, then*

$$|\Lambda_{\mathcal{T}}(f, g)| \leq C\sigma\eta|R_T|. \quad (2.22)$$

The size assumption can be replaced by the same bound on any encompassing collection.

Proof. The assertion is immediate for an empty tree, so assume the tree is nonempty. Work in centered coordinates $(x, y - tx)$ of the top, so its coordinate extents are $l \times W$. For clarity write the transverse coordinate as y' , with top center $(0, 0)$. All tile centers satisfy $|x_s| \leq l/2$ and $|y'_s| \leq W/2$. The packet envelopes are uniform in these coordinates.

Partition the horizontal line into maximal dyadic intervals I for which there is no tile with $x_s \in 3I$ and $\ell_s \leq |I|$. The tiles are finite and have positive minimum length, so these intervals cover the line up to endpoints and are pairwise disjoint. Each parent fails the condition, furnishing a tile q_I with

$$\ell_{q_I} \leq 2|I|, \quad \text{dist}(x_{q_I}, I) \leq C|I|. \quad (2.23)$$

Fix a sufficiently large absolute dyadic constant C_W . We treat lengths $\ell_s \leq C_W|I|$ and longer lengths separately on the strip above I .

For a short tile, density tested at its own box and wave decay give, for every fixed large N_0 ,

$$\int_{E \cap \{x \in I\}} |v_s b_s(u) \Psi_s| \leq C\sigma\eta|R_s| \left(1 + \frac{\text{dist}(x_s, I)}{\ell_s}\right)^{-N_0}. \quad (2.24)$$

Indeed the selector support lies in Ω_s , the density definition includes R_s with direction θ_s , and a fixed number of additional wave-decay powers extracts the last factor while retaining the density weight.

Here is an explicit summation of (2.24). At a fixed length there are $O(1)$ labels, and each horizontal column contains $O(1)$ centers within the transverse top strip. If $\ell_s \leq r := |I|$, the defining property of I gives $\text{dist}(x_s, I) \geq r$. Summing over the one-dimensional center lattice then bounds its contribution, divided by $C\sigma\eta W$, by $Cr(\ell_s/r)^{N_0-1}$. For $r < \ell_s \leq C_W r$, the same lattice sum is $O(r)$ after multiplication by ℓ_s , and there are only a fixed number of such lengths. Thus the total is $O(\sigma\eta W r)$. All intervals with $r \leq l$ lie within a fixed enlargement of the top's horizontal interval, by (2.23); their disjoint lengths sum to $O(l)$. If $r > l$, all tile lengths are at most l , all centers are at distance at least r from I , and their restriction $|x_s| \leq l/2$ improves the preceding sum to $C\sigma\eta W l(l/r)^{N_0-1}$. For each such dyadic r , only $O(1)$ intervals can occur, again by the parent witness. Summing proves a total short-tile contribution $C\sigma\eta l W$.

Now let I admit longer tiles. Choose a dyadic $l' \asymp |I|$, greater than ℓ_{q_I} and small enough that $l' < \ell_s$ for every longer tile. Increasing C_W once ensures this. It also ensures $l' \leq \tau$. By (2.23), a box of slope t , length l' , and width W can contain z_{q_I} and the portion $I \times [-W/2, W/2]$, with a fixed enlargement absorbed into the choice of l' . Dyadic nesting puts every longer selector support inside $\Omega(t, l')$. Consequently the original longer-tile sum vanishes outside $A_I = \{u \in \Omega(t, l')\}$. We perform all following freezing and error decompositions after multiplying that sum by $\mathbf{1}_{A_I}$, and retain this indicator on both the main term and the error. Individually the frozen terms need not vanish outside A_I . For the cells

$$Q_{I,k} = I \times [(k - \tfrac{1}{2})W, (k + \tfrac{1}{2})W], \quad k \in \mathbb{Z},$$

the density test at q_I therefore yields

$$\mu_{I,k} := |Q_{I,k} \cap E \cap \{u \in \Omega(t, l')\}| \leq C\eta(1 + |k|)^{D_1} |I|W, \quad (2.25)$$

where the fixed D_1 can be taken to be D_0 . The density weight on the cell is bounded below by a constant times $(1 + |k|)^{-D_0}$.

Put $a = |u - t|$ at the point of evaluation. For nonseparated tiles, selector support implies $a \asymp d_s$, so only a fixed number of dyadic lengths occur. Their pointwise sum is bounded by $C\sigma(1 + |k|)^{-N_1}$ on $Q_{I,k}$, for any fixed N_1 . For separated tiles replace $b_s(u)$ by $\mathbf{1}_{\{d_s \geq a\}} b_s(t)$. If $d_s \geq a$, the error is at most Ca/d_s , using the derivative bound on b_s . If $d_s < a$, the actual selector vanishes unless $d_s \geq a/C$, since $|t - \theta_s| \leq C_\Omega d_s$. There are only a fixed number of these remaining scales. Summing the geometric series in a/d_s , and using lattice overlap at each scale, bounds the entire error by $C\sigma(1 + |k|)^{-N_1}$. This argument also covers $a = 0$, when the separated error is zero.

The resulting main sum, at each point, is an interval sum of the separated packets with coefficients $v_s b_s(t)$: its lengths obey $C_W |I| < \ell_s \leq w/a$. Let $M_{I,k}$ be the supremum on $Q_{I,k}$ of the absolute values of all such interval sums with arbitrary upper endpoint. We claim that, for every fixed sufficiently large N_2 ,

$$\sum_{I,k} M_{I,k}^2 (1 + |k|)^{N_2} |I|W \leq C\sigma^2 |R_T|, \quad (2.26)$$

where only the I admitting longer tiles occur. To see this, let U be the maximal interval sum over all separated lengths of the tree. For each fixed I , every interval sum under consideration has horizontal t -frequency at most $C/|I|$ and vertical frequency, after the common carrier is removed, at most C/W . The same is true after multiplication by any top polynomial. Apply (2.6) to the weighted sums and dominate their absolute values by $\mathfrak{p}_{T,J_0} U$. Since the cells are disjoint, the squared cell suprema are bounded in sum by $C \|\mathcal{M}_t(\mathfrak{p}_{T,J_0} U)\|_2^2$. For all the present cells $|x| \leq Cl$, and $\mathfrak{p}_{T,J_0} \asymp (1 + |k|)^{2J_0}$ there. Taking $4J_0 \geq N_2$, (2.5) and Theorem 2.2 prove (2.26).

The intervals admitting longer tiles have total length $O(l)$, by their parent witnesses and $|I| < l/C_W$. Choose $N_2 > 2D_1 + 2$. Factoring the full density bound before Cauchy–Schwarz,

$$\begin{aligned} \sum_{I,k} M_{I,k} \mu_{I,k} &\leq C\eta \left(\sum_{I,k} M_{I,k}^2 (1+|k|)^{N_2} |I|W \right)^{1/2} \\ &\quad \cdot \left(\sum_{I,k} (1+|k|)^{2D_1-N_2} |I|W \right)^{1/2} \leq C\sigma\eta |R_T|. \end{aligned}$$

The previously bounded pointwise errors have the same total bound by (2.25), taking $N_1 > D_1 + 2$. Adding the short contribution proves (2.22). $\square$

2.6. Proof of the model estimates.

Proof of Theorem 2.1. We first prove the scalar part of (2.3). If $\|f\|_2 = 0$ or $|E| = 0$, the form is zero; assume otherwise. Process thresholds $\eta = 2^{-n}$ in decreasing order, starting high enough that no density or size condition fails. At each step first remove densities exceeding η by Theorem 2.4, then lower size to

$$\sigma_\eta = \sqrt{\eta} \|f\|_2 / |E|^{1/2}$$

by Theorem 2.3. At the start of the step, the remaining collection has density at most 2η , as well as the original bound γ , and size at most $\sqrt{2}\sigma_\eta$. These bounds persist under all removals within the step.

The density removals have total ordinary top area at most $C(1 + \eta^{-1/20})\eta^{-1}|E|$, and the size removals at most $C\sigma_\eta^{-2} \|f\|_2^2 = C\eta^{-1}|E|$. Applying Theorem 2.5 to every resulting ordinary tree and summing bounds the form by

$$C \|f\|_2 |E|^{1/2} \sum_{\eta \text{ dyadic}} \min(\gamma, 2\eta) \sqrt{\eta} (1 + \eta^{-1/20}) \eta^{-1}. \quad (2.27)$$

There is no limiting remainder to estimate for a finite collection. Every positive-density tile is eventually removed. If $D_E(s) = 0$, testing at R_s shows that $b_s(u)g = 0$ almost everywhere, since the weight is strictly positive; such a term is zero. Zero-size terms are zero as well.

For $0 < \gamma \leq 1$, the terms with $\eta \leq \gamma$ sum to $C\gamma^{9/20}$, and the terms with $\eta \geq \gamma$ do also: the relevant powers are $\eta^{1/2}, \eta^{9/20}$ below the crossover and $\gamma\eta^{-1/2}, \gamma\eta^{-11/20}$ above it. For bounded larger γ the sum is uniformly bounded. This proves the asserted scalar $\gamma^{1/4}$ estimate, with room in its exponent.

For (2.4), put $r = |E|/|F| < 1$. In addition to the threshold size estimate, every tree has size at most a fixed constant by (2.12). Its actual size at step η is therefore at most $C \min(1, \sqrt{\eta/r})$. No change in the removal algorithm is necessary. With the universal density bound, the sum of the tree contributions at that step is at most

$$C|E| \min(1, \sqrt{\eta/r}) \min(1, \eta) \eta^{-1} (1 + \eta^{-1/20}).$$

For $r \leq \eta \leq 1$, this sums to $C|E|r^{-1/20}$; the scales $\eta > 1$ cost $C|E|$. For $\eta \leq r$, the sum is at most

$$C|E|r^{-1/2} \sum_{\eta \leq r} (\eta^{1/2} + \eta^{9/20}) \leq C|E|r^{-1/20} = C\sqrt{|F||E|} r^{9/20}.$$

This proves the reverse-disparity claim, including the density-area inflation from the selection.

Finally consider $\mathcal{H} = \mathbb{C}^J$. For a scalar measurable h supported in E , write

$$\|h\|_{2,1}^\# = \int_0^\infty |\{ |h| > a \}|^{1/2} da.$$

Layer-cake decomposition, retaining the complex phase of h , extends the scalar estimate to

$$|\Lambda_j(f_j, g_j)| \leq C\gamma^{1/4} \|f_j\|_2 \|g_j\|_{2,1}^\#.$$

Every sliced support lies in E , so its density still has upper bound γ . As $|g_j| \leq 1$, split the last integral at $a_0 = (2 + m)^{-2}$. The portion below a_0 is at most $a_0|E|^{1/2}$; for the rest, Cauchy–Schwarz gives

$$\int_{a_0}^1 |\{ |g_j| > a \}|^{1/2} da \leq \left(\int_{a_0}^1 a |\{ |g_j| > a \}| da \right)^{1/2} \left(\int_{a_0}^1 \frac{da}{a} \right)^{1/2} \leq C\sqrt{1 + \log(2m)} \|g_j\|_2.$$

Since $J \leq C_{\text{bin}} m$ and $\sum_j \|g_j\|_2^2 \leq |E|$,

$$\sum_j (\|g_j\|_{2,1}^\#)^2 \leq C(1 + \log(2m))|E|.$$

Cauchy–Schwarz in j proves (2.3), even with a square root of this logarithmic factor. All arguments used only the stated packet hypotheses, which persist on every subcollection. Every choice of decay or derivative order above is fixed before the parameters and the finite tile collection are chosen. This completes the proof of [Theorem 2.1](#). $\square$

For later parameter choices we record a fixed numerical loss available from this proof. In the regime $m \asymp \exp(L^{0.9})$, with L sufficiently large, the last vector estimate implies

$$|\Lambda_S(f, g)| \leq CL^{b_4} \gamma^{c_4} |E|^{1/2} \|f\|_{L^2(\mathcal{H})}, \quad b_4 = 1, \quad c_4 = \frac{1}{4}. \quad (2.28)$$

These exponents are fixed independently of all subsequent density, forest, and mask choices.

2.7. The improvement problem and popular tops. We now turn to the improvement estimate (1.39) in the local Lipschitz setting: $|u| \leq 1$, $\text{Lip}(u) \leq C_{\text{Lip}}$, and the sets F, E have positive finite measure and lie in the horizontal testing strip $\mathcal{D} = I \times \mathbb{R}$, where $|I| \leq b_0$. Write $\mathcal{H}$ for the finite-dimensional Hilbert space of vertical bins; the input and test satisfy $\|f(z)\|_{\mathcal{H}} \leq \mathbf{1}_F(z)$ and $\|g(z)\|_{\mathcal{H}} \leq \mathbf{1}_E(z)$. We use the parameters in (1.17): L is sufficiently large, $m \asymp \exp(L^{0.9})$, $W = mw$, and every tile length is at most the dyadic cutoff $\tau \leq \exp(-L^{0.97})$. All tile collections in this part are finite. Every estimate is independent of the number of tiles, tops, scales, and bins.

2.8. The assignment to popular tops. Using the fixed exponents $b_4 = 1$ and $c_4 = 1/4$ from (2.28), choose once an integer C_2 with $c_4 C_2 \geq b_4 + 42$. Remove tiles for which $D_E(s) \leq L^{-C_2}$. By [Theorem 2.1](#), in the form (2.28), the absolute value of their contribution is at most

$$CL^{b_4 - c_4 C_2} \|f\|_2 \sqrt{|E|} \leq CL^{-40} \sqrt{|F||E|}.$$

The choice of C_2 depends only on the fixed ordinary loss and the fixed target exponent.

Apply [Theorem 2.4](#) to the remaining tiles with threshold $\eta = L^{-C_2}$, retaining its unsplit keeper boxes. Denote the resulting finite collection of tops by $\mathcal{T}$. A top T has center $z_T = (x_T, y_T)$, slope s_T , length $l_T \leq \tau$, and box

$$R_T = \left\{ (x, y) : |x - x_T| \leq \frac{l_T}{2}, \quad |(y - y_T) - s_T(x - x_T)| \leq \frac{W}{2} \right\}.$$

We put $d_T = w/l_T$, and denote the top assigned to a tile s by $T(s)$. Set $C_{\text{as}} = C_2 + 10$, increasing the absolute lower threshold for L if necessary. This exponent absorbs the assignment dilation $C(1 + L^{C_2/100})$, the witness density cL^{-C_2} , and the area bound $CL^{C_2}|E|$. The density lemma gives

$$\ell_s \leq l_{T(s)}, \quad s_{T(s)} \in \Omega_s, \quad |\theta_s - s_{T(s)}| \leq Cd_s, \quad z_s \in L^{C_{\text{as}}} R_{T(s)}. \quad (2.29)$$

There are pairwise disjoint measurable sets E_T such that

$$\begin{aligned} E_T &\subset E \cap L^{C_{\text{as}}} R_T, & |E_T| &\geq L^{-C_{\text{as}}} |R_T|, \\ |u(z) - s_T| &\leq C d_T \quad (z \in E_T), & S &:= \sum_{T \in \mathcal{T}} |R_T| \leq L^{C_{\text{as}}} |E|. \end{aligned} \quad (2.30)$$

These inequalities hold for every sufficiently large L , without additional constants in the displayed powers. The angular inclusion in (2.29) follows from the nesting of the keeper interval inside Ω_s . The direction bound on E_T follows from the length of that keeper interval. The disjointness and lower measure bound also imply the stated estimate for S .

The horizontal centers of all tops lie in a fixed slight enlargement of I . Indeed, $E_T \neq \emptyset$, and a point of E_T has horizontal coordinate in I and at distance at most $L^{C_{\text{as}}} l_T / 2$ from x_T . Since $L^{C_{\text{as}}} \tau \rightarrow 0$, this distance is smaller than any prescribed positive absolute constant for sufficiently large L .

2.9. The adjoint sum and exceptional subsets. Only the retained tiles, with the assignment $T(s)$, are used henceforth. With pairings linear in the first entry, define

$$a_{s,j} = |R_s|^{-1} \langle g_j, b_{s,j}(u) \Psi_{s,j} \rangle, \quad \phi_s = (a_{s,j} \Phi_{s,j})_j, \quad P = \sum_s \phi_s. \quad (2.31)$$

These conventions give precisely the form (1.33):

$$\langle f, P \rangle = \sum_s |R_s|^{-1} \sum_j \langle f_j, \Phi_{s,j} \rangle \langle b_{s,j}(u) \Psi_{s,j}, g_j \rangle.$$

Indeed $\langle f_j, a_{s,j} \Phi_{s,j} \rangle = \overline{a_{s,j}} \langle f_j, \Phi_{s,j} \rangle$. For fixed g satisfying $\|g(z)\|_{\mathcal{H}} \leq \mathbf{1}_E(z)$, the supremum of the absolute value of this pairing over $\|f(z)\|_{\mathcal{H}} \leq \mathbf{1}_F(z)$ equals $\int_F \|P(z)\|_{\mathcal{H}} dz$. Equality follows by taking $f = P / \|P\|_{\mathcal{H}}$ on $F \cap \{P \neq 0\}$, and zero elsewhere.

Put $N = \sqrt{|E|/|F|}$. The remaining objective, to be proved by the subsequent estimates, is

$$\int_F \|P(z)\|_{\mathcal{H}} dz \leq C L^{-20} N |F|, \quad \frac{L}{2} \leq \log N \leq 2L. \quad (2.32)$$

Applying (2.28) with its absolute density upper bound and arbitrary L^2 input, and then taking the Hilbert-space dual norm, gives the fixed exponent $C_P = b_4 = 1$ such that

$$\|P\|_{L^2(\mathbb{R}^2; \mathcal{H})} \leq C L^{C_P} \sqrt{|E|}. \quad (2.33)$$

Consequently, for every measurable $Z \subset F$,

$$\int_Z \|P(z)\|_{\mathcal{H}} dz \leq C L^{C_P} \left(\frac{|Z|}{|F|} \right)^{1/2} N |F|. \quad (2.34)$$

For any prescribed fixed $Q_0 > 0$, an exceptional subset with $|Z| \leq L^{-2(C_P+Q_0+1)} |F|$ therefore costs at most $C L^{-Q_0} N |F|$. This estimate is applied to the original sum P before any masks or other modifications. It does not require an L^2 bound for those modified sums.

2.10. A weak L^2 estimate for the top count. For a top T , use normalized coordinates

$$X_T(z) = \frac{x - x_T}{l_T}, \quad Y_T(z) = \frac{(y - y_T) - s_T(x - x_T)}{W}, \quad \text{dist}_T(z) = |X_T(z)| + |Y_T(z)|. \quad (2.35)$$

Fix a sufficiently large absolute exponent $D_2 > 4$. For now let $K_0 \geq 1$ be any fixed power of L ; the choice used with the masks is specified below. Set $C_3 = 2^{D_2}$ and

$$h_T(z) = C_3 \left(1 + \frac{\text{dist}_T(z)}{K_0} \right)^{-D_2}, \quad n(z) = \sum_{T \in \mathcal{T}} h_T(z). \quad (2.36)$$

In particular, $h_T \geq 1$ on $\{\text{dist}_T \leq K_0\}$. We use the convention

$$\|a\|_{L^{2,\infty}(\mathcal{D})} = \sup_{\lambda>0} \lambda |\{z \in \mathcal{D} : |a(z)| > \lambda\}|^{1/2}.$$

Proposition 2.6. *For the tops in (2.30),*

$$\|n\|_{L^{2,\infty}(\mathcal{D})} \leq CL^C \sqrt{S}. \quad (2.37)$$

More precisely, if

$$D_* \geq \max(2K_0, 2L^{C_{\text{as}}}, 2),$$

then one may take the right-hand side to be $CL^{C_{\text{as}}} D_*^3 \sqrt{S}$, provided D_* is chosen within an absolute factor of the displayed maximum. In particular, when $K_0 = L^k$, a permissible exponent in (2.37) is

$$c_{\text{cnt}}(k) = C_{\text{as}} + 3 \max(k, C_{\text{as}}). \quad (2.38)$$

Once the fixed power K_0 is chosen, this estimate together with $S \leq L^{C_{\text{as}}} N^2 |F|$ makes the part of F where n exceeds N times a sufficiently large fixed power of L small enough for (2.34).

Proof. If $S = 0$, the assertion is immediate, so assume $S > 0$. We first state the maximal-function input in the form used here. For a Lipschitz unit vector field v_0 , let $M_{v_0, \delta}$ be the supremum of averages over Euclidean rectangles containing the evaluation point, with long side at most $c_{\text{LL}}/\text{Lip}(v_0)$, for which the field belongs to the rectangle's direction uncertainty arc on at least a δ fraction of the rectangle. The arc is centered at the long-side direction and has total length equal to the short-to-long side ratio, as in Theorem 1.2. Theorem 1.2, namely [20, Theorem 1.4], gives

$$\|M_{v_0, \delta} a\|_{2,\infty} \leq C \delta^{-1/2} \|a\|_2, \quad 0 < \delta < 1. \quad (2.39)$$

The same estimate holds if the admissible rectangles are required to satisfy a smaller fixed uncertainty interval, since this restricts the family of rectangles. When $\text{Lip}(v_0) = 0$, the length restriction is understood as absent. We apply this result to $v_0 = (1, u)/\sqrt{1+u^2}$, whose Lipschitz constant is bounded by an absolute constant. We also use the ordinary uncentered strong maximal operator M_{str} over axis-parallel rectangles, which is bounded on $L^2(\mathbb{R}^2)$.

Let $Q \subset \mathcal{D}$ have finite measure, let $D \geq D_*$, and put

$$a_T(D) = \frac{|Q \cap DR_T|}{|DR_T|} = \frac{|Q \cap DR_T|}{D^2 |R_T|}.$$

Here and below dilation is about the box center in its slope coordinates. We prove, for $0 < \beta \leq 1$, that

$$\sum_{T: a_T(D) > \beta} |R_T| \leq \min(S, CL^{2C_{\text{as}}} D^2 \beta^{-2} |Q|). \quad (2.40)$$

Fix a top occurring on the left-hand side. Set

$$J_T = I \cap [x_T - Dl_T/2, x_T + Dl_T/2], \quad a = |J_T|, \quad b = C_w DW,$$

where $C_w \geq 1$ is a fixed absolute constant. The witness set is contained in $DR_T \cap \mathcal{D}$, so $a > 0$. The slope parallelogram

$$\tilde{R}_T = \{(x, y) : x \in J_T, \quad |(y - y_T) - s_T(x - x_T)| \leq b/2\}$$

contains both E_T and $Q \cap DR_T$. It satisfies

$$a \leq \min(|I|, Dl_T), \quad ab \leq C_w D^2 |R_T|, \quad \frac{b}{a} \geq C_w \frac{W}{l_T} = C_w m d_T. \quad (2.41)$$

Thus the possibly very long horizontal dilation has been clipped before any Euclidean enclosing rectangle is constructed.

Suppose first that $b/a \leq \varepsilon_0$, for a fixed small absolute ε_0 . Projection onto the orthonormal directions

$$e_T = \frac{(1, s_T)}{\sqrt{1 + s_T^2}}, \quad e_T^\perp = \frac{(-s_T, 1)}{\sqrt{1 + s_T^2}}$$

encloses $\tilde{R}_T$ in a Euclidean rectangle U_T whose side lengths are

$$L_T = \sqrt{1 + s_T^2} a + \frac{|s_T|}{\sqrt{1 + s_T^2}} b, \quad B_T = \frac{b}{\sqrt{1 + s_T^2}}.$$

The slopes are absolutely bounded. Consequently $L_T \asymp a$, $B_T \asymp b$, $|U_T| \leq CD^2|R_T|$, and

$$L_T \leq C|I|, \quad \frac{B_T}{L_T} \geq cmd_T.$$

On the witness set, normalization of the direction gives $\|v_0 - e_T\| \leq Cd_T$. Since $m \rightarrow \infty$, this error is smaller than any required fixed small multiple of B_T/L_T , for sufficiently large L . The fixed widening constant C_w can absorb the absolute aperture constants as well. Choose the horizontal localization constant so that $C|I|$ meets the length restriction in (2.39). The rectangle U_T is then admissible with the common popularity parameter

$$\delta_D = cL^{-C_{\text{as}}}D^{-2},$$

where $c > 0$ is sufficiently small and absolute. Indeed, $|E_T| \geq L^{-C_{\text{as}}}|R_T|$, and $|U_T| \leq CD^2|R_T|$. Moreover,

$$\frac{1}{|U_T|} \int_{U_T} \mathbf{1}_Q \geq \frac{|Q \cap DR_T|}{CD^2|R_T|} > c\beta.$$

Every point of E_T therefore belongs to $\{M_{v_0, \delta_D} \mathbf{1}_Q > c\beta\}$.

If $b/a > \varepsilon_0$, enclose $\tilde{R}_T$ in the axis-parallel rectangle with horizontal side J_T and vertical side of length $b + |s_T|a$. Its area is at most $Cab \leq CD^2|R_T|$. It contains E_T and $Q \cap DR_T$, so every point of E_T belongs to $\{M_{\text{str}} \mathbf{1}_Q > c\beta\}$. This argument also covers boxes whose transverse side exceeds the horizontal side.

The sets E_T are pairwise disjoint. Summing their lower measure bounds and using the two maximal estimates yields

$$\begin{aligned} \sum_{T: a_T(D) > \beta} |R_T| &\leq L^{C_{\text{as}}} \left| \bigcup_{T: a_T(D) > \beta} E_T \right| \\ &\leq CL^{C_{\text{as}}}(\delta_D^{-1} + 1)\beta^{-2}|Q| \\ &\leq CL^{2C_{\text{as}}}D^2\beta^{-2}|Q|. \end{aligned}$$

The bound by S is immediate, proving (2.40).

For positive numbers S_0, A_0 , direct integration, split at $\min(1, \sqrt{A_0/S_0})$, gives

$$\int_0^1 \min(S_0, A_0\beta^{-2}) d\beta \leq 2\sqrt{S_0 A_0}.$$

Layer cake in (2.40) therefore gives

$$\sum_T |R_T| a_T(D) \leq CL^{C_{\text{as}}} D \sqrt{S|Q|}, \quad (2.42)$$

$$\sum_T |Q \cap DR_T| \leq CL^{C_{\text{as}}} D^3 \sqrt{S|Q|}. \quad (2.43)$$

In the second line the additional factor D^2 is the dilation-area ratio.

Set $D_k = 2^k D_*$. The explicit formula for h_T implies the pointwise bound

$$h_T(z) \leq C_{D_2} \sum_{k \geq 0} 2^{-k D_2} \mathbf{1}_{D_k R_T}(z).$$

To verify it outside $D_* R_T$, choose the least $k \geq 1$ such that $z \in D_k R_T$. Then $\text{dist}_T(z) > 2^{k-2} D_*$, while $D_* \geq 2K_0$, so $h_T(z) \leq C_{D_2} 2^{-k D_2}$. Inside $D_* R_T$, use $h_T \leq C_3$. Applying (2.43),

$$\int_Q n \leq C L^{C_{\text{as}}} D_*^3 \sqrt{S|Q|} \sum_{k \geq 0} 2^{-k(D_2-3)} \leq C L^{C_{\text{as}}} D_*^3 \sqrt{S|Q|}. \quad (2.44)$$

The sum converges because $D_2 > 4$. Finally take $Q = \{z \in \mathcal{D} : n(z) > \lambda\}$. This set has finite measure, since the collection is finite and $D_2 > 2$. The inequality $\lambda|Q| \leq \int_Q n$ proves the asserted weak L^2 bound. All exponents in this argument are fixed: in particular the dilation cost before the final summation is exactly D^3 , independently of the number of tops and angular scales. $\square$

2.11. Parameters and masks. All logarithms below are natural. Set

$$\begin{aligned} p &= L^{1/2}, & G &= 2^{\lceil L^{0.7}/(2 \log 2) \rceil}, \\ K &= \lceil (\log L)^2 \rceil, & B &= G^K. \end{aligned} \quad (2.45)$$

Thus G and $\sqrt{G}$ are dyadic powers, $\log G \asymp L^{0.7}$, and $\log B \asymp L^{0.7}(\log L)^2 = o(L^{0.9})$. In particular,

$$\frac{B^{100}}{m} \longrightarrow 0 \quad (L \longrightarrow \infty). \quad (2.46)$$

The parameters have different tasks. The large bin number m permits fine angular phase separation while keeping changes in the packet envelopes small. The much smaller time cutoff τ pays for the commutation errors. The scale ratio G will separate prediction and averaging lengths; grouping K angular transitions into a block gives the ratio B . Their strict asymptotic ordering is what allows these tasks to coexist.

Let $A > 2$ be a fixed exponent to be chosen sufficiently large after the unmasked top estimates in (3.18). Their fixed exponents and dilation $K_1 = L^{A_1}$ are independent of A . The comparison of the original and masked sums uses Theorem 2.6 at K_1 , so its loss $c_{\text{cnt}}(A_1)$ is fixed before A is chosen. The count at the A -dependent K_0 below is used only after that choice. Constants in the mask estimates may depend on the fixed value of A . Put $r_L = 2\lceil L \rceil$, and define, in the coordinates (2.35),

$$q_T(z) = \left(\text{sinc} \frac{X_T(z)}{L^A} \text{sinc} \frac{Y_T(z)}{L^A} \right)^{r_L}, \quad \text{sinc}(a) = \begin{cases} \sin(a)/a, & a \neq 0, \\ 1, & a = 0. \end{cases} \quad (2.47)$$

The exponent is even, so $0 \leq q_T \leq 1$ and the mask is real. From now on take

$$K_0 = L^{A+2}$$

in (2.36). A top is called *significant at z* if $\text{dist}_T(z) \leq K_0$. Every significant top contributes at least one to $n(z)$.

Lemma 2.7. *The mask has Fourier support*

$$\text{supp } \hat{q}_T \subset \left\{ (\xi, \eta) : |\xi + s_T \eta| \leq \frac{r_L}{L^A L_T}, \quad |\eta| \leq \frac{r_L}{L^A W} \right\}. \quad (2.48)$$

For sufficiently large L , with all other exponents fixed,

$$|q_T(z)| \leq \exp(-L \log L) \quad \text{if } \text{dist}_T(z) > K_0, \quad (2.49)$$

$$|q_T(z)|h_T(z)^{-1} \leq C \quad \text{for all } z, \quad (2.50)$$

$$|1 - q_T(z)| \leq CL^{1-A} \quad \text{if } \text{dist}_T(z) \leq L^{A/2}. \quad (2.51)$$

In particular multiplication by q_T adds at most $O(L^{1-A})$ times the bandwidths of the top box to a Fourier support.

Proof. For the Fourier convention $e^{ix\xi}$, the transform of $\text{sinc}(x)$ is supported in $[-1, 1]$. Taking its r_L -th power convolves that transform r_L times, so the transform of $\text{sinc}(X/L^A)^{r_L}$ is supported in $[-r_L/L^A, r_L/L^A]$. The product of the two one-dimensional functions has this support bound in each normalized frequency coordinate. The affine coordinate change (2.35) sends these coordinates to $l_T(\xi + s_T\eta)$ and $W\eta$, proving (2.48). The center translation contributes only a Fourier phase. The calculation can equivalently be performed first for compactly supported Fourier distributions and then their convolutions; the resulting mask is integrable for $r_L > 1$.

Write $d = \text{dist}_T(z)$. Since at least one of $|X_T(z)|, |Y_T(z)|$ is at least $d/2$, the elementary inequality $|\text{sinc}(a)| \leq \min(1, |a|^{-1})$ gives, for $d > 0$,

$$|q_T(z)| \leq \min \left(1, \left(\frac{2L^A}{d} \right)^{r_L} \right). \quad (2.52)$$

If $d > K_0 = L^{A+2}$, this is at most $(2L^{-2})^{r_L} \leq \exp(-L \log L)$, proving (2.49). To include arbitrary distances in the weighted bound, write $d = tK_0$, with $t > 1$. Once $r_L \geq D_2$,

$$|q_T(z)|h_T(z)^{-1} \leq 2^{-D_2}(1+t)^{D_2}(2L^{-2})^{r_L}t^{-r_L} \leq (2L^{-2})^{r_L}.$$

For $d \leq K_0$, use $q_T \leq 1$ and $h_T \geq 1$. This proves (2.50), including the region arbitrarily far from the top.

Finally, if $d \leq L^{A/2}$, both arguments of the sinc functions have absolute value at most $L^{-A/2}$. They lie in $[0, 1]$ after evaluation by sinc, and

$$0 \leq 1 - \text{sinc}(X_T/L^A) \text{sinc}(Y_T/L^A) \leq C \frac{X_T^2 + Y_T^2}{L^{2A}} \leq CL^{-A}.$$

The inequality $1 - a^{r_L} \leq r_L(1 - a)$, for $0 \leq a \leq 1$, proves (2.51). Since $r_L/L^A = O(L^{1-A})$, the bandwidth assertion follows from (2.48) and addition of Fourier supports under multiplication. $\square$

For completeness, the spectral enlargement also respects the finer tile coordinates. If $T = T(s)$, an added frequency $(\zeta, \nu) \in \text{supp } \hat{q}_T$ satisfies

$$|\zeta + \theta_s \nu| \leq \frac{r_L}{L^A} \left(\frac{1}{l_T} + \frac{|\theta_s - s_T|}{W} \right) \leq \frac{CL^{1-A}}{\ell_s}, \quad |\nu| \leq \frac{CL^{1-A}}{W}.$$

Here we used $\ell_s \leq l_T$, $|\theta_s - s_T| \leq Cd_s$, and $d_s/W = 1/(m\ell_s)$. Thus all fixed spectral margins needed later remain available after taking L sufficiently large.

3. REFERENCE-LINE CONCENTRATION AND SIGNIFICANT DIRECTIONS

We use the finite packet forms, their assignment to popular tops, and the notation of the preceding sections. In particular, all estimates concern one of the separate forms in the packet expansion. Its label multiplicity and its spatial lattice multiplicity are bounded by absolute constants. The Hilbert space of bin components is denoted by $\mathcal{H}$, and $\|g(z)\|_{\mathcal{H}} \leq \mathbf{1}_E(z) \leq 1$. Every subcollection in this section is fixed before its functions are evaluated at a spatial point.

The first task is to control every interval of scales assigned to a top outside one small exceptional set. We prove moment bounds on each reference line, then obtain the uniform top and derivative bounds in [Theorem 3.6](#). [Theorem 3.7](#) uses these bounds to replace the original adjoint sum P by the masked sum P_q and impose the weighted top-count threshold.

For a slope t , put $Y_t(x, y) = y - tx$. A t -box of dimensions $r \times W$ is a rectangle in the coordinates (x, Y_t) . The affine change to these coordinates preserves area. Fix a reference length l and consider tiles satisfying

$$\ell_s \leq l, \quad |\theta_s - t| \leq C_* B^4 d_s, \quad d_s = w/\ell_s. \quad (3.1)$$

Here and below the fixed constant C_* can be increased finitely many times. This only changes absolute constants in the conclusions. The inequality $B^{100} \ll m$ is more than sufficient for all such choices.

The tiles assigned to one top satisfy the narrower bound $|\theta_s - s_T| \leq C d_s$. The wider B^4 window also covers the fixed packet families used for residual estimates, where the reference slope can be that of a different top; see (5.41) and (5.42).

For a fixed, sufficiently large integer n , write

$$\mathcal{B}_{s,n}^t(z) = \left(1 + \frac{(x - x_s)^2}{\ell_s^2}\right)^{-n} \left(1 + \frac{(Y_t(z) - Y_t(z_s))^2}{W^2}\right)^{-n}. \quad (3.2)$$

The change from the packet coordinates of slope θ_s to these coordinates has normalized shear at most $C_* B^4/m$. Thus packet bounds with any specified fixed number of derivatives and any specified fixed decay order imply the corresponding bounds by $C_n \mathcal{B}_{s,n}^t$. We always choose the available packet decay orders larger than the envelope orders used below. These are fixed integers; none grows with L , the number of scales, or $\dim \mathcal{H}$.

3.1. Positive estimates and the main coefficients. For a reference slope t , decompose the actual coefficient as $a_s = a_s^{\text{main}} + a_s^{\text{err}}$, where componentwise

$$a_{s,j}^{\text{main}} = \frac{1}{|R_s|} \int \mathbf{1}_{\{|u(z)-t| \leq d_s\}} g_j(z) \overline{b_{s,j}(t) \Psi_{s,j}(z)} \, dz. \quad (3.3)$$

This convention agrees with the pairing, linear in its first argument, used to define a_s . In particular the error analyzes $\overline{b_{s,j}(u)} g_j - \mathbf{1}_{\{|u-t| \leq d_s\}} \overline{b_{s,j}(t)} g_j$.

We first record a positive estimate which also applies when only one scale is present. For $r > 0$ let $k_{r,n}(x) = r^{-1}(1 + x^2/r^2)^{-n}$.

Lemma 3.1. *For a fixed scale $\ell = w/d$, and any subcollection of tiles satisfying (3.1) at this scale,*

$$\sum_{\ell_s=\ell} \|a_s\|_{\mathcal{H}} \mathcal{B}_{s,n}^t(z) \leq C_n. \quad (3.4)$$

There are fixed constants $0 < c < 1 < C'$ for which

$$f_d(z) = C \left(\mathbf{1}_{\{cd \leq |u(z)-t| \leq C' B^4 d\}} + \frac{|u(z)-t|}{d} \mathbf{1}_{\{|u(z)-t| \leq cd\}} \right), \quad (3.5)$$

$$\sum_{\ell_s=\ell} \|a_s^{\text{err}}\|_{\mathcal{H}} \mathcal{B}_{s,n}^t(z) \leq C_n (k_{\ell,n'} *_x k_{W,n'} *_Y f_d)(z) \quad (3.6)$$

for a fixed $n' > 2$ with $2n' < n$. The constants are independent of the number of labels permitted in (3.1).

Proof. Minkowski's inequality bounds the norm of an analyzing coefficient by the integral of its scalar analyzing envelope times the largest modulus of its multiplier among the components, divided by ℓW ; the factor $\|g\|_{\mathcal{H}}$ can be replaced by one. At a fixed source point the support

condition for $b_{s,j}(u)$ allows only a bounded number of labels at this scale. This remains true after taking the maximum over components, since the support condition is the common condition on the label slope at precision d .

Choose the available analyzing decay order $N \geq 2n$, with n, n', N all fixed. For one label, the lattice gives the unequal-exponent bound

$$\sum_{s \text{ in that lattice}} \frac{\mathcal{B}_{s,N}^t(z') \mathcal{B}_{s,n}^t(z)}{\ell W} \leq C_{n,N,n'} k_{\ell,n'}(x - x') k_{W,n'}(Y_t(z) - Y_t(z')).$$

Indeed, in each normalized coordinate, $1 + |a - b|^2 \leq 2(1 + |a - c|^2)(1 + |b - c|^2)$ extracts the relative kernel with exponent n' from the two endpoint envelopes. The remaining exponents are $N - n'$ at the source and $n - n'$ at the output. Drop the source factor, which is at most one, and sum the output envelope. Its lattice sum is uniformly bounded since $n - n' > 1/2$: first sum the vertical centers in each horizontal row, then the horizontal rows. The transverse translations of the rows do not affect these one-dimensional bounds. The comparison with the reference coordinates already gives the required analyzing envelope. Summing the bounded number of labels active at the source and integrating proves (3.4).

For the error, if $|u - t| \leq cd$ with c sufficiently small, the replacement indicator is one and smoothness gives $|b_{s,j}(u) - b_{s,j}(t)| \leq C|u - t|/d$. The union of labels on which either multiplier is nonzero is bounded. Outside this region, any nonzero original multiplier requires $|u - t| \leq C'B^4d$ by (3.1), whereas a nonzero replacement requires $|u - t| \leq d$. The multipliers are bounded there. Again, only boundedly many labels are active at the source, counting both families. Applying the same lattice calculation with the common factor (3.5) proves (3.6). $\square$

The smoothness and support assumptions on $b_{s,j}$ also show that nonzero main coefficients use only a bounded number of labels per scale. In such a component, $|t - \theta_s| \asymp d_s$. On a line of slope t , the Fourier frequency of a packet is $\xi + t\eta$, and hence its support is contained in

$$c_a \ell_s^{-1} \leq |\xi + t\eta| \leq C_a \ell_s^{-1} \quad (3.7)$$

with absolute $0 < c_a < C_a$. This uses the separation of the selector from θ_s and the small choice of the packet bandwidth c_* .

Multiplying a synthesizing packet by its top mask preserves (3.7), with slightly relaxed absolute constants. Indeed the extra frequency on a t -line has magnitude at most

$$CL^{1-A} \left(l_{T(s)}^{-1} + \frac{|t - s_{T(s)}|}{W} \right) \leq CL^{1-A} (1 + B^4/m) \ell_s^{-1}. \quad (3.8)$$

Here $\ell_s \leq l_{T(s)}$ and $|t - s_{T(s)}| \leq CB^4 d_s$. For $A \geq 2$ and sufficiently large L , this is less than $c_a/(10\ell_s)$. Moreover, for every fixed derivative order the masked packets have the same uniform envelope bounds. One can see this either from the explicit sinc product or from its compact Fourier support and bounded supremum. In particular

$$|(\partial_x + t\partial_y)q_T| \leq CL^{1-A}(l_T^{-1} + |t - s_T|/W), \quad |\partial_y q_T| \leq CL^{1-A}/W. \quad (3.9)$$

These absolute bounds are global and do not require a good-point assumption.

Lemma 3.2. *For every t -box Q of width W , the main coefficients satisfy*

$$\sum_{\substack{z_s \in Q \\ \ell_s \leq \text{length}(Q)}} \|a_s^{\text{main}}\|_{\mathcal{H}}^2 |R_s| \leq C|Q|. \quad (3.10)$$

The same estimate holds after restricting to any fixed subcollection.

Proof. Let $\mathcal{S}_Q$ be the finite set in the sum and dualize its weighted coefficient norm against vectors c_s satisfying $\sum_{s \in \mathcal{S}_Q} \|c_s\|^2 |R_s| = 1$. Using (3.3), the dual pairing is bounded by the integral of the norm of the synthesis

$$\sum_{s \in \mathcal{S}_Q} \mathbf{1}_{\{|u(z)-t| \leq d_s\}} (c_{s,j} b_{s,j}(t) \Psi_{s,j}(z))_j.$$

At each z , the indicators choose a partial interval of the dyadic lengths: $|u(z) - t| \leq w/\ell_s$ is monotone in ℓ_s . The norm of this synthesis is therefore at most its maximal partial length sum. All its nonzero component frequencies satisfy (3.7).

Write $r = \text{length}(Q) > 0$. Apply the proof of the Hilbert-valued synthesis estimate in Theorem 2.2, including a fixed polynomial growth weight P_Q about Q . This proof applies to the present reference box even if r is not dyadic or $r > \tau$. For the weighted packet bounds it uses only center containment in Q and $\ell_s \leq r$; multiplication by P_Q preserves frequency support and gives tile bounds with slower fixed decay. The annular separation and maximal partial sums depend on the dyadic tile lengths, not on the reference length r . For example, a fixed sufficiently large integer q gives the admissible polynomial

$$P_Q(z) = \left(1 + \frac{(x - x_Q)^2}{r^2}\right)^q \left(1 + \frac{(Y_t(z) - Y_t(z_Q))^2}{W^2}\right)^q, \quad \int P_Q^{-2} \leq C_q r W = C_q |Q|.$$

The main label count is bounded, so the same synthesis proof yields

$$\int \|\text{selected synthesis}\| \leq \left(\int P_Q^{-2}\right)^{1/2} \|P_Q \text{ maximal synthesis}\|_2 \leq C |Q|^{1/2}.$$

Taking the supremum over c_s proves (3.10). Deleting packets preserves both the synthesis hypotheses and this proof. $\square$

3.2. Fourier sums and a dyadic martingale. For a finite sequence (F_k) of functions with values in $\mathcal{H}$, define its interval-sum variation by

$$\mathcal{V}_3(F_k)(x) = \sup_{\mathcal{I}} \left(\sum_{I \in \mathcal{I}} \left\| \sum_{k \in I} F_k(x) \right\|_{\mathcal{H}}^3 \right)^{1/3}, \quad (3.11)$$

where $\mathcal{I}$ ranges over finite families of disjoint integer intervals. Empty intervals contribute zero. This is bounded by the usual 3-variation of partial sums with the initial zero partial sum included: insert the endpoints of the chosen intervals in order and retain the resulting nonnegative terms. This formulation allows the interval endpoints to depend on the evaluation point.

Lemma 3.3. *Suppose $F_k \in L^2(\mathbb{R}; \mathcal{H})$ has Fourier support in $\{\nu : c_a 2^k \leq |\nu| \leq C_a 2^k\}$, with fixed positive c_a, C_a . Then*

$$\|\mathcal{V}_3(F_k)\|_{L^2(\mathbb{R})} \leq C_{c_a, C_a} \left(\sum_k \|F_k\|_2^2 \right)^{1/2}. \quad (3.12)$$

The constant is independent of the finite index range and of $\dim \mathcal{H}$.

Proof. We give the smooth-convolution versus dyadic-expectation comparison explicitly; compare [19, Lemma 3.2 and Section 4]. Split the integers into a fixed number q of residue classes, large enough that the annuli in one class have gaps of ratio greater than four. The variation of the sum is at most the sum of the variations of these classes, so it suffices to treat one class. Its supports are disjoint, and $F = \sum_k F_k$ satisfies $\|F\|_2^2 = \sum_k \|F_k\|_2^2$.

Choose a smooth function χ equal to one on $[-1, 1]$ and supported in $[-2, 2]$, and let P_i have multiplier $\chi(2^{-i}\nu)$. At a sequence of integer indices lying in the annular gaps, $P_i F$ is exactly

the partial sum of all the preceding bands. Include an index below the first band, where this partial sum is zero.

For each integer i , let $\mathcal{F}_i$ be the sigma algebra generated by the intervals $[a2^{-i}, (a+1)2^{-i})$, $a \in \mathbb{Z}$, and let E_i be averaging on its atoms. The sigma algebras increase with i and $(E_i F)_i$ is a martingale. We prove the comparison bound

$$\sum_i \|(P_i - E_i)F\|_2^2 \leq C \|F\|_2^2. \quad (3.13)$$

Take a smooth dyadic partition $F = \sum_j f_j$ with f_j supported in a fixed annulus of size 2^j and $\sum_j \|f_j\|_2^2 \leq C \|F\|_2^2$. If $j \leq i - C$, then $P_i f_j = f_j$, and Poincaré's inequality on each interval of length 2^{-i} , followed by Bernstein's inequality, gives

$$\|(P_i - E_i)f_j\|_2 \leq C2^{j-i} \|f_j\|_2.$$

If $j \geq i + C$, then $P_i f_j = 0$. Define its band primitive A_j by $\hat{A}_j(\nu) = \hat{f}_j(\nu)/(i\nu)$. Thus $A'_j = f_j$ and $\|A_j\|_2 \leq C2^{-j} \|f_j\|_2$. With $h = 2^{-i}$, averaging on a cell gives $E_i f_j = (A_j((a+1)h) - A_j(ah))/h$ there. The sampling estimate

$$\sum_{a \in \mathbb{Z}} \|A_j(ah)\|_{\mathcal{H}}^2 \leq C2^j \|A_j\|_2^2, \quad h \geq c2^{-j}, \quad (3.14)$$

follows directly from smooth reproduction. Namely, take a Schwartz kernel K_j whose Fourier transform equals one on the spectrum of A_j and which satisfies $|K_j(x)| \leq C_N 2^j (1 + 2^j |x|)^{-N}$. Cauchy-Schwarz in $A_j = K_j * A_j$ gives $\|A_j(ah)\|^2 \leq C \int |K_j(ah - x)| \|A_j(x)\|^2 dx$; summing uses $\sup_x \sum_a |K_j(ah - x)| \leq C2^j$. Consequently

$$\|E_i f_j\|_2^2 \leq Ch^{-1} \sum_a \|A_j(ah)\|^2 \leq C2^{i-j} \|f_j\|_2^2.$$

For $|i - j| \leq C$ use the bounded L^2 norms of P_i, E_i . The resulting bounds have the form $\|(P_i - E_i)f_j\|_2 \leq a_{i-j} \|f_j\|_2$ with a fixed $\ell^1(\mathbb{Z})$ sequence a . The triangle inequality followed by the $\ell^1 * \ell^2$ convolution inequality proves (3.13).

For any sequence (D_i) , its usual 3-variation is bounded pointwise by $2(\sum_i \|D_i\|^2)^{1/2}$: each endpoint of an increasing sequence is used at most twice, and ℓ^3 is bounded by ℓ^2 . Thus (3.13) controls the variation of $(P_i - E_i)F$.

Lépingle's martingale variation inequality [24], in the unweighted L^2 case and at variation exponent 3, gives $\|V^3(E_i F)\|_2 \leq C \|F\|_2$; see [29, Theorem 1.1, with $p = 2, r = 3$, and weight one]. For a finite index range on $\mathbb{R}$, apply its probability-space version on every atom of the coarsest sigma algebra, normalizing Lebesgue measure there, and sum the squared inequalities. This proves the same statement on the line without a finite-measure assumption. For Hilbert-valued F , apply the scalar inequality to its real and imaginary components. Minkowski's inequality, in the form $\ell^3(\ell^2) \leq \ell^2(\ell^3)$, and the outer L^2 norm give the same constant independently of the number of components. Combining with the comparison and then taking the selected low-pass indices proves the estimate for one spaced class. Summing the fixed number of classes proves (3.12). $\square$

3.3. Every-line moment bounds. We use the following precise local form of the John-Nirenberg inequality [18]; a proof is included to specify the normalization.

Lemma 3.4. *Let I be a finite interval and let f be a locally integrable scalar function on I . Suppose that for each dyadic subinterval J of I , $|J|^{-1} \int_J |f - f_J| \leq A$, where f_J denotes its*

average. Then, for every $b \geq 1$,

$$\left(\frac{1}{|I|} \int_I |f|^b \right)^{1/b} \leq |f_I| + CbA.$$

Proof. If $A = 0$, dyadic differentiation makes f constant almost everywhere, and the assertion is immediate. Otherwise take the maximal proper dyadic subintervals J of I on which the average of $|f - f_I|$ exceeds $2A$. Their total length is at most $|I|/2$, the function has $|f - f_I| \leq 2A$ almost everywhere off their union, and dyadic parent maximality gives $|f_J - f_I| \leq 4A$ on each selected interval. Repeat the same selection inside each J , now subtracting its own mean. At generation k the selected union has measure at most $2^{-k}|I|$. Outside it, the successive differences of means and the final differentiation bound show $|f - f_I| \leq (4k + 2)A$. This proves an exponential distributional bound for $|f - f_I|/A$. Integrating that bound by layer cake gives $\|f - f_I\|_{L^b(I, dx/|I|)} \leq CbA$; adding the mean proves the lemma. Finite stopping generations and then monotone convergence justify the argument for any locally integrable f . $\square$

Fix a sufficiently large envelope exponent n . At each scale put

$$U_\ell(z) = \sum_{\ell_s=\ell} \|a_s^{\text{main}}\| \mathcal{B}_{s,n}^t(z), \quad \mathcal{E}_t(z) = \sum_s \|a_s^{\text{err}}\| \mathcal{B}_{s,n}^t(z), \quad \mathcal{S}_t(z) = \left(\sum_\ell U_\ell(z)^2 \right)^{1/2}.$$

These quantities can be formed on any fixed subcollection; it is also permissible to dominate a subcollection by the positive quantities on the full collection.

Proposition 3.5. *For every slope t , every collection satisfying (3.1), every line $Y_t = y_0$, every horizontal interval I of length l on that line, and every $b \geq 2$, the normalized $L^b(I)$ norms of $\mathcal{E}_t$, $\mathcal{S}_t$, and*

$$\mathcal{V}_3 \left(\sum_{\ell_s=2^k} \phi_s \right), \quad \mathcal{V}_3 \left(\sum_{\ell_s=2^k} q_{T(s)} \phi_s \right)$$

are at most $Cb^C(1 + \log B)^C$. The constants are uniform in the line, interval, finite collection, dimension, and number of scales. In particular the assertion holds for every fixed deterministic subcollection with the same constants.

Proof. We give the local mean and oscillation bounds, including their behavior on intervals shorter than l . All convolutions in what follows are in the (x, Y_t) coordinates.

First, (3.5) satisfies

$$0 \leq f_d \leq C, \quad \sum_{d \text{ dyadic}} f_d \leq C(1 + \log B). \quad (3.15)$$

The first indicators are nonzero for only $O(1 + \log B)$ dyadic d at a fixed point, and the second terms sum geometrically. Average first with the common vertical kernel $k_{W,n'}$. Its L^1 norm is fixed, so (3.15) persists on each reference line. The right side of (3.6), summed over scales, therefore defines a nonnegative majorant $\mathcal{A}_t$ for $\mathcal{E}_t$.

Let J be any horizontal interval of length $r \leq l$. If $\ell \leq r$,

$$\frac{1}{r} \int_J k_{\ell,n'}(x - x') dx \leq \frac{C}{r} \left(1 + \frac{\text{dist}(x', J)}{r} \right)^{-2}.$$

Hence the local mean of the short-scale part of $\mathcal{A}_t$ is at most $C(1 + \log B)$, by summing the source functions before integration. For $\ell > r$, the derivative of its convolution with a bounded source has supremum at most C/ℓ . The total oscillation of these terms on J is at most $Cr \sum_{\ell > r} \ell^{-1} \leq C$. It follows that $\mathcal{A}_t$ has local BMO norm $O(1 + \log B)$. Its mean on an interval of length l has

the same bound, since all scales then belong to the short-scale part. John–Nirenberg on that interval gives $\|\mathcal{E}_t\|_{L^b(I, dx/l)} \leq Cb(1 + \log B)$. We only use domination by the BMO majorant here; no assertion that an arbitrary function below a BMO function is BMO is required.

For the square function, fixed-scale lattice overlap gives

$$U_\ell(z)^2 \leq C \sum_{\ell_s=\ell} \|a_s^{\text{main}}\|^2 \mathcal{B}_{s,n}^t(z). \quad (3.16)$$

For the same interval J , partition center space into t -boxes Q_{ab} of dimensions $r \times W$, with horizontal offset a from J and transverse offset b from the reference line. For $z_s \in Q_{ab}$ and $\ell_s \leq r$, the envelope bound implies

$$\int_J \mathcal{B}_{s,n}^t(x, y_0 + tx) dx \leq C\ell_s(1 + |a| + |b|)^{-n_1}$$

for a fixed $n_1 > 4$, after choosing n sufficiently large. Indeed the transverse decay gives the b factor; for a distant horizontal box the integral is bounded by $Cr(\ell_s/(r|a|))^{n_1}$, which is at most $C\ell_s(1 + |a|)^{-n_1}$ because $\ell_s \leq r$. Equation (3.10) gives exactly

$$\sum_{\substack{z_s \in Q_{ab} \\ \ell_s \leq r}} \|a_s^{\text{main}}\|^2 \ell_s = W^{-1} \sum_{\substack{z_s \in Q_{ab} \\ \ell_s \leq r}} \|a_s^{\text{main}}\|^2 |R_s| \leq Cr.$$

Summing the offsets in (3.16) proves

$$\frac{1}{r} \int_J \sum_{\ell \leq r} U_\ell^2 dx \leq C. \quad (3.17)$$

This argument applies on each line directly; it is not a restriction of an almost-everywhere plane estimate to a line. Notice also that the factor W has canceled, so the estimate has no dependence on m . Since main coefficients are bounded, fixed-scale lattice bounds give $U_\ell \leq C$ and $|\partial_x U_\ell| \leq C/\ell$ on the line. The long-scale part of $\mathcal{S}_t^2$ has oscillation at most C on J . By (3.17), its local BMO norm is bounded; its mean on every length- l interval is bounded as well. John–Nirenberg applied to $\mathcal{S}_t^2$ gives $\|\mathcal{S}_t\|_{L^b(I, dx/l)} \leq C\sqrt{b}$.

It remains to prove the main-term variation bound, with or without the mask. Fix J and its center boxes Q_{ab} as above, and first restrict to scales $\ell_s \leq r$. Group the synthesis by those boxes. For a fixed box multiply its synthesis by

$$P_{ab}(x) = \left(1 + \frac{(x - x_{Q_{ab}})^2}{r^2}\right)^{q_0},$$

where q_0 is a sufficiently large fixed integer. Because centers lie in Q_{ab} and $\ell_s \leq r$, these multiplied waves still have fixed tile envelope bounds. Their Fourier supports are preserved by polynomial multiplication. Using (3.16) and integrating on the entire line therefore gives

$$\sum_{\ell} \left\| P_{ab} \sum_{\substack{\ell_s=\ell \\ z_s \in Q_{ab}}} a_s^{\text{main}} \Phi_s \right\|_{L^2(\mathbb{R}; \mathcal{H})}^2 \leq Cr(1 + |b|)^{-n_1},$$

and the same bound holds with $q_{T(s)}\Phi_s$ in place of Φ_s . Here and elsewhere $a_s^{\text{main}}\Phi_s$ denotes componentwise multiplication. The bound follows from (3.10) as above, retaining transverse decay while integrating the horizontal envelope. Apply Theorem 3.3 with the length index reversed; this preserves the interval-sum variation. The support hypotheses hold by (3.7) and

(3.8). On J , $P_{ab} \geq c(1 + |a|)^{2q_0}$, and multiplication by this one scalar commutes with every interval sum and with its norm. Thus the $L^2(J)$ variation of the synthesis from Q_{ab} is at most

$$Cr^{1/2}(1 + |a|)^{-2q_0}(1 + |b|)^{-n_1/2}.$$

Minkowski and summation over a, b give a bound $Cr^{1/2}$ for the short-scale main variation.

For each longer scale the derivative of main synthesis on the line is bounded by C/ℓ , also after masking by (3.9). The variation of the long-scale part therefore changes by at most $Cr \sum_{\ell > r} \ell^{-1} \leq C$ on J . The full main variation differs from the long-scale variation by at most the short-scale variation pointwise, by the triangle inequality for (3.11). This proves its bounded local BMO norm. Taking $r = l$ proves its bounded mean on I . John–Nirenberg gives normalized L^b norm at most Cb . The variation of the error synthesis is bounded by the positive sum $C\mathcal{E}_t$, since disjoint intervals use each scale at most once. For the masked error the same is true because $|q_T| \leq 1$. The already proved error bound completes the proposition. $\square$

All arguments in Theorem 3.5 use only fixed support, decay, and derivative bounds on the synthesizing waves. They still hold if those waves are replaced by waves with the same supports and such bounds multiplied by a fixed constant C_0 ; the conclusions are then multiplied by CC_0 . This observation includes fixed polynomial growth weights about a region containing the centers, provided the polynomial is absorbed into a slower fixed tile envelope. It also includes the componentwise operator $W(\partial_y - i\eta_j)$ on the unmasked synthesizing waves. After demodulation its multiplier has bounded magnitude on the vertical packet support. These observations will be used only at fixed orders below.

3.4. Uniform top bounds and neighborhood derivatives. Choose a fixed exponent A_1 so that $K_1 = L^{A_1}$ contains the polynomial dilation of R_T in which all assigned centers lie, with a fixed additional margin. This choice depends on the top assignment and the fixed weight exponent D_2 , and is made before choosing A . Set

$$h_T^{(1)}(z) = C_3(1 + \text{dist}_T(z)/K_1)^{-D_2}.$$

For an interval $\mathcal{J}$ of dyadic lengths define the top sum $F_{T,\mathcal{J}} = \sum_{s:T(s)=T, \ell_s \in \mathcal{J}} \phi_s$. The family of these sums includes empty intervals and is finite for our finite packet collection. Let $\mathcal{D}F = (W(\partial_y - i\eta_j)F_j)_j$.

Corollary 3.6. *For every fixed $J_* > 0$ there are a fixed exponent C_{top} and a measurable set $\mathcal{X}_{\text{top}}$ with $|\mathcal{X}_{\text{top}}| \leq Se^{-J_*L}$ such that, for all tops T and all $z \notin \mathcal{X}_{\text{top}}$,*

$$\sup_{\mathcal{J}} \sup_{|v| \leq W} (\|F_{T,\mathcal{J}}(z + (0, v))\|_{\mathcal{H}} + \|\mathcal{D}F_{T,\mathcal{J}}(z + (0, v))\|_{\mathcal{H}}) \leq L^{C_{\text{top}}}(h_T^{(1)}(z))^8. \quad (3.18)$$

Independently of this exceptional set, for every z , every $l_ > 0$, and every interval $\mathcal{J} \subset [l_*, \infty)$,*

$$\|(\partial_x + s_T \partial_y)F_{T,\mathcal{J}}(z)\|_{\mathcal{H}} \leq \frac{L^{C_{\text{top}}}}{l_*} (h_T^{(1)}(z))^8. \quad (3.19)$$

All packet orders, K_1 , and C_{top} can be fixed before A ; they are independent of the later mass cutoff and hierarchy depth.

Proof. Use Theorem 3.5 on assigned top tiles, with $t = s_T$ and $l = l_T$. These satisfy (3.1) with a fixed angular constant. Write $X = (x - x_T)/l_T$ and $Y = (Y_{s_T}(z) - Y_{s_T}(z_T))/W$. For a sufficiently large fixed integer q , let

$$P_T(z) = \left(1 + \frac{X^2 + Y^2}{K_1^2}\right)^q.$$

Its product with any assigned wave is bounded by a slower fixed tile envelope with uniform constants. Indeed at the centers X, Y are bounded by a fixed fraction of K_1 , and in moving away from a center the top scales are no smaller than the corresponding tile scales. The same statement holds for the fixed derivatives in use. Multiplication by P_T preserves spectrum. Applying the last observation after [Theorem 3.5](#) gives a uniform polynomial moment bound on every length- l_T interval of every parallel line for $\sup_{\mathcal{J}} \|P_T F_{T,\mathcal{J}}\|$ and $\sup_{\mathcal{J}} \|P_T \mathcal{D}F_{T,\mathcal{J}}\|$.

We justify the required vertical neighborhood supremum. Demodulate component j by $e^{-i\eta_j Y_{sT}}$. Each of the just described weighted sums has vertical Fourier support in $[-C/W, C/W]$. Choose one Schwartz reproduction kernel K_W with Fourier transform one on that interval. Its absolute values at all arguments shifted by at most W are bounded by $C_n W^{-1}(1 + |y|/W)^{-n}$. Reproduction, the Hilbert norm inequality, and the supremum over $\mathcal{J}$ show that its vertical neighborhood supremum is bounded by convolution of the corresponding unshifted supremum with this positive integrable kernel. Minkowski in normalized L^b on horizontal intervals preserves the uniform moment estimate. Finally $P_T(z + (0, v)) \asymp P_T(z)$ for $|v| \leq W$, uniformly in z , because $K_1 \geq 1$. We have therefore proved, on every top-scale box, polynomial moments for P_T times the entire left side of [\(3.18\)](#).

For clarity, this yields a global exceptional-area bound, not merely one inside the top box. Partition the plane into boxes of dimensions $l_T \times W$, indexed by $(a, b) \in \mathbb{Z}^2$. On each such box, $1 + \text{dist}_T / K_1 \asymp 1 + (|a| + |b|)/K_1$. Choose q with $2q > 8D_2 + 4$. If the left side of [\(3.18\)](#) exceeds its right side in that box, its product with P_T exceeds

$$cL^{C_{\text{top}}} (1 + (|a| + |b|)/K_1)^{2q-8D_2}.$$

Take moment order $b_L = \lceil L \rceil$. The available moment bound is at most a fixed power of L , since $1 + \log B$ is such a power. Choose C_{top} larger by a fixed amount. Markov's inequality and summation over (a, b) then give

$$|\mathcal{X}_T| \leq C |R_T| K_1^2 e^{-(J_*+1)L} \leq |R_T| e^{-J_*L}$$

for sufficiently large L , after increasing the threshold exponent or the fixed lower bound on L . Equivalently one can first arrange $e^{-(J_*+2)L}$ to absorb the polynomial K_1^2 . Taking the union over tops proves the asserted bound by $S = \sum_T |R_T|$. This argument uses fixed q and fixed packet orders, even though the moment order grows with L .

For [\(3.19\)](#), coefficient norms are bounded, there are boundedly many assigned labels per scale, and each differentiated wave has a bound $C\ell_s^{-1}$ times a fixed tile envelope in slope s_T . At a fixed scale, summing this envelope over assigned centers is at most $C\ell_s^{-1}(1 + \text{dist}_T / K_1)^{-8D_2}$. To see the decay, multiply by the inverse top factor, absorb it into a slower tile envelope as for P_T , and use bounded lattice overlap. Summing over $\ell_s \geq l_*$ costs $\sum_{\ell \geq l_*} \ell^{-1} \leq C/l_*$. The same bound applies to any interval of these lengths by the triangle inequality. This proves the unconditional derivative estimate and completes the corollary. $\square$

3.5. The masked sum and the count threshold. The choice of A in the preceding section can now be completed. Its dependence on the fixed constants just obtained is allowed; their construction did not depend on A .

Proposition 3.7. *Given any fixed $H > 0$, choose A sufficiently large after K_1 and C_{top} , and then choose a sufficiently large fixed count threshold exponent C_n . There is a measurable set $F_{\text{conc}} \subset F$ such that*

$$|F \setminus F_{\text{conc}}| \leq L^{-H} |F|, \quad \int_{F_{\text{conc}}} \|P - P_q\|_{\mathcal{H}} \leq L^{-H} N |F|, \quad P_q = \sum_s q_{T(s)} \phi_s,$$

and (3.18) holds there for every top. On this set,

$$n(z) = \sum_T h_T(z) \leq NL^{C_n}. \quad (3.20)$$

One has $K_0 = L^{A+2} > K_1$, so both (3.18) and (3.19) also hold with h_T in place of $h_T^{(1)}$.

Proof. First fix, for example, $J_* = 10$ in Theorem 3.6, and fix the resulting C_{top} before making any choice of A . Its exceptional area is smaller than any prescribed fixed inverse power of L times $|F|$, since $S/|F| \leq L^C N^2 \leq L^C e^{4L}$. On $\text{dist}_T \leq L^{A/2}$, the elementary estimate for the mask is $|1 - q_T| \leq CL^{1-A}$: use $|1 - \text{sinc } v| \leq Cv^2$ near zero and the integer power $2\lceil L \rceil$ in (2.47). Outside this region, $|1 - q_T| \leq 2$, whereas

$$(h_T^{(1)})^7 \leq CL^{-7D_2(A/2-A_1)}.$$

Equation (3.18), with the full length interval, therefore implies outside $\mathcal{X}_{\text{top}}$ that, for any prescribed fixed H_1 ,

$$\|P - P_q\| \leq L^{-H_1} \sum_T h_T^{(1)} \quad (3.21)$$

once A is chosen large enough. In the near region the factor $L^{C_{\text{top}}+1-A}$ suffices; in the far region use the displayed seventh power to reduce $(h_T^{(1)})^8$ to $h_T^{(1)}$.

Apply (2.37) with dilation K_1 to $n_1 = \sum_T h_T^{(1)}$. For any finite-measure set A_0 in the testing domain, weak L^2 and layer cake imply

$$\int_{A_0} n_1 \leq 2 \|n_1\|_{2,\infty} |A_0|^{1/2} \leq L^C S^{1/2} |A_0|^{1/2}.$$

Since $S \leq L^C |E|$ and $|E| = N^2 |F|$, choosing H_1 larger than H by the fixed powers in these inequalities proves the claimed mask integral even when $A_0 = F \setminus \mathcal{X}_{\text{top}}$.

After A has been fixed, (2.37) with dilation K_0 gives

$$\left| \{z \in F : n(z) > NL^{C_n}\} \right| \leq L^{C-2C_n} S/N^2 \leq L^{C-2C_n} |F|.$$

Take C_n sufficiently large and define F_{conc} by deleting this set and $\mathcal{X}_{\text{top}}$ from F . Increasing the fixed thresholds slightly makes the sum of their areas at most $L^{-H} |F|$. This proves all assertions. $\square$

The original adjoint estimate (2.3) gives $\|P\|_2 \leq L^{C_P} |E|^{1/2}$ for a fixed exponent C_P . Thus, by taking H as large as required, any discarded set above has arbitrarily small prescribed polynomial pairing with the original P . The concentration estimates above hold for finite tile sets and scale ranges, with constants independent of their cardinalities. After the later assembly proves the uniform model bound for each finite form, the estimate passes to the convergent bilinear packet representation as in Theorems 1.14 and 1.15.

3.6. A simultaneous count of separated significant directions. On the set F_{conc} from Theorem 3.7, the weighted threshold (3.20) bounds the total top mass by NL^{C_n} . Because each significant top contributes at least one, this bounds their number by the same quantity, which may grow with N . For the later finite Fourier gain we need a bound polynomial in L for significant directions that are ρ -separated inside an interval of length $B\rho$, with top precisions at most ρ/B^2 . We derive that bound from the popularity witnesses, simultaneously for every $\rho > 0$ and every such list.

We retain the parameters and the finite family of tops from the preceding sections. In particular, every top has width $W = mw$, length $l_T \leq \tau$, and precision $d_T = w/l_T$. Its witness satisfies

$$E_T \subset L^{a_0} R_T, \quad |E_T| \geq L^{-a_0} l_T W, \quad |u - s_T| \leq c_0 d_T \quad \text{on } E_T, \quad (3.22)$$

where a_0 and c_0 are fixed constants; increasing a_0 absorbs the fixed constants in the first two bounds. The dilated box in (3.22) is understood in the coordinates of R_T . The sum of the top areas is $S = \sum_T l_T W$. A top is significant at z when $\text{dist}_T(z) \leq K_0$, with $K_0 = L^{A+2}$ as before.

Proposition 3.8. *There is a fixed exponent C_D with the following property. For every fixed $J > 0$ and all sufficiently large L , there is a measurable set $\mathcal{E}_{\text{count}} \subset \mathbb{R}^2$ such that*

$$|\mathcal{E}_{\text{count}}| \leq S e^{-JL}. \quad (3.23)$$

Define

$$Z_0 = F_{\text{conc}} \setminus \mathcal{E}_{\text{count}}, \quad D = L^{C_D}. \quad (3.24)$$

For every $z \in Z_0$, every $\rho > 0$, and every list $\mathcal{L}$ of tops significant at z satisfying

$$\begin{aligned} \max_{T \in \mathcal{L}} s_T - \min_{T \in \mathcal{L}} s_T &\leq B\rho, \quad d_T \leq \rho/B^2 \quad (T \in \mathcal{L}), \\ |s_T - s_{T'}| &\geq \rho \quad (T, T' \in \mathcal{L}, T \neq T'), \end{aligned} \quad (3.25)$$

one has $\#\mathcal{L} \leq D/100$. The exponent C_D depends only on the fixed geometric constants, including the already chosen mask exponent A . It is independent of the number of tops, the number of scales, and the bin Hilbert space.

The margin in the conclusion permits a fixed number of additional directions and fixed multiplicity factors in later applications, after increasing the absolute lower bound on L . The exceptional set will be constructed from one angular bin family for each reference top. It therefore applies to all ρ and all lists simultaneously.

We transfer the other tops' popularity witnesses to a line in the direction of a longest top. Separated top directions then occupy distinct angular bins, and the following lemma bounds the number of bins each having a large average on some interval through the evaluation point. The reference top contributes one additional direction.

3.7. A counting lemma on a line. The popular-bin counting argument is inspired by Bateman's one-variable covering/BMO method [1, Section 3.3.2, Lemmas 5–6]. The moment estimate below is proved directly; the Lipschitz transfer to reference lines and the simultaneous multiresolution formulation are established here, rather than imported from Bateman's maximal theorem.

Lemma 3.9. *Let $(f_b)_{b \in \mathcal{B}}$ be a countable family of measurable functions on $\mathbb{R}$ such that $0 \leq f_b \leq 1$ and $\sum_b f_b \leq C_0$, where $C_0 \geq 1$. Fix $H > 0$ and $0 < \vartheta \leq 1$. There is a measurable function $\mathcal{N} : \mathbb{R} \rightarrow [0, \infty]$ with the following properties. If $\mathcal{B}_x$ is any collection of distinct indices such that, for every $b \in \mathcal{B}_x$, some interval I_b containing x satisfies*

$$0 < |I_b| \leq H, \quad |I_b|^{-1} \int_{I_b} f_b \geq \vartheta, \quad (3.26)$$

then $\#\mathcal{B}_x \leq \mathcal{N}(x)$. Moreover, for every bounded interval U and every $\lambda \geq 0$,

$$|\{x \in U : \mathcal{N}(x) > \lambda\}| \leq C(|U| + H) \exp(-c\vartheta\lambda/C_0), \quad (3.27)$$

where $C, c > 0$ are absolute. If the f_b depend jointly measurably on an additional parameter, $\mathcal{N}$ can be chosen jointly measurable in that parameter and x .

Proof. For $t \in \{0, 1/3, 2/3\}$ use the dyadic grid whose intervals of length 2^k are

$$\mathcal{D}_k^t = \{2^k[j + (-1)^k t, j + 1 + (-1)^k t) : j \in \mathbb{Z}\}.$$

These grids are nested: a boundary at scale $k + 1$ is a boundary at scale k because $3t$ is an integer. At a fixed scale their combined boundary points have spacing $2^k/3$. Given an interval I of positive finite length, choose the least integer k with $2^k > 3|I|$. Its closure meets boundary points from at most one of the three grids. Hence I is contained in an interval Q from one of the other grids, and $|Q| \leq 6|I|$.

Choose an integer k_H with $6H \leq 2^{k_H} < 12H$, and in each grid retain only scales $k \leq k_H$. Put $\beta = \vartheta/6$. For each b and each grid, call Q bad if

$$|Q|^{-1} \int_Q f_b \geq \beta.$$

Let $\mathcal{J}_b^t$ be the intervals maximal under inclusion among these bad intervals. Every bad interval is contained in a maximal one, since it has only finitely many ancestors up to level k_H . For fixed b, t , the members of $\mathcal{J}_b^t$ are pairwise disjoint. Define

$$\mathcal{N}_t(x) = \sum_{b \in \mathcal{B}} \sum_{Q \in \mathcal{J}_b^t} \mathbf{1}_Q(x), \quad \mathcal{N}(x) = \sum_{t \in \{0, 1/3, 2/3\}} \mathcal{N}_t(x).$$

Each interval in (3.26) produces a bad dyadic interval containing x , at an allowed scale. Thus each distinct index in $\mathcal{B}_x$ contributes at least once to $\mathcal{N}(x)$.

For every dyadic interval P at an allowed scale, disjointness within each bin gives the Carleson estimate

$$\begin{aligned} \sum_b \sum_{\substack{Q \in \mathcal{J}_b^t \\ Q \subseteq P}} |Q| &\leq \beta^{-1} \sum_b \sum_{\substack{Q \in \mathcal{J}_b^t \\ Q \subseteq P}} \int_Q f_b \\ &\leq \beta^{-1} \int_P \sum_b f_b \leq \mathcal{A}|P|, \quad \mathcal{A} = C_0/\beta. \end{aligned} \tag{3.28}$$

All sums here are of nonnegative terms, so this argument also applies to countably many bins and intervals.

For completeness, (3.28) has an elementary exponential-tail consequence. Fix a root interval P_0 of length 2^{k_H} . First take any finite subfamily of the selected interval occurrences inside P_0 , retaining multiplicities when different bins select the same interval, and let V count that subfamily. For an integer $r \geq 1$, expansion of V^r and the nesting of intersecting dyadic intervals give

$$\int_{P_0} V^r \leq r! \sum_{Q_1 \supseteq \dots \supseteq Q_r} |Q_r| \leq r! \mathcal{A}^r |P_0|. \tag{3.29}$$

The chain sum counts occurrences with their multiplicities. To obtain the second inequality, sum first over $Q_r \subseteq Q_{r-1}$ and apply (3.28); repeat until the remaining sum is bounded by $\mathcal{A}|P_0|$. Repeated intervals are allowed throughout. Ordering the r occurrences accounts for the factor $r!$ and can only overcount.

Enumerate all selected occurrences inside P_0 . Their finite initial sums increase to $\mathcal{N}_t$ on P_0 , so monotone convergence and (3.29) show

$$\int_{P_0} \exp(\mathcal{N}_t/(2\mathcal{A})) \leq \sum_{r=0}^{\infty} 2^{-r} |P_0| = 2|P_0|. \tag{3.30}$$

In particular the count is finite almost everywhere. Markov's inequality bounds its superlevel measure in P_0 by $2|P_0| \exp(-\lambda/(2\mathcal{A}))$. The root intervals of one grid meeting U have total

length at most $|U| + 2^{k_H+1}$. Sum the root estimates and observe that $\mathcal{N} > \lambda$ implies $\mathcal{N}_t > \lambda/3$ for at least one grid. Since $\mathcal{A} = 6C_0/\vartheta$ and $2^{k_H} < 12H$, this proves (3.27).

There are countably many bins and dyadic intervals. For each fixed interval its badness is determined by a measurable integral. Its maximality adds only finitely many conditions on its ancestors before level k_H . Consequently the displayed count is measurable. The same argument, using measurability of integrals with a parameter, proves the final assertion. No maximality selection or exceptional set depending on an uncountable family of intervals is required. $\square$

3.8. Transferring witnesses to a reference line. Fix a top R . We use the coordinates

$$z = (x, y_R + s_R(x - x_R) + v).$$

The significant region of R is contained in

$$\mathcal{Q}_R = \{|x - x_R| \leq K_0 l_R, |v| \leq K_0 W\}. \quad (3.31)$$

The coordinate change $(x, v) \mapsto z$ has Jacobian one.

For each integer h , partition each of the two angular annuli

$$[s_R + 2^h, s_R + 2^{h+1}), \quad (s_R - 2^{h+1}, s_R - 2^h]$$

into B^2 intervals of length $\delta_h = 2^h/B^2$. Here B is a dyadic integer, as stipulated in its definition. Denote this countable family of base intervals by $\mathcal{B}_R$. Enlarge every base interval concentrically by a fixed factor c_1 , chosen below, and write b^+ for the resulting closed interval. For sufficiently large L , and hence B , these enlarged intervals satisfy

$$\sum_{b \in \mathcal{B}_R} \mathbf{1}_{b^+}(a) \leq C_{\text{bin}} \quad (a \in \mathbb{R}), \quad (3.32)$$

with a fixed C_{bin} . Indeed, enlargement moves an endpoint by at most $c_1 2^h/B^2$. Once $B^2 \geq 4c_1$, an enlarged interval from annulus h stays at distance comparable to 2^h from s_R . A point therefore meets only a fixed number of annuli, and at most a fixed number, depending on c_1 , of the enlarged equal-length intervals in each annulus.

Choose a fixed power $P_L = L^{a_1}$ sufficiently large to dominate the dilation constants in (3.22), K_0 , and their fixed multiples. It will not depend on R or on a list. Replacing a witness for this argument by a compact subset $\tilde{E}_T \subset E_T$ of at least half its measure is permitted by inner regularity. The slope condition can first be enforced after deleting a null subset if necessary. The horizontal projection A_T of $\tilde{E}_T$ is compact, and Fubini's theorem gives

$$|A_T| \geq \frac{|\tilde{E}_T|}{2L^{a_0}W} \geq \frac{1}{4}L^{-2a_0}l_T. \quad (3.33)$$

The denominator bounds the length of each vertical section of the dilated witness box. The original witnesses themselves remain unchanged elsewhere in the proof.

Suppose now that a list satisfies (3.25) at z , and choose R from the list with maximal length. If the list is empty there is nothing to prove. For $T \neq R$ put $\Delta_T = |s_T - s_R|$. Then

$$\rho \leq \Delta_T \leq B\rho, \quad d_T \leq \Delta_T/B^2, \quad l_T \leq l_R \leq \tau. \quad (3.34)$$

There is an interval I_T containing both A_T and the horizontal coordinate x of z with

$$|I_T| \leq 2P_L l_T \leq 2P_L l_R. \quad (3.35)$$

For example use the interval centered at x_T of radius $P_L l_T$; significance gives $|x - x_T| \leq K_0 l_T$.

For $x' \in A_T$, take any $(x', y') \in \tilde{E}_T$, and let $z' = (x', y + s_R(x' - x))$ be the point with horizontal coordinate x' on the reference line through z . The top-coordinate representations of the witness point and z give

$$|y' - (y + s_R(x' - x))| \leq P_L(W + l_T \Delta_T). \quad (3.36)$$

This bound holds for every possible witness point above x' , so no measurable choice of such points is needed. By the Lipschitz bound on u ,

$$|u(z') - s_T| \leq c_0 d_T + \text{Lip}(u) P_L (W + l_T \Delta_T). \quad (3.37)$$

The parameter choices ensure

$$\text{Lip}(u) P_L m \tau \leq B^{-100}, \quad B^2 \leq m \quad (3.38)$$

for sufficiently large L . To check the first assertion, $\log(P_L m \tau) = -L^{0.97} + O(L^{0.9} + \log L)$, whereas $\log B = O(L^{0.7}(\log L)^2)$; the fixed Lipschitz constant does not affect this comparison. Since $W = m l_T d_T \leq m \tau \Delta_T / B^2$, the last two terms in (3.37) are at most

$$(\text{Lip}(u) P_L m \tau + \text{Lip}(u) P_L \tau B^2) \frac{\Delta_T}{B^2} \leq 2B^{-100} \frac{\Delta_T}{B^2}.$$

Consequently the transferred error is bounded by $(c_0 + 1) \Delta_T / B^2$.

Let $b(T)$ be the unique base interval containing s_T . If its annulus index is h , then $2^h \leq \Delta_T < 2^{h+1}$. Choose c_1 large enough in terms of c_0 so that every point within $2(c_0 + 1)\delta_h$ of any point of a base interval lies in its enlargement. Equations (3.37) and (3.38) now show

$$u(z') \in b(T)^+ \quad (x' \in A_T).$$

Define, on every reference line, the bin functions

$$f_{R,b}(x', v) = \mathbf{1}_{b^+}(u(x', y_R + s_R(x' - x_R) + v)). \quad (3.39)$$

They are jointly measurable, and their sum is bounded by C_{bin} by (3.32). Combining (3.33) and (3.35), there is a fixed exponent a_2 such that, with $\vartheta = L^{-a_2}$,

$$\frac{1}{|I_T|} \int_{I_T} f_{R,b(T)}(x', v) \, dx' \geq \vartheta. \quad (3.40)$$

For example any fixed $a_2 > 2a_0 + a_1$ suffices once L is large. All choices so far precede the eventual choice of D .

The base intervals $b(T)$ for $T \in \mathcal{L} \setminus \{R\}$ are distinct. Indeed, any one such interval has length $2^h / B^2 \leq \Delta_T / B^2 \leq \rho / B < \rho$, so it cannot contain two directions from a ρ -separated list.

3.9. The common exceptional set and the choice of D . Apply Theorem 3.9 to the full family (3.39), with

$$H_R = 2P_L l_R, \quad \vartheta = L^{-a_2}, \quad C_0 = C_{\text{bin}}.$$

Write $\mathcal{N}_R(x, v)$ for its jointly measurable count. This construction depends on R but on neither ρ nor a list. Whenever R is a longest member of an admissible list at $z = (x, y_R + s_R(x - x_R) + v)$, the preceding argument gives

$$\#\mathcal{L} \leq 1 + \mathcal{N}_R(x, v). \quad (3.41)$$

Use the threshold $\Lambda_{\text{count}} = L^{a_2+2}$ and let

$$\mathcal{E}_R = \{z \in \mathcal{Q}_R : \mathcal{N}_R(x, v) > \Lambda_{\text{count}}\}.$$

In (3.27), take $U = [x_R - K_0 l_R, x_R + K_0 l_R]$. Since P_L dominates K_0 , that estimate gives, for each v ,

$$|\{x \in U : \mathcal{N}_R(x, v) > \Lambda_{\text{count}}\}| \leq C P_L l_R e^{-cL^2}.$$

Integrating over $|v| \leq K_0 W$ and using the Jacobian-one coordinates yields

$$|\mathcal{E}_R| \leq C P_L K_0 l_R W e^{-cL^2}. \quad (3.42)$$

In particular, for every fixed $J > 0$, the right-hand side is at most $e^{-JL} l_R W$ for sufficiently large L , because $P_L K_0$ is a fixed power of L .

Set $\mathcal{E}_{\text{count}} = \bigcup_R \mathcal{E}_R$. The top family is finite, and the sets are measurable. Summing (3.42) proves (3.23). The same proof applies to any countable family with finite area sum by countable subadditivity. The angular families were already countable, and their finite approximations were handled in the proof of Theorem 3.9; thus no cardinality or depth factor is introduced by a limiting operation.

Choose

$$C_D = a_2 + 4, \quad D = L^{C_D}. \quad (3.43)$$

At a point outside $\mathcal{E}_{\text{count}}$, a longest member R of any nonempty admissible list is significant, so the point lies in $\mathcal{Q}_R$ and $\mathcal{N}_R \leq \Lambda_{\text{count}}$ there. Equation (3.41) gives $\#\mathcal{L} \leq 1 + L^{a_2+2} \leq D/100$ for sufficiently large L . This proves Theorem 3.8.

Finally, this additional discard is permissible in the original adjoint pairing. The ordinary model estimate gives $\|P\|_2 \leq L^C \sqrt{|E|}$, and $S \leq L^C |E|$. Consequently

$$\int_{F_{\text{conc}} \setminus Z_0} \|P\| \leq L^C |E| e^{-JL/2} \leq L^C e^{(2-J/2)L} N|F|, \quad (3.44)$$

where the last inequality uses $|E| = N^2|F|$ and $N \leq e^{2L}$. Taking, for example, any fixed $J > 4$ makes this smaller than any prescribed inverse power of L times $N|F|$, after increasing the absolute lower bound on L . The estimates (3.18) and (3.20), together with the already established mask replacement, continue to hold on $Z_0 \subset F_{\text{conc}}$.

4. A STRICT FINITE FOURIER INEQUALITY WITH CONSTANT ONE

The next estimate improves the sequence exponent in the usual finite Fourier bound while keeping the operator norm at most one. Keeping this constant prevents a fixed loss from accumulating over the levels of the angular hierarchy. The distinction between a dominant coefficient and a strict norm deficit is related to the study of near-extremizers for discrete Young and Hausdorff–Young inequalities. Relevant precedents include Fournier’s work on sharpness, the fourth-moment case of Eisner and Tao, and the quantitative discrete results of Charalambides and Christ [12, 11, 7]. Here the frequencies are arbitrary separated real numbers and the averaging probability is specified by its Fourier support. We prove the needed quantitative gap in that setting, including the exact constant one after the coefficient exponent is improved.

Throughout this section the Fourier transform of a finite measure is $\hat{\nu}(\omega) = \int e^{-i\omega t} d\nu(t)$.

Fix the absolute constants

$$\epsilon_0 = \frac{1}{4}, \quad \kappa_0 = 2^{-10}, \quad c_{\text{ph}} = 2^{-60}. \quad (4.1)$$

For the bound $D \geq 1$ from the preceding section and for $p \geq 4$, put

$$\alpha = 2 + \frac{c_{\text{ph}}}{\log(2D)}, \quad \mu = \frac{\alpha - 1}{p}, \quad r_* = \frac{1}{1 - \mu}. \quad (4.2)$$

In particular, $2 < \alpha < 7/3$, $0 < \mu < 1/3$, and $1 < r_* \leq 3/2$. The numerical values in (4.1) are convenient fixed choices satisfying the estimates below.

Proposition 4.1 (Finite Fourier gain). *Let $\lambda_1, \dots, \lambda_d \in \mathbb{R}$, where $d \leq D$, satisfy $|\lambda_i - \lambda_j| \geq \Delta > 0$ for $i \neq j$. Let ν be any positive probability measure on $\mathbb{R}$ such that $\text{supp } \hat{\nu} \subset (-\epsilon_0 \Delta, \epsilon_0 \Delta)$. Then, for every finite $p \geq 4$ and every complex coefficient sequence,*

$$\left(\int_{\mathbb{R}} \left| \sum_{i=1}^d a_i e^{i\lambda_i t} \right|^p d\nu(t) \right)^{1/p} \leq \left(\sum_{i=1}^d |a_i|^{r_*} \right)^{1/r_*}. \quad (4.3)$$

The same estimate holds for coefficients in a finite-dimensional Hilbert space $\mathcal{H}$ when the exponentials act diagonally: in an orthonormal basis, the ξ -th component may use frequencies $\lambda_{i,\xi}$,

provided they are Δ -separated as i varies, for each fixed ξ . The right side is then $(\sum_i \|a_i\|_{\mathcal{H}}^{r_*})^{1/r_*}$. All these estimates have constant one.

We first prove a quantitative gap in the fourth moment when no coefficient dominates. The proof compares the law of three indices subject to an approximate additive relation with the independent triple law; separated lower and upper quantiles force a definite difference between these laws. We then handle a dominant coefficient directly, interpolate, and pass to Hilbert-valued coefficients.

Lemma 4.2 (A quantitative gap away from a single coefficient). *For a separated frequency list as above, define*

$$\mathcal{Q} = \{(i, j, k, l) : |\lambda_i + \lambda_j - \lambda_k - \lambda_l| < \epsilon_0 \Delta\}.$$

Let $x_i \geq 0$, $\sum_i x_i = 1$, and put

$$Q(x) = \sum_{(i,j,k,l) \in \mathcal{Q}} (x_i x_j x_k x_l)^{3/4}.$$

Then $Q(x) \leq 1$. If $0 < \kappa < 1$ and $\max_i x_i \leq 1 - \kappa$, then

$$Q(x) \leq 1 - \delta_\kappa, \quad \delta_\kappa = \frac{\kappa^4 (1 - \kappa/2)^2}{4096} \geq 2^{-14} \kappa^4. \quad (4.4)$$

Proof. Any three specified indices have at most one completion to an element of $\mathcal{Q}$: two possible missing frequencies would be within $2\epsilon_0 \Delta < \Delta$ of each other. On the finite set of quadruples, consider the four nonnegative measures

$$A = x_i x_j x_k \mathbf{1}_{\mathcal{Q}}, \quad B = x_i x_k x_l \mathbf{1}_{\mathcal{Q}}, \quad C = x_i x_j x_l \mathbf{1}_{\mathcal{Q}}, \quad E = x_j x_k x_l \mathbf{1}_{\mathcal{Q}}.$$

Each has total mass at most one, by unique completion. Their pointwise geometric mean is the summand defining $Q(x)$, so Hölder's inequality already gives $Q(x) \leq 1$.

For nonnegative measures on a finite set write $H(f, g) = \sum \sqrt{fg}$ for their Hellinger affinity. Cauchy–Schwarz gives

$$Q(x) \leq (H(A, B)H(C, E))^{1/2} \leq H(A, B)^{1/2}. \quad (4.5)$$

Project A to the three coordinates (i, k, l) , obtaining a measure P of mass at most one, and let $P_0(i, k, l) = x_i x_k x_l$ be the independent triple law. For each (i, k, l) there is at most one completing j . Consequently

$$H(P, P_0) = H(A, B) \geq Q(x)^2. \quad (4.6)$$

This identity also holds when a factor x_i , x_k , or x_l vanishes, since the corresponding terms on both sides are zero.

Adjoin one point to the triple space and place the missing mass of P there, giving a probability P^+ . Extend P_0 by zero at that point. For any two probabilities U, V ,

$$\frac{1}{2} \sum |U - V| \leq \frac{1}{2} \left(\sum (\sqrt{U} - \sqrt{V})^2 \right)^{1/2} \left(\sum (\sqrt{U} + \sqrt{V})^2 \right)^{1/2} = \sqrt{1 - H(U, V)^2}.$$

Every event $\mathcal{A}$ in the triple space therefore satisfies

$$P_0(\mathcal{A}) - P(\mathcal{A}) \leq \sqrt{1 - H(P, P_0)^2} \leq \sqrt{1 - Q(x)^4}. \quad (4.7)$$

Thus a fourth moment close to the ordinary bound would force the completed triple law close to the independent law, quantitatively and without a factor depending on the number of frequencies.

Let X be a random frequency with probabilities x_i . Choose lower and upper $\kappa/4$ -quantile frequencies a, b , so that

$$\Pr(X \leq a), \Pr(X \geq b) \geq \kappa/4, \quad \Pr(X < a), \Pr(X > b) \leq \kappa/4.$$

They can be chosen with $a \leq b$. If $a = b$, that atom would have mass at least $1 - \kappa/2 > 1 - \kappa$, a contradiction. Hence $b - a \geq \Delta$, and $[a, b]$ has mass at least $1 - \kappa/2$. Of its two closed halves, choose one, denoted $[e, h]$, with mass at least $(1 - \kappa/2)/2$.

Consider the triple event

$$\mathcal{A} = \{\lambda_i \geq b, \lambda_k \leq a, \lambda_l \leq h\}.$$

For any completion in $\mathcal{Q}$ with these pair restrictions,

$$\lambda_l - \lambda_j > \lambda_i - \lambda_k - \epsilon_0 \Delta \geq b - a - \epsilon_0 \Delta > (b - a)/2 = h - e.$$

Thus $\lambda_l \leq h$ forces $\lambda_j < e$. Under the measure A , the free coordinates (i, j, k) carry their independent probabilities, with a possible deletion when no completion exists. It follows that

$$P(\mathcal{A}) \leq \Pr(X \geq b) \Pr(X \leq a) \Pr(X < e),$$

whereas

$$P_0(\mathcal{A}) = \Pr(X \geq b) \Pr(X \leq a) \Pr(X \leq h).$$

Their difference is at least

$$g_\kappa = (\kappa/4)^2 \frac{1 - \kappa/2}{2} = \frac{\kappa^2(1 - \kappa/2)}{32}.$$

Combining this with (4.7) gives $Q(x)^4 \leq 1 - g_\kappa^2$. Since $(1 - t)^{1/4} \leq 1 - t/4$ for $0 \leq t \leq 1$, $Q(x) \leq 1 - g_\kappa^2/4$. This is (4.4). $\square$

Proof of Theorem 4.1. The ordinary fourth-moment bound. Write $Ta(t) = \sum_i a_i e^{i\lambda_i t}$ and $s = 4/3$. The Fourier support hypothesis and positivity of ν give

$$\int |Ta|^4 d\nu \leq \sum_{\mathcal{Q}} |a_i a_j a_k a_l|.$$

Indeed every other term in the fourth-power expansion averages to zero, and the absolute value of every surviving Fourier coefficient is at most one. If $\|a\|_{\ell^s} = 1$, the last sum is $Q(x)$ for $x_i = |a_i|^s$. Thus Theorem 4.2 proves the ordinary norm-one bound

$$\|Ta\|_{L^4(\nu)} \leq \|a\|_{\ell^{4/3}}. \quad (4.8)$$

For the improved bound at four, put

$$\theta = \frac{c_{\text{ph}}}{\log(2D)}, \quad r_4 = \frac{4}{3 - \theta}.$$

Then $4/3 < r_4 \leq 3/2$ and

$$\|a\|_{\ell^{4/3}} \leq D^{\theta/4} \|a\|_{\ell^{r_4}}, \quad D^\theta \leq e^{c_{\text{ph}}}. \quad (4.9)$$

No dominant coefficient. For a nonzero sequence, normalize $x_i = |a_i|^{4/3} / \sum_k |a_k|^{4/3}$. If $\max_i x_i \leq 1 - \kappa_0$, (4.4) and (4.9) yield

$$\|Ta\|_{L^4(\nu)}^4 \leq (1 - \delta_{\kappa_0}) e^{c_{\text{ph}}} \|a\|_{\ell^{r_4}}^4 \leq \|a\|_{\ell^{r_4}}^4.$$

Here $\delta_{\kappa_0} \geq 2^{-54} > c_{\text{ph}}$, and $1 - \delta \leq e^{-\delta}$.

A dominant coefficient. Suppose instead that one coefficient has normalized mass exceeding $1 - \kappa_0$. Divide all coefficients by that coefficient and remove its carrier from Ta . The normalized sum is $1 + U(t)$, and, if b denotes the ℓ^{r_4} norm of the other coefficients,

$$b \leq \left(\frac{\kappa_0}{1 - \kappa_0} \right)^{3/4} < \frac{1}{64}. \quad (4.10)$$

Frequency separation gives the exact identities

$$\int U \, d\nu = 0, \quad \int |U|^2 \, d\nu = \sum_{i \neq i_0} |a_i/a_{i_0}|^2 \leq b^2.$$

By (4.8) and (4.9), $\int |U|^4 \, d\nu \leq e^{c_{\text{ph}}} b^4$, and hence $\int |U|^3 \, d\nu \leq e^{c_{\text{ph}}/2} b^3$. Expanding $|1 + U|^4$, the linear term has mean zero. Bounding the remaining terms gives

$$\begin{aligned} \int |1 + U|^4 \, d\nu &\leq 1 + 6b^2 + 4e^{c_{\text{ph}}/2} b^3 + e^{c_{\text{ph}}} b^4 \\ &\leq 1 + 14b^2 \leq 1 + \frac{8}{3} b^{r_4} \leq (1 + b^{r_4})^{4/r_4}. \end{aligned}$$

For the penultimate inequality, use $b^{2-r_4} \leq \sqrt{b} \leq 1/8$; for the last one, use $4/r_4 \geq 8/3$ and convexity. Restoring the dominant coefficient proves the norm-one ℓ^{r_4} -to- $L^4(\nu)$ estimate in this case too. The zero sequence is immediate.

Interpolation with constant one. The same operator has norm at most one from ℓ^1 to $L^\infty(\nu)$. For $0 < t \leq 1$, interpolation between these two bounds gives

$$\frac{1}{p_t} = \frac{t}{4}, \quad \frac{1}{r_t} = 1 - t + \frac{t}{r_4}.$$

One can see the normalization directly from the three-lines lemma [17, Lemma 3.3, p. 57]. For $\|a\|_{\ell^{r_t}} = \|g\|_{L^{p'_t}(\nu)} = 1$, initially with g simple, interpolate their nonzero magnitudes by

$$a_i(z) = \text{sgn}(a_i) |a_i|^{r_t(1-z+z/r_4)}, \quad g(z) = \text{sgn}(g) |g|^{p'_t(1-z+3z/4)}.$$

Here sgn denotes complex phase, and the phase of g is chosen for a bilinear dual pairing. The zero values stay zero. On the two vertical boundaries, these functions have respective unit norms in (ℓ^1, L^1) and $(\ell^{r_4}, L^{4/3})$. The analytic function $\int (Ta(z))g(z) \, d\nu$ is thus bounded by one there and at $z = t$. Finite sums and simple functions justify analyticity and boundedness on the strip; density and duality remove the simplicity restriction. Taking $t = 4/p$ gives

$$\frac{1}{r_t} = 1 - \frac{1+\theta}{p} = 1 - \frac{\alpha-1}{p},$$

which is precisely $r_t = r_*$. This proves the scalar form of (4.3) for every finite $p \geq 4$.

Hilbert-valued coefficients. Finally write the Hilbert space as ℓ^2 in an orthonormal basis, and let $F_\xi(t) = \sum_i a_{i,\xi} e^{i\lambda_{i,\xi} t}$. Minkowski's inequality, the scalar result, and Minkowski's inequality again give

$$\begin{aligned} \|F\|_{L^p(\nu; \ell^2)} &\leq \left(\sum_\xi \|F_\xi\|_{L^p(\nu)}^2 \right)^{1/2} \\ &\leq \left(\sum_\xi \left(\sum_i |a_{i,\xi}|^{r_*} \right)^{2/r_*} \right)^{1/2} \\ &\leq \left(\sum_i \left(\sum_\xi |a_{i,\xi}|^2 \right)^{r_*/2} \right)^{1/r_*}. \end{aligned}$$

The first inequality uses $p \geq 2$, and the last uses $r_* \leq 2$. Each has constant one and is independent of the Hilbert dimension. Only the frequency list within an individual component enters the scalar estimate, completing the proof. $\square$

5. ANGULAR HIERARCHY AND PREDICTED FACTORS

We continue with the finite popular forest and the set Z_0 from [Theorem 3.8](#), on which the top estimates, the bound (3.20), and the separated-top count hold. The parameters p, G, K, B are defined in (2.45), and α, μ, r_* in (4.2). We use i for individual angular levels and j for blocks of levels. Each shift chooses different block endpoints. The analytic family constructed below is shared by all K groupings. Its block identities explain how the phase estimate will compare a parent amplitude with its children, while its equal-coordinate value leaves the boundary weights removed in the final angular-grid average. We then prove the estimates that predict amplitudes at a finer angular level, up to a small error, from one good point on a much longer horizontal interval. No mass cutoff is used in those estimates.

All sums below are finite. We retain the strictly positive spatial weights h_T from (2.36); in particular,

$$0 < h_T \leq C_3, \quad \int_{\mathbb{R}^2} h_T \, dz \leq L^C |R_T|. \quad (5.1)$$

The integral follows by the change of variables to the normalized coordinates of R_T , since the decay exponent in that definition is greater than two. Its constant includes the fixed polynomial dilation K_0^2 . Constants denoted by L^C in this section are independent of the number of tiles, tops, levels, and Hilbert-space components.

5.1. Eligible masses and shifted blocks. Choose a dyadic number $d(0)$ such that $B^2 \leq d(0) < 2B^2$, enlarging it by a fixed absolute factor if necessary to cover the initial precision range. For a translation parameter $\vartheta \in [0, d(0))$, let

$$d(i) = d(0)G^{-i}, \quad \mathcal{G}_i^\vartheta = \{[\vartheta + qd(i), \vartheta + (q+1)d(i)) : q \in \mathbb{Z}\}. \quad (5.2)$$

Since G is an integer dyadic power, these grids are nested. We fix ϑ for now. The top slopes lie in a bounded interval, so only an absolute number of intervals in $\mathcal{G}_0^\vartheta$ contain a top slope. These intervals are the roots. The half-open convention assigns each top T a unique interval $v_i(T)$ at every level i .

For $v \in \mathcal{G}_i^\vartheta$ put

$$\mathcal{T}(v) = \{T : s_T \in v, \, d_T \leq d(i)/B^2\}, \quad n_v(z) = \sum_{T \in \mathcal{T}(v)} h_T(z). \quad (5.3)$$

We call these tops eligible at v . A node is nonempty if $\mathcal{T}(v) \neq \emptyset$. For such a node $n_v(z) > 0$ at every spatial point. An empty node has mass and amplitude zero and is omitted from every expression involving a negative power or logarithm of its mass. Thus none of the logarithms below requires a spatial lower cutoff.

If $v_1, \dots, v_G$ are the immediate children of b , then their eligible top sets are disjoint subsets of $\mathcal{T}(b)$. More precisely,

$$n_b - \sum_{v \text{ child of } b} n_v = \sum_{\substack{T \in \mathcal{T}(b) \\ d_T > d(i+1)/B^2}} h_T \geq 0, \quad b \in \mathcal{G}_i^\vartheta. \quad (5.4)$$

The same assertion holds between any two levels, with the corresponding eligibility cutoff. In particular, every positive mass difference used below is itself a sum of the same positive functions h_T .

For a packet s and a top T , define their last eligible levels by

$$k_s = \max\{i \geq 0 : d_s \leq d(i)/B^2\}, \quad k_T = \max\{i \geq 0 : d_T \leq d(i)/B^2\}. \quad (5.5)$$

The initial precision choice makes these sets nonempty. The assigned-top relation $\ell_s \leq \ell_{T(s)}$ gives $d_{T(s)} \leq d_s$, and therefore $k_s \leq k_{T(s)}$. Thus a packet is stopped only while its assigned top

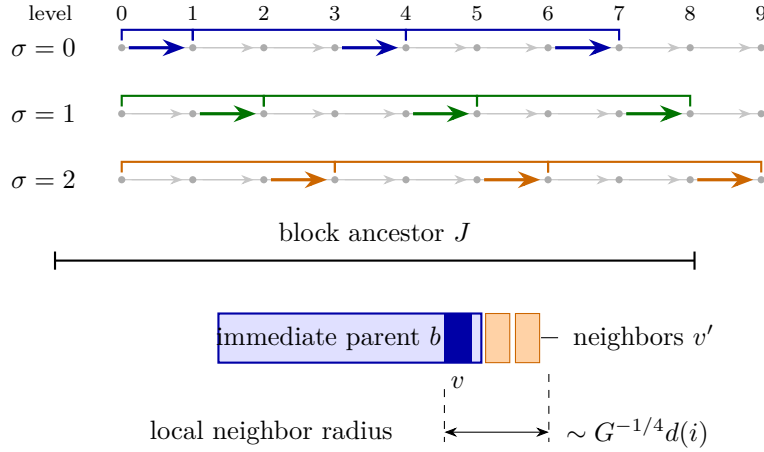


FIGURE 1. Shifted blocks and boundary masses, schematically. The upper diagram uses $K = 3$. Each shift starts with $1 + \sigma$ transitions and then uses blocks of K transitions; colored arrows are the final transitions for that shift. At a final transition $i \rightarrow i + 1$, the mass m_v includes all of n_b and the masses of neighboring children outside b but inside J . The neighbor radius is determined by the immediate scale $d(i)$, even when J begins much earlier. Neither diagram is to scale.

is still eligible. Since the collection is finite and all precisions are positive, the maximum of the k_T is finite. The grids may be defined at every level, but no amplitude requires a level beyond this maximum. We also keep the first block endpoint past this maximum, so that terminal blocks have the same notation as other blocks.

For each $\sigma \in \{0, \dots, K - 1\}$ use the endpoints

$$i_0^\sigma = 0, \quad i_j^\sigma = 1 + \sigma + (j - 1)K \quad (j \geq 1). \quad (5.6)$$

The first block has between one and K transitions; all subsequent blocks have K . The transition $i \rightarrow i + 1$ is the final transition of a block for precisely the shift $\sigma \equiv i \pmod{K}$. For this shift, let $a(i)$ be the starting level of that block. Given an immediate child $v \in \mathcal{G}_{i+1}^\vartheta$ with immediate parent $b \in \mathcal{G}_i^\vartheta$, denote its ancestor at level $a(i)$ by $J(i, v)$. Define

$$m_v = n_b + \sum_{\substack{v' \in \mathcal{G}_{i+1}^\vartheta, \ v' \subset J(i, v), \ v' \not\subset b \\ \text{dist}(v, v') < G^{-1/4}d(i)}} n_{v'}. \quad (5.7)$$

Here the distance is the distance between the angular intervals. The summands outside b have disjoint top sets, and these sets are disjoint from $\mathcal{T}(b)$. They all consist of tops eligible at $J(i, v)$. Consequently,

$$n_v \leq n_b \leq m_v \leq n_{J(i, v)}. \quad (5.8)$$

In particular, $m_v - n_v$ is a sum of positive top weights. The use of the larger block ancestor in (5.7) never changes the radius of the neighbor test: that radius is always tied to the immediate transition scale $d(i)$. Figure 1 shows the shifted block endpoints and the top sets entering m_v .

For $\zeta = (\zeta_0, \dots, \zeta_{K-1}) \in \mathbb{C}^K$ and nonempty v at level $i + 1$, put

$$w_v(\zeta, z) = \exp\left(\zeta_{i \bmod K} \log \frac{n_v(z)}{m_v(z)}\right). \quad (5.9)$$

All logarithms in this formula are real logarithms of positive numbers. Thus the weights are entire functions of ζ and have modulus at most one when all $\Re \zeta_\sigma \geq 0$.

5.2. The analytic family and its block comparisons. For a nonempty node v at level h , define the $\mathcal{H}$ -valued analytic amplitude by

$$A_v(\zeta, z) = \sum_{\substack{s: sT(s) \in v \\ k_s \geq h}} q_{T(s)}(z) \phi_s(z) n_{v_{k_s}(T(s))}(z)^{-\mu/K} \prod_{i=h}^{k_s-1} w_{v_{i+1}(T(s))}(\zeta, z). \quad (5.10)$$

The condition $k_s \geq h$ already ensures eligibility of $T(s)$ at v . The empty product is one. This single definition is used for every shift. At fixed z , it is a finite sum of entire $\mathcal{H}$ -valued functions of ζ .

We first exhibit two algebraic uses of this common family. For the block comparison, temporarily fix a shift σ and parameters satisfying

$$\Re \zeta_\sigma = \mu, \quad \Re \zeta_{\sigma'} = 0 \quad (\sigma' \neq \sigma), \quad \|\Im \zeta\|_2 \leq L^2. \quad (5.11)$$

We refer to this choice of real parts as the vertex corresponding to σ . Write $i_j = i_j^\sigma$ and $\rho_j = d(i_j)$. Thus

$$G \leq \frac{\rho_j}{\rho_{j+1}} \leq B, \quad (5.12)$$

including the first, possibly shorter, block.

For a nonempty $J \in \mathcal{G}_{i_j}^\theta$, call its nonempty descendants at level i_{j+1} its *block children*. For such a block child c , all packets entering A_c have the same transition factor from J to c . Denote it by $\lambda_c = \prod_{i=i_j}^{i_{j+1}-1} w_{v_{i+1}}$, along that descendant path. Only the last transition of this block has a nonzero real exponent. Separating packets according to whether their stopping level precedes i_{j+1} therefore gives the exact identity

$$A_J = \sum_{\substack{c \subset J \\ c \in \mathcal{G}_{i_{j+1}}^\theta}} \lambda_c A_c + B_J, \quad |\lambda_c| = \left(\frac{n_c}{m_c} \right)^\mu. \quad (5.13)$$

The remainder B_J is precisely the sum in (5.10) over $i_j \leq k_s < i_{j+1}$. Its assigned tops need not lose eligibility at the end of the block. Thus residual packets and lost top mass are different parts of the block decomposition.

For each nonempty J , we add the mass to the normalized p -th power of the amplitude. The resulting positive amount V_J and its multiplicative comparison factor β_J are

$$V_J = n_J + n_J^{\alpha-p} \|A_J\|_{\mathcal{H}}^p, \quad \beta_J = \frac{V_J}{n_J + \sum_c n_c^{\alpha-p} \|A_c\|_{\mathcal{H}}^p}. \quad (5.14)$$

We use β_J^0 for the same ratio with B_J omitted only in the numerator. The mass baseline makes V_J and every denominator positive, including when all its block-child amplitudes vanish. The sum is over the nonempty descendants at the end of the block. Equivalently, the denominator is the lost eligibility mass $n_J - \sum_c n_c$ plus the block-child amounts $\sum_c V_c$. This is the comparison later used in the exact path expansion.

At any one spatial point, abbreviate $n = n_J$ and normalize the child amplitudes by

$$l_c = \frac{n_c}{n}, \quad X_c = n_c^{\mu-1} A_c, \quad \beta_J^0 = \frac{1 + \|\sum_c l_c^{1-\mu} \lambda_c X_c\|_{\mathcal{H}}^p}{1 + \sum_c l_c \|X_c\|_{\mathcal{H}}^p}. \quad (5.15)$$

The ratio follows by dividing (5.14) by n and using $p\mu = \alpha - 1$. Also $\sum_c l_c \leq 1$. This is a pointwise identity: the masses, scalar factors, and vectors still vary with the spatial point. The

vector sum in its numerator is the weighted child sum whose phase motion we will compare with the weighted child energies in the denominator.

The boundary mass m_c is designed to pay for selecting separated children in that comparison. At the final transition of this block put $i = i_{j+1} - 1$ and

$$Q = G^{-1/4}d(i) = G^{3/4}\rho_{j+1}. \quad (5.16)$$

Fix one spatial point and choose any finite subfamily $\mathcal{C}$ of nonempty block children. Join distinct $c, c' \in \mathcal{C}$ when $\text{dist}(c, c') < Q$. Give the children independent exponential clocks of rates n_c , and retain a child when its clock rings before those of all its neighbors. The rates are positive. The clocks have no ties with probability one, and the retained intervals have pairwise distance at least Q . The retention probability of c is

$$\pi_c = \int_0^\infty n_c e^{-n_c t} \prod_{c' \sim c} e^{-n_{c'} t} dt = \frac{n_c}{n_c + \sum_{c' \sim c} n_{c'}} \geq \frac{n_c}{m_c}. \quad (5.17)$$

To see the inequality, let b be the immediate parent of c . The eligible top sets of c and its competitors inside b are disjoint subsets of $\mathcal{T}(b)$, so their masses sum to at most n_b . Every competitor outside b lies inside J and within $Q = G^{-1/4}d(i)$ of c , so its mass is included among the additional terms of m_c in (5.7). Thus m_c bounds the entire competing mass even when only a subfamily of children is used.

Multiplying the coefficient of a retained child by π_c^{-1} reconstructs its original coefficient in expectation. The corresponding expected r_* -power cost is

$$\mathbb{E}[\mathbf{1}_{\{c \text{ retained}\}} \pi_c^{-1}] = 1, \quad \mathbb{E}[\mathbf{1}_{\{c \text{ retained}\}} \pi_c^{-r_*}] = \pi_c^{1-r_*}.$$

The coefficient exponent in (5.13) matches this cost exactly. The identities $r_*(1 - \mu) = 1$ and $\mu r_* = r_* - 1$ give

$$\pi_c^{1-r_*} |\lambda_c|^{r_*} = \left(\frac{n_c/m_c}{\pi_c} \right)^{r_*-1} \leq 1. \quad (5.18)$$

The first identity also turns $l_c^{r_*(1-\mu)}$ into l_c , the weight in the denominator of (5.15). The later snapshot construction will use this selection for the child vectors it retains and check the remaining hypotheses of the finite Fourier estimate.

The same analytic family has a second algebraic use at the equal-coordinate parameter

$$\zeta^c = (\mu/K, \dots, \mu/K).$$

Let v be a root and let s be a packet contributing to A_v , with $T = T(s)$. Apart from the factor $q_T \phi_s$, the coefficient of s in $n_v^{\mu/K} A_v(\zeta^c, z)$ is

$$n_v^{\mu/K} n_{v_{k_s}(T)}^{-\mu/K} \prod_{i=0}^{k_s-1} \left(\frac{n_{v_{i+1}(T)}}{m_{v_{i+1}(T)}} \right)^{\mu/K}.$$

Cancelling consecutive eligible masses identifies it as

$$\kappa_s = \prod_{i=0}^{k_s-1} \left(\frac{n_{v_i(T)}}{m_{v_{i+1}(T)}} \right)^{\mu/K}, \quad T = T(s). \quad (5.19)$$

The empty product is one, and each factor lies in $(0, 1]$ by (5.7). Thus for each root the equal-coordinate value produces its portion of the masked packet sum with only these boundary ratios in its coefficients. The remaining tasks are to estimate the vertex values, interpolate those estimates to ζ^c , and remove the boundary ratios by averaging the angular-grid translation.

5.3. Full-root variation and amplitude bounds. We now leave the temporary vertex specialization and return to the general range $0 \leq \Re \zeta_\sigma \leq \mu$ for every coordinate, with arbitrary imaginary parts. The following bounds control the endpoint weights without a loss proportional to the number of levels. Along one eligible top path, write $n_i = n_{v_i(T)}$ and

$$P_i = \log \frac{n_i}{n_{i+1}}, \quad R_i = \log \frac{m_{v_{i+1}(T)}}{n_{i+1}}. \quad (5.20)$$

Both are nonnegative as long as the top remains eligible at level $i + 1$.

Lemma 5.1 (Full-root variation bound). *Fix an eligible path from a root $v_0(T)$ through level k . At every spatial point,*

$$\sum_{i=0}^{k-1} R_i \leq K \log \frac{n_{v_0(T)}}{n_{v_k(T)}}, \quad \sum_{i=0}^{k-1} P_i = \log \frac{n_{v_0(T)}}{n_{v_k(T)}}. \quad (5.21)$$

For a fixed starting level $h \leq k_T$, define, for $h \leq k \leq k_T$,

$$\omega_{T,h,k}(\zeta, z) = n_{v_k(T)}(z)^{-\mu/K} \prod_{i=h}^{k-1} w_{v_{i+1}(T)}(\zeta, z).$$

If $0 \leq \Re \zeta_\sigma \leq \mu$ for every σ , then the supremum plus total variation of this sequence, with the endpoint k varying, is bounded by

$$\sup_{h \leq k \leq k_T} |\omega_{T,h,k}| + \sum_{k=h}^{k_T-1} |\omega_{T,h,k+1} - \omega_{T,h,k}| \leq Ch_T^{-\mu/K} \left[1 + K(1 + \|\zeta\|_2) \log \frac{n_{v_0(T)}}{h_T} \right]. \quad (5.22)$$

The same bound holds upon restricting the endpoints to a subinterval.

Proof. Fix a shift σ . For the transitions $i \equiv \sigma \pmod{K}$ with $i < k$, (5.8) gives

$$R_i \leq \log \frac{n_{a(i)}}{n_{i+1}}.$$

For this fixed shift, the intervals of levels $[a(i), i+1]$ are successive blocks starting at zero. Their logarithms telescope. The endpoint of the last completed block is at most k , so the sum is at most $\log(n_0/n_k)$, by monotonicity of the masses. Summing over the K shifts proves the first inequality in (5.21). The second is an ordinary telescope. This is a full-root bound. In particular, it remains applicable to weights occurring on a later segment even when another shift's block starts before that segment.

Set $a = \mu/K$, solely for this proof, and use the specified logarithm

$$L_k = -a \log n_k - \sum_{i=h}^{k-1} \zeta_{i \bmod K} R_i, \quad \omega_{T,h,k} = e^{L_k}.$$

Its increment is

$$L_{k+1} - L_k = aP_k - \zeta_{k \bmod K} R_k. \quad (5.23)$$

Since $n_k \geq h_T$ and all real transition exponents are nonnegative, $|e^{L_k}| \leq h_T^{-a}$. For any complex A, B ,

$$|e^B - e^A| \leq |B - A| \int_0^1 |e^{(1-t)A + tB}| dt \leq |B - A| \max(|e^A|, |e^B|).$$

Using this inequality for adjacent logarithms and then (5.21) proves (5.22). Notice that the exponential of the sum of the absolute logarithmic increments has not been used. Such an exponential would be unsuitable when h_T is very small. $\square$

Lemma 5.2 (Amplitude bounds). *On Z_0 , for $0 \leq \Re \zeta_\sigma \leq \mu$,*

$$\|A_v(\zeta, z)\|_{\mathcal{H}} \leq L^C(1 + \|\zeta\|_2)n_v(z). \quad (5.24)$$

More precisely, the contribution of any one eligible top T to A_v , also when restricted to any interval of packet lengths, is bounded by

$$L^C(1 + \|\zeta\|_2)|q_T(z)|h_T(z)^4. \quad (5.25)$$

If all eligible tops of v are insignificant at $z \in Z_0$, then

$$\|A_v(\zeta, z)\|_{\mathcal{H}} \leq L^C(1 + \|\zeta\|_2)e^{-L \log L}n_v(z). \quad (5.26)$$

Proof. For vectors $f_h, \dots, f_k$ and scalars $\omega_h, \dots, \omega_k$, summation by parts gives

$$\left\| \sum_{i=h}^k \omega_i f_i \right\| \leq \left(|\omega_k| + \sum_{i=h}^{k-1} |\omega_{i+1} - \omega_i| \right) \max_{h \leq j \leq k} \left\| \sum_{i=h}^j f_i \right\|. \quad (5.27)$$

Group the packets assigned to T according to their stopping level. The resulting initial sums, or the sums obtained after restricting to an interval of packet lengths, are interval partial sums of the original top's length sequence. By (3.18) and the fact that the K_0 -dilated weight is at least the corresponding K_1 -dilated weight, their norm at $z \in Z_0$ is at most $L^C h_T(z)^8$. The mask $q_T(z)$ is common to this whole top. Apply (5.27) and (5.22).

To see that the logarithmic factor can be absorbed uniformly, use (3.20) and $\log N \leq 2L$. Since $n_{v_0(T)} \leq n$ and $h_T \leq C_3$,

$$0 \leq \log \frac{n_{v_0(T)}}{h_T} \leq CL + \log^+ \frac{1}{h_T}.$$

Here and below $\log^+ x = \max(0, \log x)$. Also $\mu/K < 1$. The elementary boundedness of $x^\delta \log^q(1/x)$ on $(0, 1]$, for fixed $\delta > 0$ and fixed q , shows that

$$h_T^{8-\mu/K} \left[1 + K(1 + \|\zeta\|_2) \log \frac{n_{v_0(T)}}{h_T} \right] \leq L^C(1 + \|\zeta\|_2)h_T^4.$$

This proves (5.25). Summing over the eligible tops and using $h_T^4 \leq C_3^3 h_T$ proves (5.24). For insignificant tops, the mask estimate after (2.47) gives $|q_T| \leq e^{-L \log L}$. Applying the same summation proves (5.26). $\square$

5.4. Spatial meshes for prediction. Return to a fixed shift σ and a vertex parameter satisfying (5.11). Write $i_j = i_j^\sigma$ and $\rho_j = d(i_j)$ as in the block comparison above, and take $J \in \mathcal{G}_{i_j}^\vartheta$. Its block children and their factors are those in (5.13). Let R be any top eligible in J , and parametrize a line of slope s_R by $z(x) = (x, y_* + s_R x)$. On this line use one nested dyadic grid in the parameter x , at lengths

$$H_j = \frac{w}{\rho_{j+1}\sqrt{G}}, \quad H_{-1} = \frac{w}{\rho_0\sqrt{G}}. \quad (5.28)$$

These are dyadic lengths because $w, d(0), G$, and $\sqrt{G}$ are dyadic. They increase with j . The grids can, for example, have common origin zero. Each H_j cell is a union of H_{j-1} cells.

5.5. Positive-mass motion and the locality of other shifts.

Lemma 5.3 (Mass motion). *Fix R, J, j as above and an interval I of horizontal length H_j on a line of slope s_R . Consider the masses n_J, n_c, m_c in (5.13), the masses used in the factors λ_c , and those used in any endpoint multiplier of a packet contributing to A_c . Each such mass, and each of its positive subset-complement differences used below, is a sum of weights of tops T satisfying*

$$|s_T - s_R| \leq C\rho_j, \quad d_T \leq \rho_j/B^2. \quad (5.29)$$

There is an absolute constant C such that, with $\varepsilon_{\text{mov}} = CG^{-1/2}$, every nonzero such mass a satisfies

$$|\partial_x \log a| \leq \frac{\varepsilon_{\text{mov}}}{H_j} \quad \text{almost everywhere on } I. \quad (5.30)$$

If $r = a/(a+b)$ is one of the ratios of a submass to its containing mass, then

$$|\partial_x \log r| \leq \frac{2\varepsilon_{\text{mov}}}{H_j}(1-r) \leq \frac{2\varepsilon_{\text{mov}}}{H_j}(-\log r). \quad (5.31)$$

When $b = 0$, the ratio is identically one and its derivative is zero. In particular, nonzero $n_J - n_c$ and $m_c - n_c$ have the relative motion bound (5.30). All these masses vary by a factor $1 + O(G^{-1/2})$ across I .

For the parameters (5.11), the block factors also satisfy, for $x, x_0 \in I$,

$$\left| \frac{\lambda_c(x)}{\lambda_c(x_0)} - 1 \right| \leq G^{-c_0} \quad (5.32)$$

for some absolute $c_0 > 0$ and all sufficiently large L .

Proof. All descendants of J have slopes inside J , so their top slopes are within ρ_j of s_R . Their eligibility cutoffs are at most ρ_j/B^2 . A mass in an additional boundary term (5.7) at a transition $i \geq i_j$ can come from an interval outside J , because the block associated to another shift can start before i_j . However, that interval is within $G^{-1/4}d(i)$ of the relevant child and has width $d(i+1)$. The relevant child is a descendant of J , and $d(i) \leq \rho_j$. Hence every added slope is within $C\rho_j$ of s_R . The tops in an added interval have precision at most $d(i+1)/B^2$; those in the immediate-parent mass have precision at most $d(i)/B^2$. This proves (5.29), including for weights below the block. Taking a difference of the relevant nested top sets preserves the same property. This argument uses the neighbor radius at the immediate transition, rather than the possibly larger width of the other shift's block ancestor.

In normalized coordinates of a top T , horizontal motion along s_R has velocities $1/l_T$ and $(s_R - s_T)/W$. Differentiating (2.36), using the Lipschitz property of the absolute value at almost every point, gives

$$|\partial_x \log h_T| \leq C \left(\frac{1}{l_T} + \frac{|s_R - s_T|}{W} \right) \leq C \left(\frac{\rho_j}{B^2 w} + \frac{\rho_j}{mw} \right).$$

The factor K_0^{-1} available from the dilation is not needed. Multiplying by H_j and using (5.12) yields

$$H_j |\partial_x \log h_T| \leq \frac{C}{\sqrt{G}} \left(\frac{B}{B^2} + \frac{B}{m} \right) \leq CG^{-1/2}.$$

For a positive sum $a = \sum_T h_T$, its logarithmic derivative is the weighted average of the logarithmic derivatives of the summands. This proves (5.30), also for positive differences, which are such sums by their set interpretation.

If $a, b > 0$, direct differentiation gives

$$\partial_x \log \frac{a}{a+b} = \frac{b}{a+b} (\partial_x \log a - \partial_x \log b).$$

This proves (5.31); the inequality $1 - r \leq -\log r$ holds for $0 < r \leq 1$. If the complementary top set is empty, $r = 1$ everywhere. In particular, writing $q_r = -\log r \geq 0$,

$$|\partial_x q_r| \leq \frac{2\varepsilon_{\text{mov}}}{H_j} q_r, \quad e^{-2\varepsilon_{\text{mov}}} q_r(x_0) \leq q_r(x) \leq e^{2\varepsilon_{\text{mov}}} q_r(x_0) \quad (x, x_0 \in I). \quad (5.33)$$

The inequalities remain valid for an identically zero loss.

Finally, a block factor is a product of at most K transition factors. The simpler bound $|\partial_x \log r| \leq 2\varepsilon_{\text{mov}}/H_j$ shows that its specified complex logarithm changes by at most $CK(1 + \|\zeta\|_2)G^{-1/2}$. For (5.11) this is $L^C G^{-1/2}$ and tends to zero faster than any negative fixed power of L . Exponentiating proves (5.32) after decreasing c_0 . This last argument does not require any bound on the size of a logarithmic loss itself. $\square$

5.6. Variation of moving endpoint weights. The next estimate makes explicit why the motion of the analytic weights can be controlled from a single point, even for very small h_T . For a finite scalar sequence $a = (a_k)$ let

$$\|a\|_{\text{BV}} = \sup_k |a_k| + \sum_k |a_{k+1} - a_k|.$$

Only this elementary finite-sequence norm is used here.

Lemma 5.4 (Moving weights). *In the setting of Theorem 5.3, fix a top T contributing to A_c , and take endpoint weights $\omega_k = \omega_{T, i_{j+1}, k}$. If $z_0 \in I \cap Z_0$ and the parameters satisfy (5.11), then, uniformly for $z' \in I$,*

$$\|(\omega_k(z'))_k\|_{\text{BV}} \leq L^C h_T(z_0)^{-4}, \quad (5.34)$$

$$\|(\omega_k(z') - \omega_k(z_0))_k\|_{\text{BV}} \leq G^{-1/2} L^C h_T(z_0)^{-4}. \quad (5.35)$$

Both conclusions hold for any interval of endpoints. Neither assumes that z' belongs to Z_0 .

Proof. Parametrize the interval by its horizontal coordinate x , and write $h = i_{j+1}$ and $a = \mu/K$ in this proof. Use P_i, R_i from (5.20) for $h \leq i < k_T$. By (5.33), each of these nonnegative losses is comparable throughout I to its value at x_0 , with factor $e^{O(\varepsilon_{\text{mov}})}$, and its absolute derivative is bounded by $C\varepsilon_{\text{mov}}/H_j$ times itself. Define

$$\mathcal{B}(x) = \sum_{i=h}^{k_T-1} (aP_i(x) + |\zeta_{i \bmod K}| R_i(x)).$$

It follows that $\mathcal{B}(x) \leq e^{C\varepsilon_{\text{mov}}} \mathcal{B}(x_0)$. By the full-root bound,

$$\mathcal{B}(x_0) \leq CK(1 + \|\zeta\|_2) \log \frac{n_{v_0(T)}(z_0)}{h_T(z_0)}. \quad (5.36)$$

The full-root mass is used only at the snapshot in this inequality; its spatial derivative is not needed.

Let $L_k = \log \omega_k$ be the specified logarithm in the proof of Theorem 5.1, and put $q_k = \partial_x L_k$. Equation (5.23) and the mass derivative bound imply

$$\begin{aligned} \sup_k |q_k| &\leq \frac{C\varepsilon_{\text{mov}}}{H_j} (a + \mathcal{B}(x)), \\ \sum_k |q_{k+1} - q_k| &\leq \frac{C\varepsilon_{\text{mov}}}{H_j} \mathcal{B}(x). \end{aligned}$$

For example, the initial term is $q_h = -a\partial_x \log n_h$, and subsequent differences are the derivatives of $aP_i - \zeta_{i \bmod K} R_i$. At every point of I ,

$$\sup_k |\omega_k(x)| \leq h_T(x)^{-a} \leq e^{a\varepsilon_{\text{mov}}} h_T(z_0)^{-a}.$$

The complex-exponential estimate used for (5.22) therefore gives

$$\|\omega(x)\|_{\text{BV}} \leq C h_T(z_0)^{-a} (1 + \mathcal{B}(x_0)).$$

For finite sequences the product rule for differences gives

$$\|ab\|_{\text{BV}} \leq \sup_k |a_k| \|b\|_{\text{BV}} + \sup_k |b_k| \sum_k |a_{k+1} - a_k|.$$

Using $\partial_x \omega_k = \omega_k q_k$ and the preceding estimates yields

$$\|\partial_x \omega(x)\|_{\text{BV}} \leq \frac{C\varepsilon_{\text{mov}}}{H_j} h_T(z_0)^{-a} (1 + \mathcal{B}(x_0))^2. \quad (5.37)$$

The functions involved are absolutely continuous on the line. Integrate (5.37) between x_0 and x' , over distance at most H_j , and use the triangle inequality for the finite-sequence norm. This proves the difference bound with right side $C\varepsilon_{\text{mov}} h_T(z_0)^{-a} (1 + \mathcal{B}(x_0))^2$.

At the snapshot, (3.20), (5.11), and (5.36) bound the last expression by $G^{-1/2} L^C h_T(z_0)^{-4}$. Indeed the only possible nonpolynomial quantity is $\log^+(1/h_T(z_0))$, whose square is absorbed by the difference between exponents $a < 1$ and 4. The same argument proves (5.34). There is no assumption that $G^{-1/2} \mathcal{B}(x_0)$ is small: very small h_T is allowed, and only a polynomial in its logarithm has appeared. $\square$

5.7. Amplitude motion from one good snapshot.

Lemma 5.5 (Predictable amplitude motion). *Fix a block child c of J , a path top R eligible in J , and an H_j cell on a line of slope s_R . Let t_c be the angular center of c . If $z_0 \in Z_0$ lies in that cell and $z' = z_0 + (b, s_R b)$ is any other point in the cell, then, for the parameters (5.11),*

$$\|A_c(\zeta, z') - \mathcal{D}_c(b) A_c(\zeta, z_0)\|_{\mathcal{H}} \leq G^{-c_0} n_c(z_0), \quad \mathcal{D}_c(b)_{\nu\nu} = \exp(i(s_R - t_c)\eta_\nu b). \quad (5.38)$$

Here $c_0 > 0$ is absolute and may also be used, after decreasing it, in (5.32). The estimates hold at all displaced points in the cell, not just its points in Z_0 .

Proof. Fix one top T contributing to A_c . Its contributing packets have $k_s \geq i_{j+1}$, hence

$$\ell_s \geq l_* := \frac{B^2 w}{\rho_{j+1}}, \quad l_T \geq l_*. \quad (5.39)$$

Also $s_T \in c$, so $|s_T - t_c| \leq \rho_{j+1}/2$, while $|s_T - s_R| \leq C\rho_j$.

Consider any interval partial sum S of these unweighted packets on T . The good-snapshot estimates (3.18), including their demodulated vertical-derivative and vertical-neighborhood versions, bound S and $W(\partial_y - i\eta_\nu)S_\nu$ by $L^C h_T(z_0)^8$ throughout the vertical neighborhood of radius W at z_0 . Put $v = (s_R - s_T)b$ and first move to $z_1 = z_0 + (0, v)$. Since

$$|v| \leq C\rho_j H_j \leq \frac{CBw}{\sqrt{G}} \ll W,$$

the fundamental theorem of calculus after componentwise demodulation gives

$$\|S(z_1) - \text{diag}(e^{i\eta_\nu v})S(z_0)\| \leq L^C h_T(z_0)^8 \frac{|v|}{W} \leq G^{-1/2} L^C h_T(z_0)^8.$$

Next move from z_1 to z' along slope s_T by horizontal distance b . The nonexceptional horizontal derivative estimate in [Theorem 3.6](#), applied to lengths at least l_* , bounds this change by

$$L^C h_T(z_0)^8 \frac{|b|}{l_*} \leq G^{-1/2} L^C h_T(z_0)^8.$$

To justify its spatial weight here, the two displacements change the normalized coordinates of T by at most $C/(B^2\sqrt{G}) + CB/(m\sqrt{G})$. Its weight at every intermediate point is therefore comparable to $h_T(z_0)$. If the derivative estimate is expressed using the smaller dilation K_1 , that weight is bounded above by the K_0 -dilated weight used here. No good-point estimate at z_1 or z' has been invoked for this horizontal move.

The replacement of s_T by t_c in the diagonal phase costs at most

$$C|s_T - t_c| \frac{|b|}{w} \|S(z_0)\| \leq G^{-1/2} L^C h_T(z_0)^8,$$

because $\eta_\nu \asymp w^{-1}$ uniformly in ν . This proves, uniformly over the interval partial sum,

$$\|S(z') - \mathcal{D}_c(b)S(z_0)\| \leq G^{-1/2} L^C h_T(z_0)^8. \quad (5.40)$$

The same bound holds with the mask q_T multiplying S at its own evaluation point. Indeed $|q_T| \leq 1$, and its formula (2.47), together with boundedness of the derivative of sinc, gives

$$|q_T(z') - q_T(z_0)| \leq L^C \left(\frac{|b|}{l_T} + \frac{|s_R - s_T| |b|}{W} \right) \leq G^{-1/2} L^C.$$

Combining this with the snapshot bound for S proves the masked version of (5.40). This estimate uses absolute motion of the mask and does not divide by q_T , which may vanish.

Group the masked packets by stopping level and denote their sums by $f_k(z)$. Their weighted contribution to the difference in (5.38) is

$$\begin{aligned} & \sum_k \omega_k(z') (f_k(z') - \mathcal{D}_c(b)f_k(z_0)) \\ & + \sum_k (\omega_k(z') - \omega_k(z_0)) \mathcal{D}_c(b)f_k(z_0). \end{aligned}$$

All interval partial sums of the first parenthesis have norm at most $G^{-1/2} L^C h_T(z_0)^8$, by the masked version of (5.40). The snapshot partial sums in the second line have norm at most $L^C h_T(z_0)^8$. Apply (5.27) to both lines, then (5.34) and (5.35). The result for this top is at most $G^{-1/2} L^C h_T(z_0)^4$. Sum over the tops eligible in c . Since $h_T^4 \leq C_3^3 h_T$, this is at most $G^{-1/2} L^C n_c(z_0)$. Finally $\log G \asymp L^{0.7}$ absorbs every fixed power of L into a smaller positive power of G , proving (5.38). $\square$

5.8. One deterministic residual sequence for each path. We now refine Z_0 by a set which depends on the angular translation but not on ζ . Its purpose is to make the later residual estimates available on most of the path weight. The subsets of packets used in its definition must be fixed before evaluating a spatial point or choosing a mass cutoff.

For each possible path top R , let

$$\mathcal{P}_R = \{s : \ell_s \leq l_R, \quad |\theta_s - s_R| \leq CB^4 d_s\}, \quad (5.41)$$

where the absolute constant is large enough to cover the assigned-top comparisons below. Apply the main/error decomposition of [Theorem 3.5](#) with reference slope s_R to this fixed collection. Denote the resulting positive error sum by $\mathcal{E}_R$ and positive main square function by $\mathcal{S}_R$. The absolute envelopes can use fixed decay exponents slow enough to dominate the norms of the unweighted synthesis packets. There is no loss in doing so because the packet bounds are available with arbitrarily large fixed decay exponents.

For each shift σ , define a second fixed collection

$$\mathcal{Q}_{R,\sigma} = \bigcup_{\substack{j \geq 0 \\ i_{j+1}^\sigma \leq k_R}} \left\{ s : \begin{array}{l} i_j^\sigma \leq k_s < i_{j+1}^\sigma, \\ T(s) \in \mathcal{T}(v_{i_{j+1}^\sigma}^\sigma(R)) \end{array} \right\}. \quad (5.42)$$

For blocks at whose endpoint R is still eligible, these are the packets that stop within the block while their assigned tops remain eligible in the child $v_{i_{j+1}^\sigma}^\sigma(R)$. The union takes all such blocks at once, fixing one packet collection before any spatial point or mass cutoff is chosen. Let $\mathcal{V}_{R,\sigma}$ be the 3-variation of the sequence formed by summing $q_{T(s)}\phi_s$ from this collection at each packet length. It permits any finite list of disjoint intervals of that one length sequence. Subsequent choices of those intervals may depend on the evaluation point; the collection and its length sequence remain fixed.

Every packet in $\mathcal{Q}_{R,\sigma}$ satisfies the hypotheses of [Theorem 3.5](#) with $t = s_R$ and $l = l_R$. Indeed, $k_s < i_{j+1}^\sigma \leq k_R$ gives

$$d_s > \frac{\rho_{j+1}}{B^2} \geq d_R, \quad \ell_s < l_R. \quad (5.43)$$

Since $s_{T(s)}$ and s_R are in the same block-child interval of width ρ_{j+1} ,

$$|\theta_s - s_R| \leq |\theta_s - s_{T(s)}| + |s_{T(s)} - s_R| \leq C d_s + \rho_{j+1} \leq C B^2 d_s \leq C B^4 d_s.$$

For later use, if a residual packet only lies in the larger path node $J = v_{i_j}(R)$ and has $\ell_s \leq l_R$, then $|s_{T(s)} - s_R| \leq \rho_j \leq B \rho_{j+1} < B^3 d_s$; thus it belongs to $\mathcal{P}_R$ as well. These facts explain both choices of fixed collections.

Lemma 5.6 (Good paths). *There is a fixed exponent C_{path} , chosen independently of the mass cutoff below, such that the following construction is valid. Call R bad at z if any of $\mathcal{E}_R(z), \mathcal{S}_R(z), \mathcal{V}_{R,\sigma}(z)$, $0 \leq \sigma < K$, exceeds $L^{C_{\text{path}}}$. Otherwise call it good at z . For every fixed angular translation,*

$$\sum_R \int_{\mathbb{R}^2} h_R(z) \mathbf{1}_{\{R \text{ bad at } z\}} \, dz \leq S e^{-10L}. \quad (5.44)$$

The constant 10 may be replaced by any prescribed absolute constant by increasing the fixed exponent. The measurable set

$$Z = Z_\vartheta = \left\{ z \in Z_0 : \sum_{R \text{ bad at } z} h_R(z) \leq e^{-L} \right\} \quad (5.45)$$

is independent of ζ and satisfies

$$|Z_0 \setminus Z| \leq S e^{-9L} \leq L^C e^{-5L} |F|. \quad (5.46)$$

Its removal is therefore allowable in the target estimate (2.32), using the original adjoint sum's L^2 bound.

Proof. Fix R , a line of slope s_R , and any horizontal interval I of length l_R on that line. By [Theorem 3.5](#), the normalized $L^b(I)$ norm of each indicated quantity is at most $C b^C (1 + \log B)^C$. The same bound holds for $\mathcal{V}_{R,\sigma}$ because (5.42) is a fixed subset satisfying (3.1). As $\log B = K \log G$ is bounded by a fixed power of L , this bound is polynomial in L when b is a fixed positive multiple of L .

For clarity, fix an arbitrarily large absolute number J_0 . Choose $b = \lceil L \rceil$ and then a fixed exponent C_{path} sufficiently larger than those in the moment bound. Markov's inequality gives,

for all large L ,

$$|\{x \in I : \mathcal{E}_R(x) > L^{C_{\text{path}}}\}| + |\{x \in I : \mathcal{S}_R(x) > L^{C_{\text{path}}}\}| \\ + \sum_{\sigma=0}^{K-1} |\{x \in I : \mathcal{V}_{R,\sigma}(x) > L^{C_{\text{path}}}\}| \leq e^{-J_0 L} |I|.$$

The factor $K + 2$ is polynomial and is absorbed by the same choice. Alternatively one may increase the fixed multiple in the choice of b ; neither choice depends on the number of scales or tops.

Partition the whole reference line into intervals of length l_R . Along this line h_R changes by at most an absolute factor on each such interval, directly from (2.36). Hence the preceding estimate implies

$$\int_I h_R \mathbf{1}_{\{R \text{ bad}\}} dx \leq C e^{-J_0 L} \int_I h_R dx.$$

Sum over the intervals and integrate in the transverse coordinate $y - s_R x$; this change of coordinates has determinant one. Using (5.1) gives

$$\int_{\mathbb{R}^2} h_R \mathbf{1}_{\{R \text{ bad}\}} dz \leq L^C e^{-J_0 L} |R_R|.$$

Sum over R and take J_0 large enough to obtain (5.44). This establishes the estimate with its full spatial weight, not only on the central part of R .

Markov's inequality applied to the nonnegative sum defining (5.45) gives its first measure bound. Since $S \leq L^C |E| = L^C N^2 |F|$ and $N \leq e^{2L}$, the second follows. For example, Cauchy-Schwarz and the previously proved adjoint estimate give

$$\int_{Z_0 \setminus Z} \|P\| \leq |Z_0 \setminus Z|^{1/2} \|P\|_2 \leq L^C e^{-5L/2} N |F|,$$

which is smaller than any required fixed negative power of L times $N |F|$. The estimate is applied to the original sum P , as required for exceptional discards.

Finally, all packet collections used here depend only on the finite geometric data, the angular translation, and the shift. They do not depend on z , ζ , significance, or a mass cutoff. Their finite length variations and the other thresholded quantities are measurable in z . They are measurable jointly with ϑ as well: at the finitely many relevant angular levels, membership of each of the finitely many top slopes in its interval changes at finitely many translation endpoints. On the complementary parameter intervals all the relevant membership data are fixed. This also proves the asserted independence from ζ . $\square$

5.9. Mass stopping and the exact path expansion. Choose a cutoff

$$M = L^{A_M}, \tag{5.47}$$

where the fixed exponent A_M will be taken sufficiently large in the residual estimates. Every threshold and exponent in the preceding lemmas has already been fixed without reference to A_M . Statements below that use M hold for each such fixed exponent after increasing the lower threshold for L .

For convenience set

$$C_M = L^{Cp} (1 + M)^{\alpha-1}, \tag{5.48}$$

with a fixed sufficiently large C . Equation (5.24) and $p\mu = \alpha - 1$ imply that a node of mass $n_J \leq M$ satisfies

$$V_J \leq C_M n_J. \tag{5.49}$$

Indeed $V_J/n_J \leq 1 + L^{Cp} n_J^{\alpha-1}$ for the parameters.

Lemma 5.7 (Expansion into top paths). *Fix $z \in Z$, a root, a shift, and parameters as above. Starting from a node with $n_J > M$, expand*

$$V_J = \beta_J \left(n_J - \sum_c n_c + \sum_c V_c \right), \quad (5.50)$$

and stop a branch on first reaching a node of mass at most M . At each expanding node, split $n_J - \sum_c n_c$ into the individual weights of the tops which do not remain eligible at the block endpoint. At a stopped node split V_J proportionally among its eligible top weights. This gives exactly one terminal path for each eligible root top R , with a terminal amount e_R satisfying

$$0 < e_R \leq C_M h_R, \quad V_{\text{root}} = \sum_R e_R \prod_{J \text{ active on } R's \text{ path}} \beta_J. \quad (5.51)$$

If $n_{\text{root}} > M$, at least one half of this sum comes from tops which are good at z . Consequently,

$$V_{\text{root}}(z) \leq 2C_M \sum_{\substack{R \text{ eligible at root} \\ R \text{ good at } z}} h_R(z) \prod_{J \text{ active on } R's \text{ path}} \beta_J(z). \quad (5.52)$$

If $n_{\text{root}} \leq M$, (5.49) applies instead.

Proof. Identity (5.50) follows directly from (5.14), because $V_c - n_c = n_c^{\alpha-p} \|A_c\|^p$. The term $n_J - \sum_c n_c$ is a sum of top weights by (5.4), now between the endpoints of the whole block. Every top either remains eligible in exactly one block child or loses eligibility and terminates at J . The construction therefore branches by disjoint top sets, assigning each top exactly one path. It terminates after finitely many blocks even if no mass stop occurs, since no top is eligible beyond the final level. At such a final expanding node the block-child sum is empty and all tops terminate directly.

If R terminates directly at an expanding node, its terminal amount is h_R . If it reaches a mass-stopped node c , the terminal amount is $(V_c/n_c)h_R$, which is at most $C_M h_R$ by (5.49). Repeated use of (5.50) proves the equality in (5.51).

To verify the claim about bad paths, it is useful to normalize the expansion by actual probabilities. At an active node define

$$D_J = n_J - \sum_c n_c + \sum_c V_c = n_J + \sum_c (V_c - n_c) > M.$$

Choose a block child c with probability V_c/D_J , or a directly terminating top R with probability h_R/D_J . These probabilities sum to one. At a mass-stopped node choose an eligible top with probability h_R/n_c . The probability of a given complete terminal path is exactly its summand in (5.51) divided by V_{root} . In fact, for an active edge $J \rightarrow c$, $V_c/D_J = \beta_J V_c/V_J$, and at a direct terminal edge $h_R/D_J = \beta_J h_R/V_J$. The successive V_c factors cancel; at a mass stop the last factor h_R/n_c leaves the amount $(V_c/n_c)h_R$. This proves the asserted probability identity without an independence assumption on the amplitudes.

For a directly terminating top, its last conditional probability is $h_R/D_J \leq h_R/M \leq h_R$. If a top terminates in the next node by a mass stop, the combined last choice of that child and then that top has conditional probability

$$\frac{V_c}{D_J} \frac{h_R}{n_c} \leq C_M \frac{h_R}{M} \leq C_M h_R.$$

Earlier conditional probabilities are at most one. Thus the unconditional probability of any specified terminal top is at most $C_M h_R$. This estimate also covers tops which terminate after

many active ancestors; it uses only the last active step. By (5.45), the total probability of bad terminal tops is at most

$$C_M \sum_{R \text{ bad at } z} h_R(z) \leq C_M e^{-L} < \frac{1}{2}$$

for all sufficiently large L . Here $\log C_M = O(p \log L + \log(1 + M)) = o(L)$ for each fixed A_M . The probability identity now implies that good tops carry at least half of the total amount, and the bound on e_R proves (5.52). $\square$

The remaining path estimate, proved below in Theorem 6.3, is

$$\int_{Z \cap \{R \text{ good}\}} h_R \prod_{J \text{ active on } R's \text{ path}} \beta_J \leq L^{Cp} (1 + M)^C \int_{\mathbb{R}^2} h_R. \quad (5.53)$$

Sections 5.11 and 6 prove (5.53) from the estimates established above. Once that estimate is available, (5.52) and (5.1) control the integral of V_{root} by summing over the eligible root tops.

5.10. Path drops and marked spatial cells. Fix a top R and one shift. Let J_R be the last block index with $i_{J_R} \leq k_R$. For $0 \leq j \leq J_R$ put

$$J_j = v_{i_j}(R), \quad n_j = n_{J_j}, \quad n_{J_R+1} = 0, \quad h_j^* = n_j - n_{j+1}, \quad u_j = \frac{h_j^*}{n_j}. \quad (5.54)$$

For $j < J_R$, the path block child is $v_{i_{j+1}}(R)$. For $j = J_R$ there is no path child: R ceases to be eligible at the next block endpoint. Other tops in J_j may still continue, and all their children are retained in (5.13) and (5.14). Only the path notation sets the next mass to zero. In particular, the last possible path drop has $h_j^* = n_j$ and $u_j = 1$.

The actual active indices form the prefix before the first mass at most M , equivalently the indices with $n_j > M$. A path which stops by mass need not reach its final possible block J_R . At the last active index of such a path we retain the actual eligible next mass in (5.54); it is not set to zero merely because the expansion stops there.

Lemma 5.8 (Drops and marking). *At a point of Z_0 , the active path drops satisfy*

$$\sum_{j: n_j > M} u_j \leq 1 + \log^+ \frac{n_0}{M} \leq CL. \quad (5.55)$$

On each line of slope s_R , mark an H_j cell I when

$$\inf_{z \in I} n_j(z) > M, \quad \sup_{z \in I} h_j^*(z) \geq \frac{1}{2}. \quad (5.56)$$

Every point of a marked cell is active and has $h_j^ \geq c$ for an absolute $c > 0$. At an active point in an unmarked cell with $h_j^* \geq 1$, one has $M < n_j < M + 1$ for all sufficiently large L . There is at most one such index on each active path, and its factor is bounded at points of Z_0 by*

$$\beta_{J_j} \leq L^{Cp} (1 + M)^C. \quad (5.57)$$

If an index is inactive at a point, the cell containing that point is unmarked. Finally, even in the last possible block,

$$H_j \leq \frac{l_R}{B\sqrt{G}}. \quad (5.58)$$

Proof. If there are no active indices there is nothing to prove in (5.55). Otherwise let q be the last active index. For $j < q$, the next mass is also greater than M , and

$$u_j = 1 - \frac{n_{j+1}}{n_j} \leq \log \frac{n_j}{n_{j+1}}.$$

The last term u_q is at most one, whether its next mass is positive or zero. Therefore the sum is at most $1 + \log(n_0/n_q) \leq 1 + \log(n_0/M)$. This argument deliberately does not telescope through a possibly tiny mass below the stopping cutoff. Equation (3.20) gives the last inequality.

Both n_j and h_j^* are positive sums of weights of tops eligible at J_j , except that the difference can be identically zero. In the last possible block it is simply n_j . Thus Theorem 5.3 gives relative variation $1 + O(G^{-1/2})$ across the whole H_j cell. A marked cell has positive supremum of h_j^* , so this difference is not identically zero there, and its infimum is at least $\frac{1}{2}e^{-CG^{-1/2}} \geq \frac{1}{4}$ for large L . The first marking condition supplies activity everywhere on the cell.

Suppose instead that z is active, its cell is unmarked, and $h_j^*(z) \geq 1$. The second marking condition then holds, so the first one fails: the cell has infimum mass at most M . The relative-motion bound implies

$$M < n_j(z) \leq e^{CG^{-1/2}} M < M + 1,$$

where the last inequality holds because M is a fixed polynomial in L . Its drop of at least one gives $n_{j+1}(z) < M$ (or zero at the last possible block), and all subsequent masses are at most that mass. This is consequently the last active index and can occur at most once. At a point of Z_0 , since the denominator of (5.14) is at least n_j , (5.24) gives

$$\beta_{J_j} \leq \frac{V_{J_j}}{n_j} \leq 1 + L^{Cp} n_j^{\alpha-1} \leq L^{Cp} (1 + M)^C,$$

proving (5.57). If $n_j(z) \leq M$, its cell cannot have infimum greater than M , proving the assertion about inactive indices. This observation will justify assigning the value one to the auxiliary phase factor on every unmarked cell.

Finally, eligibility at the starting level gives $d_R \leq \rho_j/B^2$, equivalently $l_R \geq B^2 w/\rho_j$. Together with (5.12), this yields

$$\frac{H_j}{l_R} \leq \frac{\rho_j}{B^2 \rho_{j+1} \sqrt{G}} \leq \frac{1}{B \sqrt{G}}.$$

This only uses eligibility at the start of the block, and therefore applies to the final possible block as well. $\square$

All constructions above are finite and measurable. Their constants are uniform in the finite depth, so no lower bound on a nonzero top weight, or on the number of angular levels retained, is part of the argument. The limits from finite packet collections and the angular-translation average are taken only after the subsequent sections have obtained the uniform estimates.

5.11. Prediction and averaging on marked cells. Fix a shift at a vertex of the analytic parameter simplex, with imaginary parameters of Euclidean norm at most L^2 . Retain the block notation, path top R , and meshes from the preceding construction. Thus J is the path interval at level i_j , its block children c have width ρ_{j+1} , and

$$G\rho_{j+1} \leq \rho_j \leq B\rho_{j+1}, \quad H_j = \frac{w}{\rho_{j+1}\sqrt{G}}, \quad H_{j-1} = \frac{w}{\rho_j\sqrt{G}}.$$

The first inequalities include the initial, possibly shorter, block. The path child is denoted by a when it exists. At the last path block there is no path child, even when some other tops continue; in that case the relative path drop is $u_j = 1$.

Proposition 5.9 (Averaging factors on marked cells). *There is an absolute $c_{av} > 0$ with the following property, for all sufficiently large L . On a fixed line of slope s_R , each marked H_j -cell I*

meeting Z admits a nonnegative function Γ_j which is constant on the H_{j-1} -subcells of I , satisfies $|I|^{-1} \int_I \Gamma_j \leq 1$, and obeys

$$\beta_j^0(z) \leq \exp(G^{-c_{\text{av}}} u_j(z)) \Gamma_j(z) \quad (z \in I). \quad (5.59)$$

Set $\Gamma_j = 1$ on unmarked cells and on cells not meeting Z . The functions can be chosen measurably on each fixed line. No choice jointly measurable in the family of parallel lines is needed.

We first construct a normalized prediction $\tilde{\beta}$ whose average is at most one for suitable probability measures. We compare it with β_j^0 , then blend it with one to obtain $\mathcal{B}$, whose comparison error contains the relative path drop. Finally we pass to uniform cell measure, take the finer-cell supremum $\mathcal{B}^\sharp$, and normalize it to obtain Γ_j . The losses in these last operations also contain the relative path drop.

5.12. Snapshot data and denominator shrinkage. Fix a marked cell I meeting Z , and choose one snapshot $z_0 \in I \cap Z$. Parameterize the line by horizontal displacement b from z_0 , and also write I for the translated parameter interval. All unprimed masses and amplitudes below are evaluated at z_0 ; a prime denotes evaluation at $z_0 + (b, s_R b)$. Let $n = n_J(z_0)$ and $v = u_j(z_0)$. If the path child exists, then $v = 1 - n_a/n$; otherwise $v = 1$. The positive-mass motion bounds give

$$\frac{1}{2}v \leq u_j(z) \leq 2v \quad (z \in I) \quad (5.60)$$

for large L . Marking ensures $v > 0$. Only nonempty children are used. Their masses are positive at every point, since the sets of eligible tops are fixed and their weights are strictly positive.

Use the pointwise normalization (5.15), with l_c and X_c evaluated at the snapshot and the primed quantities evaluated at the displaced point. That identity expresses β_j^0 in either set of data. The next construction shrinks snapshot vectors to obtain one-sided denominator comparisons for those children. When $v < 1/2$, the path vector will instead remain unchanged so that its numerator and denominator variations can be compared together. All vector norms in this section are in $\mathcal{H}$. The weights satisfy $\sum_c l_c \leq 1$, and, when there is a path child, $\sum_{c \neq a} l_c \leq v$. When there is none, an off-path sum means the sum over all children.

Choose a fixed exponent C_* , large enough for the estimates below, and put

$$K_* = L^{C_*}(1 + N^\mu). \quad (5.61)$$

Equations (3.20) and (5.24) imply, at the snapshot,

$$\|X_c\| \leq K_*, \quad n^\mu \leq K_*, \quad \sum_{c \neq a} l_c^{1-\mu} \|X_c\| \leq K_* v. \quad (5.62)$$

Enlarging the fixed exponent absorbs absolute constants here and below. For the last estimate use

$$l_c^{1-\mu} X_c = n^{\mu-1} A_c = n^\mu l_c (A_c/n_c).$$

The sum over all children in this estimate is at most K_* .

Let $\gamma > 0$ be a sufficiently small fixed exponent below the motion exponent in Theorem 5.5, and set $\varepsilon = G^{-\gamma}$. This choice is independent of the depth, dimension, and parameters L, p, D . Decreasing γ , if necessary, absorbs the fixed powers of L in the following consequences of the motion estimates:

$$\left\| l_c^{1/p} X'_c - \mathcal{D}_c(b) l_c^{1/p} X_c \right\| \leq \varepsilon l_c^{1/p} n_c^\mu, \quad (5.63)$$

$$\left\| n'^{\mu-1} \lambda'_c A'_c - n^{\mu-1} \lambda_c \mathcal{D}_c(b) A_c \right\| \leq \varepsilon K_* l_c. \quad (5.64)$$

Both estimates hold throughout I , without a good-point hypothesis at the displaced point. For the scalar rescaling in the first one, write $l_c^{1/p} X_c = n^{-1/p} n_c^{\alpha/p-1} A_c$. The exponents $-1/p$ and

$\alpha/p - 1$ are absolutely bounded. Their factors have the stated relative mass motion; the error in A_c is bounded by (5.38), and its snapshot size is at most $L^C n_c$. This proves (5.63). For the second estimate use $n^{\mu-1}$, the relative complex motion of λ_c , $|\lambda_c| \leq 1$, and $n^{\mu-1} n_c = n^\mu l_c \leq K_* l_c$.

Shrink A_c radially toward zero by a distance at most $2\varepsilon n_c$, choosing

$$\tilde{A}_c = \begin{cases} (1 - 2\varepsilon n_c / \|A_c\|)_+ A_c, & A_c \neq 0, \\ 0, & A_c = 0. \end{cases}$$

If no eligible top in c is significant at z_0 , set $\tilde{A}_c = 0$. This fits within the same error: Theorem 5.2 and the mask suppression give

$$\|A_c\| \leq L^C e^{-L \log L} n_c \leq \varepsilon n_c$$

for such a child. There is one exception: when $\nu < 1/2$, leave the path vector unchanged, setting $\tilde{A}_a = A_a$. Define $\tilde{X}_c = n_c^{\mu-1} \tilde{A}_c$. For every child to which shrinkage was applied,

$$l'_c \|X'_c\|^p \geq l_c \|\tilde{X}_c\|^p \quad (b \in I). \quad (5.65)$$

Indeed radial shrinkage reduces the norm of $l_c^{1/p} X_c$ by the smaller of its whole norm and $2\varepsilon l_c^{1/p} n_c^\mu$, whereas (5.63) permits a reduction of at most $\varepsilon l_c^{1/p} n_c^\mu$ at a displaced point.

The bounds in (5.62) continue to hold with $\tilde{X}_c$. Including the shrinkage error in (5.64) gives

$$\left\| l_c^{1-\mu} \lambda'_c X'_c - l_c^{1-\mu} \lambda_c \mathcal{D}_c(b) \tilde{X}_c \right\| \leq C \varepsilon K_* l_c. \quad (5.66)$$

5.13. Exact averaging of the prediction. On the whole parameter line, define

$$F(b) = \sum_c l_c^{1-\mu} \lambda_c \mathcal{D}_c(b) \tilde{X}_c, \quad \tilde{\beta}(b) = \frac{1 + \|F(b)\|^p}{1 + \sum_c l_c \|\tilde{X}_c\|^p}. \quad (5.67)$$

The masses, scalar weights, and vectors are fixed snapshot data. Only the diagonal unitaries $\mathcal{D}_c(b)_{\xi\xi} = e^{i(s_R - t_c)\eta_\xi b}$ vary.

Lemma 5.10 (Fourier-supported averaging). *There is a fixed sufficiently small $\epsilon' > 0$ such that, with*

$$Q = G^{3/4} \rho_{j+1}, \quad \delta = \epsilon' Q/w, \quad (5.68)$$

every positive probability ν satisfying $\text{supp } \hat{\nu} \subset (-\delta, \delta)$ obeys $\int \tilde{\beta} \, d\nu \leq 1$.

Proof. Use only children for which $\tilde{X}_c \neq 0$; an empty collection gives the assertion immediately. Apply the clock selection from (5.17) to this finite subfamily at the snapshot, with Q as in (5.68). The clock rates are fixed as b varies. Each retained set has pairwise interval distance at least Q , and the retention probabilities satisfy $\pi_c \geq n_c/m_c$.

Every used child other than the possible unshrunk path child contains a significant eligible top at z_0 . From each retained child other than that possible exception, select one such top. Their slopes are Q -separated, their precisions satisfy

$$d_T \leq \rho_{j+1}/B^2 \leq Q/B^2,$$

and their diameter is at most $\rho_j \leq B\rho_{j+1} \leq BQ$. The simultaneous count in Theorem 3.8, with D as chosen in Equation (3.43), bounds this list by $D/100$. Adding the possible retained path child leaves at most D members. Clock exclusion separates that extra child too; it does not require a significant representative for that one child.

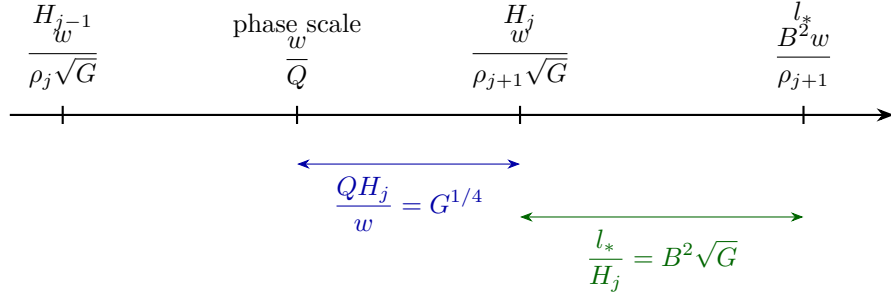


FIGURE 2. The horizontal-length axis (schematic) for a marked block. Here $Q = G^{3/4}\rho_{j+1}$ is the retained angular separation and $G\rho_{j+1} \leq \rho_j \leq B\rho_{j+1}$, including the initial block. The quantity $l_* = B^2w/\rho_{j+1}$ is the lower bound for predictable packet lengths in (5.39). On an interval of length H_j , the separated relative phases oscillate many times while the packet envelopes move little. The preceding mesh also satisfies $H_{j-1}\rho_j/w = G^{-1/2}$, which controls the step supremums. Absolute frequency constants are suppressed.

In each Hilbert component, the retained carrier frequencies are separated by at least cQ/w , with $c > 0$ absolute, since $\eta_\xi \asymp w^{-1}$. Choose $\epsilon' \leq \epsilon_0 c$. For a retained set $\mathcal{S}$, let

$$F_{\mathcal{S}}(b) = \sum_{c \in \mathcal{S}} \pi_c^{-1} l_c^{1-\mu} \lambda_c \mathcal{D}_c(b) \tilde{X}_c.$$

Expectation over the clocks gives $\mathbb{E}_{\mathcal{S}} F_{\mathcal{S}} = F$; no independence of retention indicators is asserted or needed. Put $r = r_*$. Minkowski in that expectation, (4.3), and Jensen for the concave function $t^{1/r}$ give

$$\begin{aligned} \|F\|_{L^p(\nu; \mathcal{H})} &\leq \mathbb{E}_{\mathcal{S}} \left(\sum_{c \in \mathcal{S}} \pi_c^{-r} l_c |\lambda_c|^r \|\tilde{X}_c\|^r \right)^{1/r} \\ &\leq \left(\sum_c \pi_c^{1-r} l_c |\lambda_c|^r \|\tilde{X}_c\|^r \right)^{1/r} \\ &\leq \left(\sum_c l_c \|\tilde{X}_c\|^r \right)^{1/r} \leq \left(\sum_c l_c \|\tilde{X}_c\|^p \right)^{1/p}. \end{aligned} \quad (5.69)$$

The power of l_c uses $r(1-\mu) = 1$. The third line uses the retention cancellation (5.18) for the present child subfamily. The last inequality in (5.69) is Hölder with weights of total mass at most one and $r \leq p$; the missing mass can be placed at a zero vector. Raise to power p , add one, and divide by the constant denominator in (5.67). $\square$

Figure 2 summarizes the horizontal scales used in the comparison and discretization below.

5.14. Comparison to the actual factor. On the underlying real Hilbert space put $\mathcal{L}(Y) = \log(1 + \|Y\|^p)$. For $p \geq 4$,

$$\|\nabla \mathcal{L}(Y)\| \leq p, \quad \|D^2 \mathcal{L}(Y)\| \leq Cp^2. \quad (5.70)$$

Indeed its radial first derivative is $pr^{p-1}/(1+r^p)$, its radial second derivative is

$$\frac{p(p-1)r^{p-2}}{1+r^p} - \frac{p^2r^{2p-2}}{(1+r^p)^2},$$

and each tangential Hessian eigenvalue is $pr^{p-2}/(1+r^p)$. Splitting into $r \leq 1$ and $r \geq 1$ proves the bounds; the formulas extend continuously at zero. No dimension enters these estimates.

If $v \geq 1/2$, every child was shrunk. The denominator comparison (5.65), the sum of (5.66), and (5.70) yield

$$\log \beta_J^0(z) - \log \tilde{\beta}(b) \leq CpK_*\varepsilon v, \quad v \geq 1/2. \quad (5.71)$$

The factor v here costs only an absolute factor of two.

For $0 < v < 1/2$, retaining that factor requires the unshrunk path vector. Write $\sigma_a = \lambda_a/|\lambda_a|$ and remove $\mathcal{D}_a(b)$ and σ_a from the predicted numerator. With

$$\begin{aligned} Y &= l_a^{1/p} X_a, \\ C_a &= l_a^{1-\mu-1/p} |\lambda_a| = l_a^{1-\alpha/p} (n_a/m_a)^\mu, \\ Z_1(b) &= \sigma_a^{-1} \mathcal{D}_a(b)^{-1} \sum_{c \neq a} l_c^{1-\mu} \lambda_c \mathcal{D}_c(b) \tilde{X}_c, \\ R_1 &= \sum_{c \neq a} l_c \|\tilde{X}_c\|^p, \end{aligned}$$

the prediction becomes

$$\tilde{\beta}(b) = \frac{1 + \|C_a Y + Z_1(b)\|^p}{1 + \|Y\|^p + R_1}. \quad (5.72)$$

Here Y, C_a , and the denominator are constant on the whole line. Since $n_a \leq m_a \leq n$, both l_a and n_a/m_a lie in $[1-v, 1]$. As $1 - \alpha/p \geq 0$, we have

$$0 < C_a \leq 1, \quad 1 - C_a \leq Cv, \quad \|Z_1(b)\| \leq K_*v. \quad (5.73)$$

In the actual numerator remove $\mathcal{D}_a(b)$ and $\sigma'_a = \lambda'_a/|\lambda'_a|$. Define

$$Y' = l_a'^{1/p} \mathcal{D}_a(b)^{-1} X'_a, \quad C'_a = l_a'^{1-\mu-1/p} |\lambda'_a|, \quad R'_1 = \sum_{c \neq a} l'_c \|X'_c\|^p,$$

and let Z'_1 be the remaining, similarly rotated off-path sum. The actual ratio has the form (5.72) with primes. We have

$$\begin{aligned} \|Y' - Y\| &\leq C\varepsilon K_*, & |C'_a - C_a| &\leq C\varepsilon v, \\ \|Z'_1 - Z_1\| &\leq C\varepsilon K_*v, & R'_1 &\geq R_1. \end{aligned} \quad (5.74)$$

The first and last statements follow from (5.63) and (5.65). For the off-path sum, sum (5.66) over $c \neq a$, whose weights sum to at most v . Changing the removed scalar phase costs at most $C\varepsilon$ times the off-path size, because $|\sigma'_a - \sigma_a| \leq C\varepsilon$.

The extra v in the coefficient estimate follows from the complementary-mass assertion of Theorem 5.3. If $r = A/(A+B)$, with $A > 0$, $B \geq 0$, and the positive summands change relatively by at most $C\varepsilon$, then

$$|r' - r| \leq C\varepsilon r(1-r), \quad |\log r' - \log r| \leq C\varepsilon(1-r).$$

If $B = 0$, the ratio is identically one. Apply this to n_a/n , with complement $n - n_a$, and to n_a/m_a , with complement $m_a - n_a$. Both values of $1 - r$ are at most v . The bounded nonnegative powers defining C_a give its claimed variation. This concerns the modulus of the path weight; its scalar phase has already been removed.

Set $\mathfrak{r} = R_1/(1 + \|Y\|^p + R_1)$. We claim

$$\begin{aligned} \log \beta_J^0(z) - \log \tilde{\beta}(b) &\leq Cp^2 K_*^2 \varepsilon v + CpK_*\varepsilon \mathfrak{r}, \\ -\log \tilde{\beta}(b) &\geq \mathfrak{r} - CpK_*v. \end{aligned} \quad (5.75)$$

Hold $Z_1 = Z_1(b)$ fixed when comparing Y with Y' . The derivative in V of $\mathcal{L}(C_a V + Z_1) - \mathcal{L}(V)$, along that segment, is bounded by

$$p|1 - C_a| + Cp^2(|1 - C_a|\|V\| + \|Z_1\|) \leq Cp^2 K_* v.$$

Equation (5.74) bounds its change by $Cp^2 K_*^2 \varepsilon v$. The changes in C_a, Z_1 fit the same bound by (5.70).

For the residual denominator, set

$$J(V) = -\log(1 + R_1 e^{-\mathcal{L}(V)}).$$

Its derivative has norm at most $pR_1/(e^{\mathcal{L}(V)} + R_1)$. On the segment from Y to Y' , $|\mathcal{L}(V) - \mathcal{L}(Y)| \leq CpK_* \varepsilon$. This tends to zero as L increases, so that residual ratio is at most $2\mathfrak{r}$, and $J(Y') - J(Y) \leq CpK_* \varepsilon \mathfrak{r}$. The actual residual $R'_1 \geq R_1$ can only decrease the actual ratio. This proves the first inequality of (5.75). For the second, use $C_a \leq 1$, (5.70), and $\log(1 + t) \geq t/(1 + t)$:

$$\begin{aligned} -\log \tilde{\beta} &= \mathcal{L}(Y) - \mathcal{L}(C_a Y + Z_1) + \log(1 + R_1 e^{-\mathcal{L}(Y)}) \\ &\geq -p\|Z_1\| + \mathfrak{r} \geq \mathfrak{r} - CpK_* v. \end{aligned}$$

5.15. Absorbing the baseline error by blending. The parameter sizes give

$$pK_* = G^{o(1)} \quad \text{as } L \longrightarrow \infty, \quad (5.76)$$

with every preceding constant fixed. Indeed $\log N \leq 2L$, $\mu = O(L^{-1/2})$, and $\log G \asymp L^{7/10}$, so $\log(pK_*) = O(\sqrt{L} + \log L) = o(\log G)$. All the preceding smallness requirements consequently follow from a single sufficiently large lower bound on L .

Set $\theta_{\text{bl}} = \sqrt{\varepsilon}$ and blend the prediction with one:

$$\mathcal{B}(b) = (1 - \theta_{\text{bl}})\tilde{\beta}(b) + \theta_{\text{bl}}. \quad (5.77)$$

It retains average at most one for every probability in Theorem 5.10. Concavity of the logarithm gives

$$\log \mathcal{B} - \log \tilde{\beta} \geq -\theta_{\text{bl}} \log \tilde{\beta}.$$

For $v < 1/2$, Equation (5.75) yields

$$\begin{aligned} \log \beta_J^0 - \log \mathcal{B} &\leq (Cp^2 K_*^2 \varepsilon + CpK_* \theta_{\text{bl}})v + (CpK_* \varepsilon - \theta_{\text{bl}})\mathfrak{r} \\ &\leq (Cp^2 K_*^2 \varepsilon + CpK_* \sqrt{\varepsilon})v. \end{aligned}$$

The last coefficient of $\mathfrak{r}$ is nonpositive for large L . For $v \geq 1/2$, use (5.71) and $\log \tilde{\beta} \leq \mathcal{L}(F) \leq p\|F\| \leq CpK_*$ to obtain the same last bound. Therefore

$$\beta_J^0(z) \leq e^{a_L v} \mathcal{B}(b), \quad a_L = Cp^2 K_*^2 \varepsilon + CpK_* \sqrt{\varepsilon} \leq G^{-\gamma/3} \quad (5.78)$$

for sufficiently large L . The baseline has absorbed the term proportional to $\mathfrak{r}$; the remaining error is proportional to v .

5.16. Changing to uniform measure without a loss per level. Choose a fixed nonnegative Schwartz function k of integral one with $\text{supp } \hat{k} \subset [-1/4, 1/4]$. For example, square the absolute value of the inverse Fourier transform of a nonzero smooth function supported in $(-1/8, 1/8)$ and normalize its integral. Let $H = H_j$, $h = G^{-1/16}$, $s = HG^{-1/8}$, and $k_s(t) = s^{-1}k(t/s)$. Let I^+ be the concentric enlargement of I of length $(1 + h)H$, and define

$$d\nu_I(t) = \left(\frac{\mathbf{1}_{I^+}}{|I^+|} * k_s \right)(t) dt. \quad (5.79)$$

This is a positive probability whose Fourier transform is supported in $[-1/(4s), 1/(4s)]$. Since $H\delta = \epsilon' G^{1/4}$, this support lies strictly inside $(-\delta, \delta)$ for large L .

Every point of I is at distance at least $hH/2$ from the complement of I^+ . The Schwartz tail beyond $(hH)/(2s) = G^{1/16}/2$ is $O(G^{-1/8})$, by a fixed tail estimate. Thus the density in (5.79) is at least $H^{-1}(1 - CG^{-1/16})$ on I . Writing ω_I for uniform probability on I , choose $0 < \eta \leq CG^{-1/16} < 1/2$ so that

$$\nu_I \geq (1 - \eta)\omega_I, \quad \nu_I = (1 - \eta)\omega_I + \eta\nu_{\text{rem}} \quad (5.80)$$

for a positive probability ν_{rem} .

The bound $\int \mathcal{B} \, d\nu_I \leq 1$ alone would lose an amount of order η on changing measure. We need a bound proportional to v . If $\tilde{\beta} \leq 1$ on the whole line, then $\mathcal{B} \leq 1$ and its uniform average is already at most one. Otherwise choose b_0 with $\tilde{\beta}(b_0) > 1$. When $v < 1/2$, the denominator and path vector in (5.72) are fixed. For all $b \in \mathbb{R}$,

$$\begin{aligned} \log \tilde{\beta}(b) &\geq \log \tilde{\beta}(b_0) - p \|Z_1(b) - Z_1(b_0)\| \\ &\geq -2pK_*v. \end{aligned}$$

Consequently

$$(1 - \tilde{\beta}(b))_+ \leq CpK_*v \quad (b \in \mathbb{R}). \quad (5.81)$$

This does not assume an upper bound on R_1 : its contribution cancels in the logarithmic difference. Equivalently, a value exceeding one already forces $R_1/(1 + \|Y\|^p) \leq e^{CpK_*v} - 1$. For $v \geq 1/2$, Equation (5.81) follows from nonnegativity since $pK_*v \geq 1$. The blend inherits the same lower bound, as $(1 - \mathcal{B})_+ = (1 - \theta_{\text{bl}})(1 - \tilde{\beta})_+$.

In the case where the prediction exceeds one somewhere, (5.80) gives

$$1 \geq \int \mathcal{B} \, d\nu_I \geq (1 - \eta) \int \mathcal{B} \, d\omega_I + \eta(1 - CpK_*v).$$

Including the easier case $\mathcal{B} \leq 1$, this proves

$$\int \mathcal{B} \, d\omega_I \leq 1 + CG^{-1/16}pK_*v. \quad (5.82)$$

The relative mass drop has survived the change of probability.

5.17. Step supremums and normalization. The points s_R, t_c lie in J , so all component frequencies of $\mathcal{D}_c$ have magnitude at most $C\rho_j/w$. For $v \geq 1/2$, differentiating (5.67) and using (5.62) gives $\|\partial_b F(b)\| \leq C(\rho_j/w)K_*$. For $v < 1/2$, remove the path carrier as in (5.72). The off-path frequencies are $(t_a - t_c)\eta_\xi$, again of magnitude at most $C\rho_j/w$, so $\|\partial_b Z_1(b)\| \leq C(\rho_j/w)K_*v$. In both cases,

$$\left| \frac{d}{db} \log \tilde{\beta}(b) \right| \leq CpK_*v \frac{\rho_j}{w}.$$

Blending cannot increase the absolute logarithmic derivative, because

$$\frac{d}{db} \log \mathcal{B} = \frac{(1 - \theta_{\text{bl}})\tilde{\beta}}{\mathcal{B}} \frac{d}{db} \log \tilde{\beta}, \quad 0 \leq \frac{(1 - \theta_{\text{bl}})\tilde{\beta}}{\mathcal{B}} \leq 1.$$

On each H_{j-1} -subcell $I' \subset I$, let $\mathcal{B}^\#$ equal $\sup_{b \in \overline{I'}} \mathcal{B}(b)$. The prediction is positive and continuous, and these compact-cell supremums are finite. The meshes are nested. Since $H_{j-1}\rho_j/w = G^{-1/2}$,

$$\mathcal{B} \leq \mathcal{B}^\# \leq e^{bLv} \mathcal{B}, \quad b_L = CpK_*G^{-1/2}. \quad (5.83)$$

Use the fixed half-open mesh convention at endpoints; taking supremums over closed subcells preserves the inequalities. Define

$$m_I = \max \left(1, \int \mathcal{B}^\# \, d\omega_I \right), \quad \Gamma_j = \mathcal{B}^\# / m_I \quad \text{on } I. \quad (5.84)$$

This function is nonnegative, constant on the finer cells, and has average at most one. Equations (5.82) and (5.83) yield

$$\log m_I \leq (b_L + CG^{-1/16}pK_*)v. \quad (5.85)$$

Proof of Theorem 5.9. Perform the construction on each marked cell meeting Z . Equations (5.78), (5.83), and (5.84) give

$$\beta_J^0(z) \leq e^{aLv} \mathcal{B}(b) \leq e^{aLv} \mathcal{B}^\sharp(b) = e^{aLv} m_I \Gamma_j(b).$$

By (5.85), the exponent is at most

$$(G^{-\gamma/3} + CpK_*G^{-1/2} + CpK_*G^{-1/16})v.$$

Equations (5.76) and (5.60) bound this by $G^{-c_{\text{av}}}u_j(z)$, for a fixed positive exponent: choose $c_{\text{av}} < \min(\gamma/4, 1/64)$ and then take L sufficiently large. This proves (5.59) with no error independent of the mass drop.

There are countably many cells on each fixed line and finitely many levels in the current truncation. Choose one snapshot in every relevant cell, use (5.84) there, and set $\Gamma_j = 1$ elsewhere. This is a measurable step function of the line parameter. The proof of Theorem 6.3 integrates the original integrand on lines where its slice is measurable, as holds for almost every transverse parameter. It therefore needs no jointly measurable snapshot choices across parallel lines. $\square$

6. RESIDUAL TERMS AND INTEGRATION ALONG PATHS

We work with a finite tile collection and a fixed angular translation. Fix a shift σ and a parameter ζ whose real part is the corresponding vertex: $\Re \zeta_\sigma = \mu$, all other real parts vanish, and $\|\Im \zeta\|_2 \leq L^2$. All estimates in this section are uniform in these choices, the number of tiles, the angular depth, and $\dim \mathcal{H}$. We take L sufficiently large that

$$p > \alpha, \quad 0 < \mu \leq \frac{1}{12}. \quad (6.1)$$

We retain the stopping threshold $M = L^{A_M}$ from (5.47); its fixed exponent will be chosen at the end of the residual estimates.

Fix a path top R . We retain the block notation of the hierarchy: $J_j = v_{i_j}(R)$, $n_j = n_{J_j}$, and $A_j = A_{J_j}$. The last possible block is the last one whose starting node contains R as an eligible top. At this last block there is no path child, even if other tops continue; the next path mass is defined to be zero. At every other block write J_{j+1} for the path child and set

$$h_j^* = n_j - n_{j+1}, \quad u_j = \frac{h_j^*}{n_j}.$$

An index is active if $n_j > M$. The active indices form a prefix. On Z , the mass bound (3.20) and monotonicity give

$$\sum_{j \text{ active}} u_j \leq CL. \quad (6.2)$$

Indeed, except for the last active index, use $1 - n_{j+1}/n_j \leq \log(n_j/n_{j+1})$ and telescope down to a mass larger than M . The final summand is at most one. This proof also applies when the path stops by ineligibility.

6.1. Marked residuals. For $n > 0$ and $A \in \mathcal{H}$, put

$$V(n, A) = n + n^{\alpha-p} \|A\|^p = n(1 + \|n^{\mu-1} A\|^p).$$

The function $\mathcal{L}_p(Y) = \log(1 + \|Y\|^p)$ is globally p -Lipschitz on the underlying real Hilbert space. Away from zero its derivative has norm $p\|Y\|^{p-1}/(1 + \|Y\|^p) \leq p$, and the assertion at zero follows by continuity. Since β_J and β_J^0 have the same denominator, the decomposition (5.13) therefore implies

$$\log \beta_{J_j} - \log \beta_{J_j}^0 \leq Cp n_j^{-1+\mu} \|B_{J_j}\|. \quad (6.3)$$

Lemma 6.1 (Square summation of residual blocks). *At a point $z \in Z$ where R is a good path,*

$$\left(\sum_j \|B_{J_j}(z)\|^2 \right)^{1/2} \leq L^{C_b}. \quad (6.4)$$

Here the sum can be restricted to any subset of the possible blocks. The fixed exponent C_b is independent of M .

Proof. A residual packet in block j has

$$\frac{\rho_{j+1}}{B^2} < d_s \leq \frac{\rho_j}{B^2}, \quad i_j \leq k_s < i_{j+1}. \quad (6.5)$$

Because $\rho_j/\rho_{j+1} \leq B$, this block contains at most $N_{\text{len}} := C(1 + \log B)$ dyadic packet lengths. Also $s_{T(s)}, s_R \in J_j$, so the top assignment and (6.5) imply

$$|\theta_s - s_R| \leq Cd_s + \rho_j \leq CB^3 d_s \leq CB^4 d_s.$$

Thus the reference-line estimates apply whenever $\ell_s \leq l_R$.

The coefficient multiplying $q_{T(s)}\phi_s$ in B_{J_j} is a product of transition weights of modulus at most one, followed by $n_{v_{k_s}(T(s))}^{-\mu/K}$. The ending mass is at least $h_{T(s)}$. Consequently the modulus of the coefficient multiplying ϕ_s is at most $|q_{T(s)}| h_{T(s)}^{-\mu/K} \leq C$. This last bound follows from the mask estimate (2.47): on the significant region $h_T \geq 1$, while off that region the mask has faster decay than this fixed small negative power of h_T .

At z let e_ℓ and a_ℓ be the sums of the absolute error and main envelopes, respectively, at length ℓ in the fixed family $\mathcal{P}_R$ of (5.41). The preceding bounds place every residual packet with $\ell_s \leq l_R$ in this family, so its absolute envelopes are included in these positive sums. Goodness of R , as specified in Theorem 5.6, gives

$$\sum_{\ell \leq l_R} e_\ell \leq L^C, \quad \left(\sum_{\ell \leq l_R} a_\ell^2 \right)^{1/2} \leq L^C.$$

The contribution of such lengths to $\|B_{J_j}\|$ is bounded by $C \sum_{\ell \in \mathcal{I}_j} (e_\ell + a_\ell)$, where $\mathcal{I}_j$ is the set of residual lengths in that block. The sets $\mathcal{I}_j$ are disjoint. Positivity and Cauchy-Schwarz within each block therefore give

$$\begin{aligned} \left\{ \sum_j \left(\sum_{\ell \in \mathcal{I}_j, \ell \leq l_R} e_\ell \right)^2 \right\}^{1/2} &\leq \sum_{\ell \leq l_R} e_\ell, \\ \sum_j \left(\sum_{\ell \in \mathcal{I}_j, \ell \leq l_R} a_\ell \right)^2 &\leq N_{\text{len}} \sum_{\ell \leq l_R} a_\ell^2. \end{aligned}$$

The factor N_{len} is polynomial in L .

Before the last possible block, R remains eligible at the child level. Hence $d_R \leq \rho_{j+1}/B^2 < d_s$, and consequently $\ell_s < l_R$. Lengths larger than l_R can occur only in the last possible block. At each of its at most N_{len} lengths, the total absolute coefficient envelope is bounded by a constant, by [Theorem 3.1](#), applied with reference length equal to that scale. To recall the reason this bound has no factor counting the available labels, the condition $|\theta_s - s_R| \leq CB^4 d_s$ makes the analyzing and synthesizing box envelopes comparable to common s_R -coordinate envelopes, since $W \gg B^4 w$. Summing centers first produces a common positive convolution kernel of integral $O(1)$. At each source point only boundedly many selector labels are nonzero. Summing these labels then uses $\|g\|_{\mathcal{H}} \leq 1$, and costs $O(1)$. The uniformly bounded multipliers just considered preserve this absolute estimate. The last block therefore contributes at most CN_{len} in norm. Combining the estimates proves (6.4). $\square$

By [Theorem 5.8](#), at any marked point one has $h_j^* \geq c_m$ for a fixed $c_m > 0$. Monotonicity of the masses therefore implies

$$\#\{j : j \text{ marked at } z, t \leq n_j < 2t\} \leq C(1+t), \quad t \geq M. \quad (6.6)$$

For completeness, all but the final marked index in this shell account for disjoint mass drops of size at least c_m while the mass lies in an interval of length at most t ; the final index costs one even if its drop leaves the shell. Dyadic summation gives

$$\sum_{j \text{ marked at } z} n_j^{-2+2\mu} \leq C \sum_{r \geq 0} (2^r M)^{-1+2\mu} \leq CM^{-1+2\mu}. \quad (6.7)$$

Combining (6.3), [Theorem 6.1](#), and Cauchy–Schwarz yields

$$\sum_{j \text{ marked at } z} (\log \beta_{J_j} - \log \beta_{J_j}^0) \leq CpL^{C_b} M^{-1/2+\mu}. \quad (6.8)$$

6.2. Unary segments and their energy telescope. Call an active index *unary* at z if its cell is unmarked and $h_j^*(z) < 1$. At such an index the path child exists: at the last possible block the drop would instead be $n_j > M$. A significant top has mass at least one. Thus every significant top eligible in J_j must remain eligible in the path child, since the total mass removed from that child is $h_j^* < 1$.

At a unary index write

$$A_j = \lambda_j A_{j+1} + U_j + W_j. \quad (6.9)$$

Here λ_j is the path-child coefficient in (5.13); U_j consists of the residual packets assigned to tops which remain eligible in the path child. These are exactly the packets of the fixed subset $\mathcal{Q}_{R,\sigma}$ in (5.42) which stop in block j . The term W_j collects all contributions of tops eligible in J_j which do not continue in the path child, including their residual packets and, when present, their contributions through other children.

Those latter tops are all insignificant at z . Their contributions within A_{J_j} are whole interval sums on individual tops, so [Theorem 5.2](#) bounds the contribution of each by $L^C(1 + \|\zeta\|)|q_T|h_T^4$. Since h_T is bounded above by an absolute constant and $|q_T| \leq e^{-L \log L}$ there, this is at most $Ce^{-L}h_T$ for sufficiently large L . The masses of these tops sum to h_j^* . Therefore

$$\|W_j\| \leq Ce^{-L}h_j^*. \quad (6.10)$$

This estimate uses a sum of top masses, not the number of such tops.

Partition each consecutive run of unary indices into consecutive segments $[e, f)$. Starting at e , end a completed segment as soon as its accumulated drop is at least one. Each individual

drop is less than one, so every completed segment has total drop in $[1, 2)$. At the end of the run, retain the final remainder as one segment if it is nonempty. Hence every segment satisfies

$$\Delta_{e,f} := \sum_{e \leq j < f} h_j^* = n_e - n_f < 2, \quad n_f > M - 2, \quad n_j \asymp n_e \quad (e \leq j \leq f). \quad (6.11)$$

The endpoint remains an eligible path node, including when it is the node where the mass stopping rule first applies.

The number of segment starts in a mass shell obeys

$$\#\{[e, f) : t \leq n_e < 2t\} \leq C(1+t), \quad t \geq M. \quad (6.12)$$

Completed segments account for disjoint drops of size at least one. For each remainder other than the final remainder on the path, charge it to the next interruption of a unary run. Such an interruption is either marked, with drop at least c_m , or is an active unmarked index with drop at least one. The mass at this interruption differs from the start mass of the remainder by less than two. Thus the interruptions charged by starts in $[t, 2t)$ lie in the enlarged shell $[t-2, 2t)$ and their fixed positive drops give $O(1+t)$ choices. The one possible final remainder adds one.

For one segment define

$$A_i^* = \left(\prod_{i \leq j < f} \lambda_j \right) A_f, \quad V_i^* = V(n_i, A_i^*), \quad e \leq i \leq f,$$

with the empty product equal to one. Since $n_i \geq n_{i+1}$, $\alpha - p < 0$, and $|\lambda_i| \leq 1$, we have

$$V_i^* \leq h_i^* + V_{i+1}^*, \quad V_f^* = V_f. \quad (6.13)$$

Indeed, the baseline terms differ by h_i^* , while the energy term on the left is at most $n_{i+1}^{\alpha-p} \|A_{i+1}^*\|^p$. The denominator of β_{J_i} contains the energy of the path child and the nonnegative energies of all other children. It is therefore at least $h_i^* + V_{i+1}$. It follows that

$$\sum_{e \leq i < f} \log \beta_{J_i} \leq \log V_e - \log V_f - \sum_{e \leq i < f} \log(1 + h_i^*/V_{i+1}).$$

On the other hand, (6.13) gives

$$\log V_e^* - \log V_f \leq \sum_{e \leq i < f} \log(1 + h_i^*/V_{i+1}^*).$$

Subtracting these inequalities produces the useful estimate

$$\begin{aligned} \sum_{e \leq i < f} \log \beta_{J_i} &\leq \log(V_e/V_e^*) + \sum_{e \leq i < f} [\log(1 + h_i^*/V_{i+1}^*) - \log(1 + h_i^*/V_{i+1})] \\ &\leq Cp n_e^{-1+\mu} \max_{e \leq i \leq f} \|A_i - A_i^*\|. \end{aligned} \quad (6.14)$$

To justify the last line, first use the Lipschitz estimate for $\mathcal{L}_p$ to obtain

$$|\log V(n, A) - \log V(n, A')| \leq pn^{\mu-1} \|A - A'\|.$$

For $h \geq 0$, the derivative of $x \mapsto \log(1 + he^{-x})$ has absolute value $h/(e^x + h)$. Along the interval between $\log V_{i+1}$ and $\log V_{i+1}^*$, both endpoint amounts, and therefore every intermediate amount, are at least n_{i+1} . The corresponding correction term is bounded by

$$\frac{h_i^*}{n_{i+1}} pn_{i+1}^{\mu-1} \|A_{i+1} - A_{i+1}^*\|.$$

Now use (6.11) and $\sum_{e \leq i < f} h_i^*/n_{i+1} \leq 2/(M-2)$. This proves (6.14), including its endpoint case.

6.3. Common residual weights and variation. We give the coefficient calculation needed to apply the good-path variation estimate. It is important here that the same angular grid underlies all shifts.

For integer levels k along the path put $v_k = v_k(R)$ and $a_k = n_{v_k}$, so $a_{i_j} = n_j$. For a fixed segment $[e, f)$ and a block index $j \in [e, f)$, define

$$\omega_j(k) = a_k^{-\mu/K} \prod_{r=i_j}^{k-1} \left(\frac{a_{r+1}}{m_{v_{r+1}}} \right)^{\zeta_r \bmod K}, \quad i_j \leq k < i_f. \quad (6.15)$$

Here i_j denotes the angular level at the start of block j .

Every packet in U_b , for $j \leq b < f$, is assigned to a top in $v_{i_{b+1}}(R)$. It follows by nesting that its top lies in $v_k(R)$ at every intermediate level $k \leq i_{b+1}$, in particular at its stopping level k_s . Multiplying its coefficient in U_b by the path propagation factor $\prod_{j \leq r < b} \lambda_r$ therefore gives exactly $\omega_j(k_s) q_{T(s)} \phi_s$. This identity holds for every contributing top, whether or not that top is significant. Thus the scalar coefficient depends only on the stopping level, rather than on the packet or top inside that level.

Lemma 6.2 (Variation of the common coefficients). *Uniformly over the segments and their starting indices,*

$$\sup_{i_j \leq k < i_f} |\omega_j(k)| + \sum_{k=i_j}^{i_f-2} |\omega_j(k+1) - \omega_j(k)| \leq L^{C_w}. \quad (6.16)$$

The exponent C_w is fixed before M is chosen.

Proof. Every ending mass in this sequence is at least $a_{i_f} = n_f > M - 2$. Since all real parts of ζ are nonnegative, $|\omega_j(k)| \leq a_k^{-\mu/K} \leq 1$ when $M \geq 3$. Using the real logarithms of positive masses, a logarithmic increment of the coefficient is

$$\log \omega_j(k+1) - \log \omega_j(k) = \frac{\mu}{K} \log \frac{a_k}{a_{k+1}} - \zeta_{k \bmod K} \log \frac{m_{v_{k+1}}}{a_{k+1}}.$$

The logarithm here is the specified linear expression defining the coefficient, so no branch of a complex logarithm is being chosen. The first sum of absolute increments telescopes. For the second use [Theorem 5.1](#) on the entire root-to-endpoint path. All its summands are nonnegative, whence

$$\begin{aligned} \sum_{k=i_j}^{i_f-2} |\log \omega_j(k+1) - \log \omega_j(k)| &\leq \left(\frac{\mu}{K} + K \max_{\sigma} |\zeta_{\sigma}| \right) \log \frac{n_{\text{root}}}{n_f} \\ &\leq L^{C_w}. \end{aligned}$$

The last step uses $n_{\text{root}} \leq NL^C$ on Z , $\log N \leq 2L$, $n_f > M - 2 \geq 1$, and $\max_{\sigma} |\zeta_{\sigma}| \leq \mu + L^2$. The full-root bound is needed: a denominator m_v belonging to a different shift may involve a block whose starting level precedes the segment. No estimate using only $\log(n_e/n_f)$ is asserted.

For two successive logarithmic values z_0, z_1 in this sequence, both $|e^{z_0}|$ and $|e^{z_1}|$ are at most one. Along the straight segment between them, $|e^{(1-t)z_0+tz_1}| \leq 1$. Integration of the derivative of the exponential consequently yields $|e^{z_1} - e^{z_0}| \leq |z_1 - z_0|$. The logarithmic-increment bound proves (6.16). $\square$

Let $\mathcal{Q}_{R,\sigma}$ denote the fixed residual subset in (5.42). For each unary segment $[e, f)$ define

$$\Lambda_{e,f}(z) = \max_{i_e \leq a \leq b < i_f} \left\| \sum_{\substack{s \in \mathcal{Q}_{R,\sigma} \\ a \leq k_s \leq b}} q_{T(s)}(z) \phi_s(z) \right\|.$$

An empty sum has value zero. Finite depth makes the maximum finite. The interval of stopping levels $[a, b]$ corresponds to an interval of dyadic packet lengths. The latter follows directly from the definition $k_s = \max\{k : d_s \leq d(k)/B^2\}$, which is monotone in ℓ_s . Since distinct unary segments have disjoint level intervals $[i_e, i_f)$, one may choose a maximizing interval in each segment and apply the single good-path 3-variation bound to all those disjoint intervals. Thus

$$\left(\sum_{[e,f)} \Lambda_{e,f}(z)^3 \right)^{1/3} \leq L^{C_v}. \quad (6.17)$$

The segments may depend on z ; the variation is a supremum over all choices of disjoint intervals, so this dependence does not alter the inequality. The underlying packet subset itself is the single deterministic set $\mathcal{Q}_{R,\sigma}$.

Abel summation for a finite scalar sequence (ω_k) and vector sequence (F_k) gives

$$\left\| \sum_{k=a}^b \omega_k F_k \right\| \leq \left(|\omega_b| + \sum_{k=a}^{b-1} |\omega_k - \omega_{k+1}| \right) \max_{a \leq t \leq b} \left\| \sum_{k=a}^t F_k \right\|.$$

Applying this identity with (6.15), and then iterating (6.9), yields

$$\max_{e \leq i \leq f} \|A_i - A_i^*\| \leq L^{C_w} \Lambda_{e,f} + C e^{-L} \Delta_{e,f}. \quad (6.18)$$

Indeed the propagated W_j terms have modulus at most their unpropagated bounds because $|\lambda_j| \leq 1$. There are no residual terms beyond i_f in this calculation: U_j uses only $i_j \leq k_s < i_{j+1}$ with $j < f$. When f is a mass-stopping node it is still an eligible node and (6.11) applies. A final block ending by ineligibility cannot be unary and is excluded from this calculation.

Using (6.12) and dyadic mass shells, we get

$$\begin{aligned} \sum_{[e,f)} n_e^{(-1+\mu)3/2} &\leq C \sum_{r \geq 0} (2^r M)^{-1/2+3\mu/2} \leq C M^{-1/2+3\mu/2}, \\ \left(\sum_{[e,f)} n_e^{(-1+\mu)3/2} \right)^{2/3} &\leq C M^{-1/3+\mu}. \end{aligned} \quad (6.19)$$

Hölder with exponents $3/2$ and 3 , followed by (6.14), (6.17), and (6.18), therefore bounds the sum of the unary log factors by

$$\sum_{j \text{ unary at } z} \log \beta_{J_j} \leq C p L^{C_u} M^{-1/3+\mu} + C p e^{-L} \left(\max_{j \text{ active}} n_j^\mu \right) \sum_{j \text{ active}} u_j. \quad (6.20)$$

For the error term, use $n_e \geq n_j$ and $\mu - 1 < 0$ within a segment to obtain $n_e^{\mu-1} h_j^* \leq n_j^{\mu-1} h_j^* = n_j^\mu u_j$. By (3.20), $\max n_j^\mu \leq \exp(C\sqrt{L})$. Together with (6.2), this makes the second term in (6.20) tend to zero.

Choose a fixed exponent C_{res} larger than all polynomial exponents in (6.8) and (6.20), and then choose

$$A_M > 4(C_{\text{res}} + 2), \quad M = L^{A_M}. \quad (6.21)$$

By (6.1), $-1/3 + \mu \leq -1/4$ and $-1/2 + \mu \leq -5/12$. Since $p = L^{1/2}$, this choice makes both principal residual costs bounded by an absolute constant for sufficiently large L . None of the good-path thresholds or the exponents used to select A_M depended on M .

6.4. The path product and its integral.

Theorem 6.3 (Integrated path product). *With M chosen by (6.21), the estimate (5.53) holds for every path top R , every shift, and every vertex parameter with $\|\Im \zeta\|_2 \leq L^2$. Its constants are independent of the tile collection, angular depth, and Hilbert-space dimension.*

Proof. Fix a line of slope s_R , and on that line use the functions Γ_j constructed in Theorem 5.9. At a point of Z where R is good, the active indices fall into the marked indices, the unary indices, and at most one active unmarked index with $h_j^* \geq 1$. For the marked indices, combine (5.59), (6.8), and (6.2). Their phase-domination losses have sum at most CLG^{-c} , which is bounded. The unary factors have bounded log product by (6.20) and (6.21).

At the possible remaining index, Theorem 5.8 gives $M < n_j < M + 1$. From (5.24) and the baseline n_j in the denominator of β_{J_j} ,

$$\beta_{J_j} \leq 1 + L^{Cp} n_j^{\alpha-1} \leq L^{Cp} (1 + M)^C.$$

Here C is fixed, since α stays in a fixed compact interval below three. We have therefore proved the pointwise domination

$$\prod_{j \text{ active at } z} \beta_{J_j}(z) \leq L^{Cp} (1 + M)^C \prod_{j=0}^{J_R} \Gamma_j(z), \quad z \in Z \cap \{R \text{ good}\}, \quad (6.22)$$

where J_R is the last possible block index for R . It is legitimate to include every possible index on the right: if $n_j(z) \leq M$, its mesh cell cannot satisfy $\inf n_j > M$, so it is unmarked and $\Gamma_j(z) = 1$. Unmarked active indices likewise have $\Gamma_j = 1$.

Recall that Γ_j is constant on the mesh of length H_{j-1} and has average at most one on every cell of length H_j . The meshes are nested. On any cell I of the largest mesh H_{J_R} , successive averaging from the smallest mesh upward therefore gives

$$\int_I \prod_{j=0}^{J_R} \Gamma_j(x) \, dx \leq |I|. \quad (6.23)$$

More explicitly, all later factors Γ_k , $k > j$, are constant on each H_j cell. Integrating Γ_j on those cells can thus only decrease the integral of the nonnegative product. Iterate this observation for $j = 0, 1, \dots, J_R$.

The weight h_R is uniformly comparable on I . This remains true for a terminal block and for the short first block: eligibility of R at i_j gives $\rho_j \geq B^2 d_R$, and every block has $\rho_j / \rho_{j+1} \leq B$. Thus

$$H_j = \frac{w}{\rho_{j+1} \sqrt{G}} \leq \frac{l_R}{B \sqrt{G}} \leq l_R. \quad (6.24)$$

Along the line of slope s_R , the transverse normalized coordinate of R is fixed and the horizontal normalized coordinate changes by at most H_j / l_R . The definition (2.36) consequently gives $\sup_I h_R \leq C \inf_I h_R$, with an absolute constant depending only on the fixed decay exponent. By (6.23),

$$\int_I h_R \prod_{j=0}^{J_R} \Gamma_j \, dx \leq (\sup_I h_R) |I| \leq C \int_I h_R \, dx.$$

Sum over the largest-mesh cells and use (6.22). Whenever the original integrand in (5.53) has a measurable slice on this line, we obtain

$$\int_{Z \cap \{R \text{ good}\} \text{ on the line}} h_R \prod_{j \text{ active}} \beta_{J_j} dx \leq L^{Cp}(1+M)^C \int_{\text{line}} h_R dx.$$

The snapshots used to construct the Γ_j need only be chosen on this one line. There are countably many mesh cells, so these choices give measurable step functions on that line. No jointly measurable choice of snapshots across different lines is required. The original integrand in (5.53), expressed entirely through h_R , the good sets, and the finite hierarchy factors β_{J_j} , is nonnegative and Lebesgue measurable. In the coordinates $(x, y - s_R x)$, it has a measurable slice for almost every transverse parameter. The preceding line estimate therefore holds for almost every such parameter. Integrating it and applying Tonelli, using the Jacobian-one coordinate change, proves (5.53). $\square$

All sums and products in this section were finite. The shell counts, Abel estimates, and mesh averaging bounds contain no factor equal to the number of scales, tiles, or hierarchy levels. They therefore apply with identical constants to every finite truncation required in the packet reduction.

7. INTERPOLATION, BOUNDARY AVERAGING, AND COMPLETION

We continue with a finite popular collection and its adjoint sum $P = \sum_s \phi_s$. Thus $N = (|E|/|F|)^{1/2}$, $L/2 \leq \log N \leq 2L$, and $S \lesssim L^C |E|$. All statements in this section are uniform in the finite collection, its depth, and the dimension of $\mathcal{H}$. We write ϑ for the translation of the angular grid. The set Z_0 and the functions P, P_q, h_T, n do not depend on ϑ , whereas the smaller set $Z = Z_\vartheta$, the angular masses, and the analytic sums do. Until the boundary-averaging step, we fix ϑ and suppress it in the notation.

7.1. From the path estimate to analytic interpolation.

Proposition 7.1 (Vertex estimate). *For every nonempty root v , every shift σ , and every parameter with*

$$\Re \zeta = \mu e_\sigma, \quad \|\Im \zeta\|_2 \leq L^2,$$

one has

$$\|n_v^{\alpha/p-1} A_v(\zeta)\|_{L^p(Z; \mathcal{H})} \leq L^C S^{1/p}. \quad (7.1)$$

The constant and exponent are independent of the shift and the grid translation.

Proof. Recall the nonnegative amount

$$V_v = n_v + n_v^{\alpha-p} \|A_v\|^p.$$

On the set where $n_v \leq M$, Equation (5.24) gives

$$V_v \leq L^{Cp}(1+M)^{\alpha-1} n_v.$$

Here and below the bound on the imaginary parts is absorbed into L^{Cp} . On the remaining set, apply Lemma 5.7. At least half of the root amount is carried by good paths, and the terminal amount of a path ending at R is at most $L^{Cp}(1+M)^{\alpha-1} h_R$. Consequently,

$$\begin{aligned} \int_Z V_v &\leq L^{Cp}(1+M)^{\alpha-1} \int n_v \\ &\quad + 2L^{Cp}(1+M)^{\alpha-1} \sum_{R: s_R \in v} \int_{Z \cap \{\text{good } R\}} h_R \prod_{\text{active path of } R} \beta_{J_j}. \end{aligned}$$

The root and its descendants contain only their eligible tops; sums over tops here and subsequently are understood with that convention. Theorem 6.3, namely Equation (5.53), bounds each integral in the second line by $L^{Cp}(1+M)^C \int h_R$. Moreover,

$$\sum_R \int h_R \lesssim L^C S,$$

by the definition of the dilated decaying weights. We conclude that

$$\int_Z V_v \leq L^{Cp}(1+M)^C S.$$

Since M is a fixed power of L , taking a p -th root and using $n_v^{\alpha-p} \|A_v\|^p \leq V_v$ proves the result. $\square$

Lemma 7.2 (Interpolation of the shift parameters). *Equation (7.1) also holds at*

$$\zeta^c = (\mu/K, \dots, \mu/K).$$

Proof. Let

$$\mathfrak{S}_\mu = \{a \in [0, \infty)^K : \sum_\sigma a_\sigma = \mu\}, \quad \mathcal{F}_v(\zeta, z) = n_v(z)^{\alpha/p-1} A_v(\zeta, z).$$

Every nonempty mass is positive. For each fixed z , the defining sum is a finite sum of exponentials of linear functions of ζ , with real logarithms of positive ratios as coefficients. In particular it is entire in ζ .

We first give a uniform bound on the tube over $\mathfrak{S}_\mu$, without using the vertex estimate. From Equations (5.24) and (3.20), and from $\int n_v \leq \int n \lesssim L^C S$, we have, for $a \in \mathfrak{S}_\mu$ and $\eta \in \mathbb{R}^K$,

$$\begin{aligned} \|\mathcal{F}_v(a + i\eta)\|_{L^p(Z)} &\leq L^C (1 + \mu + \|\eta\|_2) \left(\int_Z n_v^\alpha \right)^{1/p} \\ &\leq L^C (1 + \|\eta\|_2) (NL^C)^{(\alpha-1)/p} S^{1/p}. \end{aligned} \quad (7.2)$$

The factor involving N is at most $\exp(C\sqrt{L})$, since $\log N \leq 2L$, $p = L^{1/2}$, and $2 < \alpha < 3$ for sufficiently large L .

Multiply the analytic family by

$$\mathcal{D}(\zeta) = \exp \left(\sum_\sigma \zeta_\sigma^2 \right).$$

On the tube under consideration,

$$|\mathcal{D}(a + i\eta)| = \exp(\|a\|_2^2 - \|\eta\|_2^2) \leq \exp(\mu^2 - \|\eta\|_2^2).$$

At a vertex, Proposition 7.1 controls $\mathcal{DF}_v$ whenever $\|\eta\|_2 \leq L^2$. For $\|\eta\|_2 > L^2$, Equation (7.2) gives the same bound: indeed

$$(1 + \|\eta\|_2) \exp(\mu^2 - \|\eta\|_2^2 + C\sqrt{L})$$

is bounded by an absolute constant in this range, after increasing the fixed lower threshold for L . Thus the damped family has norm at most $L^C S^{1/p}$ at every vertex, uniformly in all imaginary parts.

Here are details that justify applying scalar three-lines to this family. Test against any element b of the dual unit ball $L^{p'}(Z; \mathcal{H})$. One may first restrict the pairing to a bounded set on which every nonempty mass used in the finite family is bounded below by a positive constant. Such sets exhaust Z , and the restriction does not increase the norm of b . On each such set, integration defines an entire scalar function; logarithms of the finitely many masses are bounded there. The crude bound above and the damping make this scalar function bounded on every interpolating

strip. If two real vectors a_0, a_1 have established uniform bounds for all imaginary parts, apply the bounded scalar three-lines lemma [17, Lemma 3.3, p. 57] to

$$z \mapsto \int_Z \mathcal{D}((1-z)a_0 + za_1 + i\eta) \langle \mathcal{F}_v((1-z)a_0 + za_1 + i\eta, \cdot), b \rangle dx,$$

with the indicated preliminary restriction understood. The boundary lines have the required bounds because their imaginary vectors are still arbitrary. The real parts throughout the strip remain in the simplex. Starting with its vertices and successively taking convex combinations proves the same bound at their barycenter, with no factor depending on K . Exhaustion of Z , followed by duality, removes the preliminary restriction. Finally $\mathcal{D}(\zeta^c) = \exp(\mu^2/K)$ is an absolute factor. This proves the lemma. $\square$

7.2. The central weight and the exponent budget. The central coefficient identity (5.19) gives weights $\kappa_s \in (0, 1]$. Define

$$\mathcal{W}_\vartheta = \sum_s \kappa_s q_{T(s)} \phi_s.$$

By that identity, its portion in root v is $n_v^{\mu/K} A_v(\zeta^c)$.

Put $\gamma = 1 - \alpha/p + \mu/K$, which is positive for the values of L under consideration. Interpolation and Hölder's inequality give, for each root,

$$\begin{aligned} \int_Z \|n_v^{\mu/K} A_v(\zeta^c)\| &\leq \|n_v^{\alpha/p-1} A_v(\zeta^c)\|_{L^p(Z)} \left(\int_Z n_v^{\gamma p'} \right)^{1/p'} \\ &\leq L^C S^{1/p} |F|^{1-1/p} (NL^C)^\gamma. \end{aligned}$$

Only an absolute number of roots meet the bounded set of top slopes. Since $S \lesssim L^C |E| = L^C N^2 |F|$, summing over them proves

$$\int_{Z_\vartheta} \|\mathcal{W}_\vartheta\| \leq L^C |F| N^{1+(2-\alpha)/p+\mu/K}. \quad (7.3)$$

We record explicitly why this improves by every fixed negative power of L . The net decrease of the exponent of N is

$$\delta_L = \frac{\alpha-2}{p} - \frac{\mu}{K} = \frac{1}{p} \left(\alpha - 2 - \frac{\alpha-1}{K} \right).$$

The separated-count bound is a fixed power of L , so the definition of α gives $\alpha - 2 \geq c_1/\log L$ with a fixed $c_1 > 0$. Also $\alpha - 1 \leq 2$ and $K \geq (\log L)^2$. For sufficiently large L , therefore,

$$\delta_L \geq \frac{c_1}{2\sqrt{L} \log L}, \quad \delta_L \log N \geq \frac{c_1 \sqrt{L}}{4 \log L}.$$

In particular, for every fixed $Q > 0$, after increasing the absolute lower threshold for L , Equation (7.3) implies, uniformly in ϑ ,

$$\int_{Z_\vartheta} \|\mathcal{W}_\vartheta\| \leq L^{-Q} N |F|. \quad (7.4)$$

This conclusion absorbs any fixed polynomial losses in the preceding estimates; none of their exponents is changed here.

7.3. Boundary averaging with the eligibility condition. We now restore the dependence on the uniformly translated angular grid. The next estimate is pointwise on the fixed set Z_0 .

Lemma 7.3 (Expected boundary loss). *Let $\epsilon = 4G^{-1/4}$. For $z \in Z_0$,*

$$\mathbb{E}_\vartheta \|P_q(z) - \mathcal{W}_\vartheta(z)\| \leq L^C \epsilon n(z). \quad (7.5)$$

Here the expectation is normalized Lebesgue measure for the common translation modulo $d(0)$; independent translations at different levels are neither assumed nor needed.

Proof. Fix $z \in Z_0$ and a top T , and suppress z from the masses. For $i < k_T$ let

$$r_i(\vartheta) = \log \frac{m_{v_{i+1}(T)}}{n_{v_i(T)}} \geq 0, \quad M_i(b) = \sum_{\substack{U: |s_U - s_T| \leq b d(i) \\ d_U \leq d(i)/B^2}} h_U.$$

Unlike r_i , the latter mass does not depend on the grid translation. Each mass in this display is at least h_T when $i < k_T$, and is at most n .

Let $b_i(\vartheta) \in [0, 1/2]$ be the distance from s_T to the nearer boundary of its level- i interval, divided by $d(i)$. An extra top counted in $m_{v_{i+1}(T)} - n_{v_i(T)}$ has slope within $\epsilon d(i)$ of s_T . Indeed the neighbor distance in Equation (5.7) is $G^{-1/4}d(i)$, and the two level- $(i+1)$ interval lengths add at most $2d(i)/G$. Such a top is eligible at level $i+1$, hence also at level i . It lies outside the immediate parent. It follows that $r_i = 0$ if $b_i > \epsilon$. If $0 < b_i \leq \epsilon$, the immediate parent contains every top counted by $M_i(b_i)$, whereas its extra mass is disjoint from these inner tops and is at most $M_i(\epsilon) - M_i(b_i)$. Consequently

$$r_i(\vartheta) \leq \log \frac{M_i(\epsilon)}{M_i(b_i(\vartheta))}. \quad (7.6)$$

Endpoint coincidences can be omitted: for the finite set of slopes and levels they form a null set of translations.

For a fixed number $0 < b < \epsilon$, set $r = \lceil \log_G(\epsilon/b) \rceil$. The inclusion

$$M_{i+r}(\epsilon) \leq M_i(b) \quad (7.7)$$

uses both conditions in the definition of M_i : the radius satisfies $\epsilon d(i+r) \leq b d(i)$, and the precision threshold $d(i+r)/B^2$ is no larger than $d(i)/B^2$. To handle the end of the path without losing a term, define

$$a_i = \begin{cases} \log(M_i(\epsilon)/h_T), & 0 \leq i < k_T, \\ 0, & i \geq k_T. \end{cases}$$

If $i+r < k_T$, Equation (7.7) applies. If $i+r \geq k_T$, use $M_i(b) \geq h_T$ instead. In both cases

$$\log \frac{M_i(\epsilon)}{M_i(b)} \leq a_i - a_{i+r}.$$

Summing and cancelling gives

$$\sum_{i < k_T} \log \frac{M_i(\epsilon)}{M_i(b)} \leq \sum_{0 \leq i < \min(r, k_T)} a_i \leq r \log \frac{n}{h_T}. \quad (7.8)$$

This argument also covers $k_T = 0$, with empty sums.

Because $d(0)/d(i) = G^i$ is an integer, the common uniform translation is uniform modulo every $d(i)$. Thus each b_i has density 2 on $[0, 1/2]$. Linearity of expectation, Equation (7.6), and

Equation (7.8) imply

$$\begin{aligned} \mathbb{E}_\vartheta \sum_{i < k_T} r_i(\vartheta) &\leq 2 \int_0^\epsilon \sum_{i < k_T} \log \frac{M_i(\epsilon)}{M_i(b)} db \\ &\leq 2 \log(n/h_T) \int_0^\epsilon \lceil \log_G(\epsilon/b) \rceil db \leq C\epsilon \log(n/h_T). \end{aligned} \quad (7.9)$$

For completeness the integral in the last line equals $\epsilon/(1 - G^{-1})$: express the ceiling as a sum of indicators of $b < \epsilon G^{-j}$, $j \geq 0$, and integrate. The dependence of the different b_i 's is absent from this calculation.

On the fixed top T , the value of κ_s depends only on the stopping level k_s . As packet length increases, k_s increases; Equation (5.19) shows that these values are nonincreasing. Their deficits from one are nondecreasing and satisfy

$$0 \leq 1 - \kappa_s \leq \frac{\mu}{K} \sum_{i < k_T} r_i(\vartheta),$$

using $k_s \leq k_T$ and $1 - e^{-x} \leq x$. Group the packets by stopping level. Every interval of these levels is an interval of packet lengths, so Equation (3.18) bounds its unweighted sum by $L^C h_T^8$. Summation by parts against the monotone deficits, and $|q_T| \leq 1$, therefore give

$$\left\| \sum_{s: T(s)=T} (1 - \kappa_s) q_T \phi_s \right\| \leq L^C h_T^8 \frac{\mu}{K} \sum_{i < k_T} r_i(\vartheta).$$

For example, write the deficits as sums of their nonnegative increments and use the bound on the corresponding suffix sums; the sum of these increments is their largest value. Taking expectations, using $\mu/K \leq 1$, and summing over tops proves

$$\mathbb{E}_\vartheta \|P_q - \mathcal{W}_\vartheta\| \leq L^C \epsilon \sum_T h_T^8 \log(n/h_T).$$

The weights have a fixed absolute upper bound. For every such weight,

$$h_T^8 \log(n/h_T) \leq C h_T (1 + \log_+ n),$$

because $h^7 \log_+(1/h)$ is bounded on a fixed bounded interval. Equation (3.20) gives $1 + \log_+ n \lesssim L$ on Z_0 . This proves Equation (7.5). $\square$

7.4. Recovering the original adjoint sum.

Theorem 7.4 (Improvement for the popular model). *The popular adjoint sum satisfies Equation (2.32); namely,*

$$\int_F \|P\| \lesssim L^{-20} N |F|.$$

Proof. We make the exceptional-set comparison explicit because the good set depends on the grid. Write $\|P\|_2 \leq L^{C_P} \sqrt{|E|}$, as supplied by Equation (2.3). Proposition 3.7 and Proposition 3.8 allow the fixed polynomial thresholds to be chosen so that

$$\int_{F \setminus Z_0} \|P\| \leq L^{-30} N |F|, \quad \int_{Z_0} \|P - P_q\| \leq L^{-30} N |F|. \quad (7.10)$$

Indeed a discarded area at most $L^{-2C_P-60} |F|$ has the stated first bound by Cauchy–Schwarz. The exponentially small concentration and count exceptions meet this area bound for sufficiently large L .

For each fixed translation the good-path construction gives

$$|Z_0 \setminus Z_\vartheta| \leq S e^{-9L}.$$

This is Markov's inequality applied to the weighted bad-path mass: its integral is at most Se^{-10L} , and the deletion threshold is e^{-L} . Since $S \lesssim L^C N^2 |F|$ and $N^2 \leq e^{4L}$,

$$\int_{Z_0 \setminus Z_\vartheta} \|P\| \leq L^{C_P} \sqrt{|E|} (Se^{-9L})^{1/2} \leq L^C e^{-5L/2} N |F| \leq L^{-30} N |F|. \quad (7.11)$$

These are bounds for each translation; no union over translations is taken.

For every ϑ , the triangle inequality gives the following decomposition, whose left side is independent of ϑ :

$$\int_F \|P\| \leq \int_{F \setminus Z_\vartheta} \|P\| + \int_{Z_\vartheta} \|P - P_q\| + \int_{Z_\vartheta} \|P_q - \mathcal{W}_\vartheta\| + \int_{Z_\vartheta} \|\mathcal{W}_\vartheta\|. \quad (7.12)$$

The first two terms are controlled by Equations (7.10) and (7.11). The final term is controlled by Equation (7.4) with $Q = 30$. Average the remaining term over translations. It is nonnegative and $Z_\vartheta \subset Z_0$, so Tonelli's theorem and Lemma 7.3 yield

$$\begin{aligned} \mathbb{E}_\vartheta \int_{Z_\vartheta} \|P_q - \mathcal{W}_\vartheta\| &\leq \int_{Z_0} \mathbb{E}_\vartheta \|P_q - \mathcal{W}_\vartheta\| \\ &\leq L^C \epsilon \int_{Z_0} n \leq L^C \epsilon N |F| \leq L^{-30} N |F|. \end{aligned}$$

The penultimate inequality uses Equation (3.20); the last uses $\epsilon = 4G^{-1/4}$ and $\log G \asymp L^7$. All functions being averaged are measurable: for finitely many tops, packets, and levels, the angular assignments change only at finitely many translation values, and the good sets were defined by measurable quantities. Thus this averaging is legitimate even though Z_ϑ and $\mathcal{W}_\vartheta$ use the same translation. Averaging Equation (7.12) now proves the asserted estimate. In particular, on every discarded set we have estimated the original sum P . $\square$

7.5. Choice of parameters and completion of the reductions. We collect the order of the absolute choices. First fix the packet adaptation, separation, and overlap constants and the fixed geometric enlargements used with Theorem 1.2. Fix the local slope bound C_{Lip} , strip length b_0 , and outer length $a = a_*$ as in (1.2), with the required room for those enlargements. Choose the low-density exponent C_2 by (2.28), before constructing the popular forest.

Next fix the decay exponent in the top-count weight and the concentration orders that determine K_1 . Choose the fixed mask-removal and exceptional-area powers, then choose A using the count at dilation K_1 , whose constants do not depend on A . The count at the larger dilation $K_0 = L^{A+2}$ is used only after this comparison. Fix the threshold in (3.20), the separated-count exponent defining D , and the good-path thresholds. These choices determine the polynomial powers in the motion, prediction, and residual estimates.

Now choose $M = L^{A_M}$ as in (6.21). For large L , $\mu < 1/12$, so this fixed exponent controls both

$$pL^C M^{-1/2+\mu} \quad \text{and} \quad pL^C M^{-1/3+\mu}$$

in the marked and unary residual estimates. The good-path thresholds were fixed before A_M , and their required proportion remains valid:

$$e^{-L} L^{C_p} (1 + M)^{\alpha-1} = \exp(-L + O(\sqrt{L} \log L) + O(\log L)) < \frac{1}{2}$$

for sufficiently large L .

Finally choose L_0 after the outer length and all fixed exponents. It ensures $\tau < a$, the geometric smallness requirements, and all the inequalities used above, because

$$\log B = O(L^7 (\log L)^2) = o(L^9), \quad \log(N^\mu) = O(\sqrt{L}) = o(\log G),$$

and, for every fixed exponent C ,

$$\log(m^C \tau) = O(L^9) - \Theta(L^{.97}) \longrightarrow -\infty.$$

It also enforces the saving in (7.4). All choices are independent of the depth, tile count, and dimension.

Theorem 7.5 (The three model estimates). *For the packet forms in (1.33), let u be a bounded measurable real slope, let $F, E \subset \mathbb{R}^2$ be measurable sets of positive finite measure, and suppose $\|f\|_{\mathcal{H}} \leq \mathbf{1}_F$ and $\|g\|_{\mathcal{H}} \leq \mathbf{1}_E$ pointwise. The following estimates hold uniformly over every finite subcollection.*

(i) For $m \geq 1$,

$$|\Lambda(f, g)| \lesssim (1 + \log(2m))^C \sqrt{|F||E|}.$$

(ii) Suppose $L \geq L_0$, with τ, m, W chosen by (1.17), and $L/2 \leq \log \sqrt{|E|/|F|} \leq 2L$. If $|u| \leq 1$, $\text{Lip}(u) \leq C_{\text{Lip}}$, and $F, E \subset I \times \mathbb{R}$ for an interval I of length at most b_0 , then

$$|\Lambda(f, g)| \lesssim L^{-20} \sqrt{|F||E|}.$$

(iii) For the scalar model with $m = 1$ and $|E| \leq |F|$, there is an absolute $c > 0$ such that

$$|\Lambda(f, g)| \lesssim \sqrt{|F||E|} \left(\frac{|E|}{|F|} \right)^c.$$

Proof. The first and third assertions are precisely the ordinary estimates proved with (2.3). For the second, the fixed deletion of tiles with $D_E(s) \leq L^{-C_2}$ in Section 2.7 contributes at most $CL^{-40} \sqrt{|F||E|}$, by (2.28). Assign the remaining tiles to the popular tops as already constructed. Their adjoint sum is P , and the coefficient convention in Equation (1.33) gives

$$|\Lambda_{\text{popular}}(f, g)| \leq \int_F \|P\|.$$

Theorem 7.4 bounds this by $L^{-20} N|F| = L^{-20} \sqrt{|F||E|}$. Adding the deleted contribution proves the second assertion. All preceding arguments apply to arbitrary finite tile sets and thus give the claimed uniformity. $\square$

Completion of the proofs of Theorems 1.1 and 1.4. Theorem 7.5 supplies exactly the three hypotheses of Proposition 1.15. That proposition therefore proves the band restricted estimate (1.4). The dependence here is forward: the reductions established the implication from the model estimates, which have now been proved without using the band restricted theorem.

All estimates were obtained first for finite packet collections and positive inner truncations, with constants independent of both. The summable smooth expansions and translated-lattice averages in the packet reduction consequently pass to their limits as specified in Proposition 1.15. The amplitude summation, vertical Littlewood–Paley reassembly, and spatial localization proved in the reductions give a uniform strong L^2 estimate for the original fixed-field transform with a positive inner cutoff. For a Schwartz input its symmetric principal value exists pointwise; Fatou’s lemma gives the same strong bound for the principal-value operator, which extends by density. For the already chosen outer length $a = a_*$, Chebyshev’s inequality gives the weak (2, 2) assertion of Theorem 1.1. The constants are independent of the vector field, the input, and the level in that inequality. The dilation calculation following Theorem 1.1 gives the outer length $a_*/\text{Lip}(v)$ for positive Lipschitz seminorm; constant directions have already been handled by the one-dimensional theorem. $\square$

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